

# A FINITELY PRESENTED NON-MF GROUP

ANONYMOUS

**ABSTRACT.** A countable discrete group is MF if it embeds in the unitary group of a quotient  $\mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$ ; equivalently, if it admits finite-dimensional approximate unitary representations that are asymptotically multiplicative in operator norm and separate every nontrivial element from the identity. We construct a finitely presented group  $E$ , on eight generators and forty-one relators (equivalently, on six generators with thirty-two), that is not MF. This refutes the conjecture that every countable group is MF.

The mechanism is a finiteness principle. If  $\pi$  maps a group onto a finite group and  $t\Gamma t^{-1} \subseteq \Gamma$ , then  $\pi(t\Gamma t^{-1}) = \pi(\Gamma)$ : the left-hand side is a conjugate of the right-hand side contained in it, and the two are finite of the same cardinality. We prove the analogue for  $\mathcal{Q}$ , with finiteness of an ultraproduct as a  $C^*$ -algebra — every isometry in it is unitary — in place of the count.

Let  $\Gamma \leq H$  have property (T), let  $t\Gamma t^{-1} \subseteq \Gamma$ , let  $c$  centralize  $\Gamma$ , and put  $d = tct^{-1}$  and  $u = [d, a]$  for  $a \in \Gamma$ . We prove that in every approximate unitary representation  $(U_n)$  of  $H$ , conjugation by  $t$  preserves the asymptotic commutant of  $\Gamma$  in normalized Hilbert–Schmidt norm; hence  $\|U_n(u) - 1\|_2 \rightarrow 0$ . In  $E$  the element  $w = u^2$  is central and satisfies  $w^2 = 1$ , and a quotient of  $E$  onto a Clifford group carries  $w$  to the central element  $-1$ ; so  $w \neq 1$ , while  $w$  lies in the kernel of every homomorphism  $E \rightarrow \mathcal{U}(\mathcal{Q})$ . The construction applies to any finitely presented property-(T) group with a proper injective self-embedding, and  $E$  is its instance at  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ .

The reduced group  $C^*$ -algebra  $C_r^*(E)$  is separable and carries a faithful tracial state, hence is stably finite, but is not MF.

---

2020 *Mathematics Subject Classification.* Primary 46L05; Secondary 20F05, 22D25, 20E26.

*Key words and phrases.* MF groups, approximate unitary representations, property (T), group  $C^*$ -algebras, Clifford groups.

## CONTENTS

|                                                                              |    |
|------------------------------------------------------------------------------|----|
| 1. Introduction                                                              | 4  |
| The result                                                                   | 4  |
| The mechanism                                                                | 4  |
| Relation to prior work                                                       | 6  |
| Formalization and computational data                                         | 9  |
| 2. Matrix quotients                                                          | 9  |
| 3. One-sided Kazhdan transport                                               | 10 |
| The exact case                                                               | 11 |
| 4. The central involution obstruction                                        | 13 |
| 5. The Kazhdan–Clifford construction                                         | 15 |
| 6. A finitely presented non-MF group                                         | 16 |
| 7. The base, and a smaller presentation                                      | 19 |
| 7.1. Property (T) of the base                                                | 19 |
| 8. Nontriviality of $w$                                                      | 22 |
| 9. Consequences of the example                                               | 23 |
| The quotient by the central involution                                       | 24 |
| A stably finite non-MF group algebra                                         | 24 |
| A sofic example                                                              | 24 |
| 10. Transport at every scale                                                 | 26 |
| 11. Obstructions from a one-sided conjugation datum                          | 27 |
| 11.1. An intrinsic form                                                      | 29 |
| 11.2. The normal-Kazhdan form of the obstruction                             | 30 |
| 12. The finite-dimensional obstruction                                       | 32 |
| 13. The cyclic comparison                                                    | 34 |
| 14. The scaling family                                                       | 36 |
| 15. Consequences                                                             | 36 |
| 15.1. The MF radical                                                         | 36 |
| 15.2. Permanence and its failure                                             | 39 |
| 15.3. Matricial stability and the radical                                    | 39 |
| 15.4. Orbit collapse                                                         | 40 |
| 15.5. The reduced group algebra of $E$                                       | 48 |
| 15.6. A sofic non-MF group                                                   | 48 |
| 15.7. Marked-limit closure                                                   | 50 |
| 15.8. A finite universal Horn obstruction                                    | 50 |
| 15.9. Undecidability of MF recognition                                       | 51 |
| 16. Limitations of the operator-norm method                                  | 53 |
| Questions                                                                    | 54 |
| Appendix A. The MF equivalences                                              | 55 |
| Appendix B. The finite-stage transport proof                                 | 57 |
| Appendix C. The maximal group $C^*$ -algebra                                 | 59 |
| Appendix D. The thirteen-relator presentation of $\mathrm{SL}_3(\mathbb{Z})$ | 60 |
| Acknowledgements                                                             | 62 |



## 1. INTRODUCTION

Can every countable discrete group be realized by matrices, if multiplication is only required to hold up to operator-norm errors that vanish along a sequence of finite dimensions? Shulman calls the positive assertion the *MF conjecture* [Sh], and no group was known to fail it [Sc, Sh]. We prove that the answer is no: there is an explicit finitely presented group, on eight generators and forty-one relators, that is not MF. Residually finite groups satisfy the conjecture through their finite quotients, as do LEF groups [CDE] and, by quasidiagonality [TWW], all amenable groups. Recent work establishes it for many amalgamated free products [Sc, Sh, GKEMP]. Throughout,  $[g, h] = ghg^{-1}h^{-1}$  denotes the commutator of two group elements. The additive commutator of two operators is written  $[x, y]_- = xy - yx$ , so that the two brackets are never confused where a group element and its matrix image occur together.

**The result.**

**Theorem A** (an explicit finitely presented non-MF group). *Let  $E$  be the group on the eight generators  $v_1, v_2, v_3, x, y, z, t, c$  with the forty-one relators displayed in Definition 6.1, and put*

$$w = [tct^{-1}, v_1(tct^{-1})v_1^{-1}].$$

*The defining relations make  $w$  central; together with  $c^2 = 1$ , centrality implies  $w^2 = 1$ . Then the following statements hold:*

- (1)  *$w \neq 1$  in  $E$ , so  $w$  is a central involution;*
- (2) *for every sequence  $(d_n)$  of positive integers and every homomorphism  $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$  into the unitary group of the corresponding matrix quotient, one has  $\Theta(w) = 1$ .*

*In particular  $E$  is not MF, and neither  $C_{\max}^*(E)$  nor  $C_r^*(E)$  is an MF  $C^*$ -algebra. A Tietze reduction presents the same group on six generators with thirty-two relators (Section 7); the eight-generator form is displayed because it exhibits the construction.*

The group is written out in Definition 6.1; Section 6 proves the theorem.

**The mechanism.** Operator norm is the approximation norm throughout. The normalized Hilbert–Schmidt norm enters only as the Hilbert-space norm on which property (T) acts, and the passage back from it to the operator norm is what the central involution performs. Let  $\pi$  be a homomorphism to a finite group and let  $t\Gamma t^{-1} \subseteq \Gamma$ . Then  $\pi(t\Gamma t^{-1}) = \pi(\Gamma)$ : the left side is a conjugate of the right side contained in it, and the two are finite of the same cardinality. So an element  $c$  centralizing  $\Gamma$  has  $\pi(tct^{-1})$  centralizing  $\pi(\Gamma)$ , and every commutator  $[tct^{-1}, a]$  with  $a \in \Gamma$  lies in  $\ker \pi$ .

Every finite-dimensional representation  $\pi$  of the ambient group over every field gives the same conclusion, with the commutant in place of the subgroup. Let  $C$  be the commutant of  $\pi(\Gamma)$  in  $\text{End}_k(V)$ . The inclusion  $t\Gamma t^{-1} \subseteq \Gamma$

implies  $C \subseteq \pi(t) C \pi(t)^{-1}$ , the two sides have equal finite dimension, and therefore

$$C = \pi(t) C \pi(t)^{-1}$$

(Theorem 3.3 and Remark 12.2). Since  $\pi(c) \in C$ , also  $\pi(tct^{-1}) \in C$ , so it commutes with every  $\pi(a)$ ,  $a \in \Gamma$ . The rest of the paper carries out this commutant argument for  $\mathcal{Q}$ : the commutant becomes an asymptotic commutant in normalized Hilbert–Schmidt norm, equality of dimensions becomes finiteness of an ultraproduct as a  $C^*$ -algebra, and property (T) supplies the spectral projection onto the invariant vectors.

Operator norm defines both the approximation and the quotient  $\mathcal{Q}$ ; the normalized Hilbert–Schmidt norm enters only as the Hilbert-space norm on which property (T) acts. Regard  $M_d(\mathbb{C})$  as the Hilbert space  $L^2(M_d(\mathbb{C}), \text{tr}_d)$ , on which conjugation by a unitary acts unitarily. The passage between the two norms is dimension-free: for unitaries  $u_1, u_2$ ,

$$\|\text{Ad } u_1 - \text{Ad } u_2\|_{B(L^2(M_d(\mathbb{C}), \text{tr}_d))} \leq 2 \|u_1 - u_2\|,$$

so an operator-norm asymptotic representation induces one by conjugation on the Hilbert–Schmidt spaces.

Let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and let  $t, c \in H$  satisfy

$$t \iota(\Gamma) t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

Passing to a norm ultraproduct  $B_\omega$  of the blocks turns the induced conjugation action into a unitary representation  $\pi$  of  $H$ . Property (T) isolates the eigenvalue 1 of a Kazhdan average, so functional calculus produces the orthogonal projection  $P \in B_\omega$  onto the  $\iota(\Gamma)$ -invariant vectors. Writing  $V = \pi(t)$ , the inclusion above implies  $P \leq V P V^*$ ; the two projections are Murray–von Neumann equivalent, and  $B_\omega$  contains no proper isometry, so  $P = V P V^*$ . Hence conjugation by  $t$  preserves the asymptotic commutant of  $\iota(\Gamma)$  in normalized Hilbert–Schmidt norm (Theorem 3.4).

Put  $d = tct^{-1}$  and  $u = [d, \iota(a)]$  for  $a \in \Gamma$ . Since  $c$  commutes with  $\iota(\Gamma)$ , its lifts lie in that asymptotic commutant, and the theorem shows that the lifts of  $d$  lie there as well. Consequently, in every operator-norm asymptotic representation  $(U_n)$  of  $H$ ,

$$\|U_n(u) - 1\|_2 \rightarrow 0.$$

Suppose now that  $w = u^2$  is central in  $H$  with  $w^2 = 1$ , and let  $\Theta$  be a homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  with  $\Theta(w) \neq 1$ . Then  $\Theta(w)$  is a self-adjoint unitary commuting with  $\Theta(H)$ , so

$$q = \frac{1}{2}(1 - \Theta(w))$$

is a nonzero projection commuting with  $\Theta(H)$ . Choosing unitary lifts of  $\Theta$  and rounding the corresponding coordinate averages to genuine projections represents  $q$  by projections  $q_n \in M_{d_n}(\mathbb{C})$ , nonzero for infinitely many  $n$  and asymptotically commuting with the lifts. Compressing to  $q_n$ , and giving each corner its own normalized trace, yields a second operator-norm asymptotic

representation of  $H$ , one in which  $w$  goes to  $-1$ . There the previous display gives  $\|U_n(u)^2 - 1\|_2 \rightarrow 0$ , while  $\|-1 - 1\|_2 = 2$ . Hence  $\Theta(w) = 1$  for every such  $\Theta$ , and if  $w \neq 1$  then  $H$  is not MF (Theorem 4.1).

To obtain  $w \neq 1$ , adjoin  $t$  and  $c$  to a finitely presented Kazhdan group  $\Gamma$  that is not co-Hopfian, along a proper injective endomorphism  $\alpha$ , and take  $a \in \Gamma \setminus \alpha(\Gamma)$ . In the direct limit of  $\alpha$  the cosets  $t\Gamma$  and  $at\Gamma$  are distinct, since  $t^{-1}at \in \Gamma$  would place  $a$  in  $\alpha(\Gamma)$ . Let the ambient group permute the generators of a Clifford group indexed by that coset space, and send  $c$  to the generator at the base coset; then  $d$  and  $ada^{-1}$  are carried to the generators at the two distinct cosets above. Distinct Clifford generators anticommute, so  $u$  is carried to a product of two anticommuting involutions and  $w = u^2$  to  $-1$  (Theorem 5.1).

Taking

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \quad \alpha(v, A) = (2v, A), \quad a = (e_1, I)$$

gives the group  $E$  of Theorem A.

Only the subgroup  $\iota(\Gamma)$  is assumed to have property (T). The group  $E$  surjects onto  $\mathbb{Z}$ , by sending  $t$  to 1 and every other generator to 0, so  $E$  itself does not have property (T).

Three consequences deserve mention here. The reduced group  $C^*$ -algebra  $C_r^*(E)$  is separable and stably finite but not MF (Theorem C). The Clifford quotient  $W$  used to prove  $w \neq 1$  is itself finitely generated, sofic and not MF, so neither soficity nor hyperlinearity implies the MF property (Theorem D);  $E$  itself is sofic as well, by an argument owing nothing to the rest of this paper, so the separation is already witnessed by a finitely presented group (Theorem E). The quotient  $E/\langle w \rangle$  is also not MF, so  $\{1, w\}$  is a proper subgroup of the MF radical of  $E$  (Theorem B). The operator-norm method of this paper decides soficity for neither group on its own (Section 16).

**Relation to prior work.** The corresponding  $C^*$ -algebra question is older. Blackadar and Kirchberg, who introduced MF algebras, remarked that no stably finite separable  $C^*$ -algebra was known not to be MF [BK, CDE]. Their natural candidate was  $C_r^*(F_2)$ , which Haagerup and Thorbjørnsen later proved to be MF [HT]. The negative solution of the Connes embedding problem subsequently supplied abstract stably finite non-MF algebras [JNVWY, FGH]. No countable group was known whose reduced or maximal group  $C^*$ -algebra fails to be MF [Sh].

In his 2018 ICM address, Thom asked for which Schatten  $p$ -norms there exist non-approximable groups [Th, Question 3.11]. The case  $p = 2$  was settled by De Chiffre–Glebsky–Lubotzky–Thom [DGLT], the cases  $1 < p < \infty$  by Lubotzky–Oppenheim [LuO], and the case  $p = 1$  by Bachner–Dogon–Lubotzky [BDL]. Three approximation problems were left by that work: whether every countable group is sofic (permutations in the Hamming metric), hyperlinear (unitaries in the normalized Hilbert–Schmidt norm), and MF (unitaries in operator norm). The first was answered negatively in 2026:

a nonsofic group was constructed by OpenAI [OAI], Fournier-Facio then gave a torsion-free example [FFF], and Kun–Thom developed further wreath-product examples [KT26]. The present paper answers the third negatively; at the time of writing the hyperlinear problem is open. Our construction begins, as those arguments do, with a relation  $t\Gamma t^{-1} \subseteq \Gamma$  for a property-(T) subgroup  $\Gamma$ . Because the approximation is by unitaries in operator norm rather than by permutations in the Hamming metric, the rigidity input is a different one: conjugation by  $t$  preserves the asymptotic commutant of  $\Gamma$  in normalized Hilbert–Schmidt norm, in every operator-norm asymptotic representation. In the language of approximate representations, a countable discrete group  $G$  is *MF* if, for every finite subset  $F \subseteq G$  and every  $\varepsilon > 0$ , there is a unitary matrix model  $V: G \rightarrow \mathcal{U}(r)$  that is multiplicative within  $\varepsilon$  in operator norm on  $F$  and separates distinct elements of  $F$  by operator-norm distance at least 1. The notion was introduced, in the equivalent form below, by Carrión–Dadarlat–Eckhardt [CDE], and is used in that form by Shulman and by Bachner–Dugan–Lubotzky [Sh, BDL]. Let  $(d_n)_{n \in \mathbb{N}}$  be a strictly increasing sequence of positive integers and let

$$(1) \quad \mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$$

be the associated matrix quotient: bounded sequences of matrices modulo sequences converging to 0 in operator norm. Following Blackadar–Kirchberg [BK], a separable  $C^*$ -algebra is *MF* if it embeds into some algebra of the form (1); the group  $G$  is *MF* if there is an injective homomorphism  $G \rightarrow \mathcal{U}(\mathcal{Q})$ . The two formulations are equivalent (Proposition 1.1). The unitaries of  $\mathcal{Q}$  admit a direct description. Write

$$(2) \quad \mathcal{N} = \{(u_n) \in \prod_n \mathcal{U}(d_n) : \|u_n - 1\| \rightarrow 0\},$$

a normal subgroup of  $\prod_n \mathcal{U}(d_n)$ . Polar decomposition shows that the canonical map  $\kappa_{(d_n)}: \prod_n \mathcal{U}(d_n) / \mathcal{N} \rightarrow \mathcal{U}(\mathcal{Q})$  is an isomorphism (Lemma 2.2). An *MF* embedding may therefore be given either as a homomorphism into  $\mathcal{U}(\mathcal{Q})$  or as a sequence of unitaries for each group element.

**Proposition 1.1** (Equivalent formulations of the MF property). *For every countable group, the  $C^*$ -algebra definition above is equivalent to the definition by sequences of unitaries. Both are equivalent to the existence of approximate unitary representations with separation constant 1, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.*

Section A defines the approximate unitary representations and proves the proposition.

Throughout, “MF” means this property, in the operator-norm sense of Carrión–Dadarlat–Eckhardt. Two stronger conventions occur in the literature and are not used here: in [GKEMP] the approximating maps must also recover the canonical trace and the left-regular norm on every group-ring

element, and the purely matricial field (PMF) property of [MdlS, Definition 1.2] requires finite-dimensional representations whose group-ring norms converge to the regular norms.

The closest analogue is the recent work on nonsofic groups. One-sided conjugation of a property-(T) subgroup, together with an element centralizing that subgroup, appears in [OAI, KT26], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; Alekseev–Thom prove a related centralizer rigidity for sofic embeddings [AT]. Those results concern Hamming or tracial approximations. Here the group acts by conjugation on the Hilbert–Schmidt space of each matrix block, and finiteness of the norm ultraproduct as a  $C^*$ -algebra forbids a proper one-sided compression of the Kazhdan fixed space; Section B proves that step quantitatively at finite stages.

Slofstra’s hyperlinear-profile construction uses a Clifford group, a distinguished central involution, a shift, and an HNN doubling map [Slo18, Section 2 and Remark 3.3], and his quotients by the elements sent to the identity by all exact, respectively by all approximate, finite-dimensional representations [Slo17, Definitions 2.5–2.7] are the exact and the approximate analogues of the MF radical used below. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Fritz’s marked  $BS(2, 3)$  example is the context for the cyclic comparison of Section 13 [Fri, SV]. Bachner–Dogon–Lubotzky develop the operator-norm rounding, compression, and polar-correction procedures used in Section 11 [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5]; in the proof of their Proposition 1.6 they treat the finite-central case by a character-isotypic corner and reduce the general case of a finite normal subgroup to it through the finite-index centralizer. Theorem 11.3 below combines that corner with the one-sided conjugation of a Kazhdan subgroup, and its hypotheses constrain only the subgroup, not the quotient. Central extensions and rigidity of approximate representations also drive the Frobenius obstruction of De Chiffre–Glebsky–Lubotzky–Thom [DGLT]; finite-target approximation and extension results for weak soficity appear in [GR, Gl].

The passage from operator norm to the Hilbert–Schmidt conjugation representation belongs to the  $\pi \otimes \bar{\pi}$  framework of Bekka and Bekka–Valette [Be, BV], and Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. The group-MF framework is that of Carrión–Dadarlat–Eckhardt, who also give, through Abels’ group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18].

The group property is weaker than the algebra property. For the algebra one asks for an embedding  $C_r^*(E) \hookrightarrow \mathcal{Q}$  of  $C^*$ -algebras; for the group, only for an injective homomorphism  $E \rightarrow \mathcal{U}(\mathcal{Q})$ , which need not extend to the algebra. Failure of MF for a group  $C^*$ -algebra therefore leaves open whether the group itself embeds. For  $E$  both fail, and for the same reason.

**Formalization and computational data.** A Lean 4 development accompanies the paper, supplied with this submission as supplementary material. It proves Theorems A, B, C, D and E, the forty-one-relator presentation of Definition 6.1, and the finite-stage form of the transport theorem of Section B; the correspondence between numbered statements and declarations is recorded there. Results quoted from the literature are cited where they are used; the Adian–Rabin construction and the Novikov–Boone theorem appear as explicit hypotheses of Corollary 15.24 rather than as citations inside its proof. Two steps rest on finite computations in exact arithmetic — the rational spectral certificate of Section 7 and the matrix identities of Section 8 — whose data are in the same repository.

## 2. MATRIX QUOTIENTS

All groups are discrete and countable. We use the standard formulation of property (T) for unitary representations on complex Hilbert spaces. A  $C^*$ -homomorphism is unital only when this is stated; in particular an MF embedding may be nonunital. For a unital  $C^*$ -algebra  $A$  we write  $\mathcal{U}(A)$  for its unitary group. On  $M_r(\mathbb{C})$  we use the normalized trace  $\mathrm{tr}_r$ , the operator norm  $\|\cdot\|$ , and the normalized Hilbert–Schmidt norm  $\|x\|_2 = \mathrm{tr}_r(x^*x)^{1/2}$ . Several arguments below pass to a corner and renormalize, so where the trace has to be named we write  $\|x\|_{2,\mathrm{tr}_r}$  for the same quantity. The unnormalized trace is  $\mathrm{Tr} = r \cdot \mathrm{tr}_r$  on  $M_r(\mathbb{C})$ . Convergence in normalized Hilbert–Schmidt norm is always asserted of sequences of matrices, never of group elements. We use without comment:  $\|x\|_2 \leq \|x\|$ ;  $|\mathrm{tr}_r(x)| \leq \|x\|$ ; and  $\|uxv\|_2 = \|x\|_2$  for unitaries  $u, v$ .

Given a sequence  $(d_n)_{n \in \mathbb{N}}$  of positive integers, write

$$\mathcal{Q}((d_n)_{n \in \mathbb{N}}) = \prod_{n \in \mathbb{N}} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in \mathbb{N}} M_{d_n}(\mathbb{C}),$$

where the product is the bounded  $C^*$ -product and  $\bigoplus_n$  is the  $c_0$ -sum, the ideal of sequences with  $\|x_n\| \rightarrow 0$ ; equivalently,  $\mathcal{Q}$  is the corona algebra of  $\bigoplus_n M_{d_n}(\mathbb{C})$ . Write  $q: \prod_n M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$  for the quotient map.

The following lifting facts are standard perturbation theory; see Loring [Lo] for a systematic treatment.

**Lemma 2.1** (lifting). *Every unitary  $u \in \mathcal{Q}$  lifts to a sequence of unitaries.*

*Proof.* Lift  $u$  to a bounded sequence  $(x_n)$ ; unitarity of  $q((x_n))$  gives  $\|x_n^*x_n - 1\| \rightarrow 0$ . Once  $\|x_n^*x_n - 1\| \leq \frac{1}{2}$ , the matrix  $x_n$  is invertible and the polar correction  $u_n = x_n(x_n^*x_n)^{-1/2}$  is unitary with  $\|u_n - x_n\| \leq 2\|x_n\|\|x_n^*x_n - 1\| \rightarrow 0$  by continuous functional calculus; set  $u_n = 1$  at the finitely many remaining indices.  $\square$

**Lemma 2.2** (unitaries in the quotient). *For every sequence  $(d_n)$ , with  $\mathcal{N}$  as in (2), the map*

$$\kappa_{(d_n)}: \prod_n \mathcal{U}(d_n)/\mathcal{N} \longrightarrow \mathcal{U}(\mathcal{Q}), \quad [(u_n)] \longmapsto q((u_n)),$$

*is a canonical group isomorphism. Thus a homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  is the same thing as a sequence of unitaries for each element of  $H$ , multiplicative up to operator-norm errors tending to 0, taken modulo sequences converging to 1.*

*Proof.* The kernel consists exactly of those unitary sequences  $(u_n)$  for which  $\|u_n - 1\| \rightarrow 0$ , which is injectivity, and surjectivity is Lemma 2.1.  $\square$

An *operator-norm asymptotic representation* of  $H$  is a sequence of maps  $U_n: H \rightarrow \mathcal{U}(d_n)$  such that  $\|U_n(g)U_n(h) - U_n(gh)\| \rightarrow 0$  for all  $g, h \in H$ . Taking  $g = h = 1$  gives  $\|U_n(1) - 1\| \rightarrow 0$  automatically, since  $U_n(1)$  is unitary. By Lemma 2.2, choosing unitary lifts  $U_n(g)$  of  $\Theta(g)$  turns a homomorphism  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  into such a sequence, and every such sequence arises this way. A countable group is *MF* if it admits an injective homomorphism to  $\mathcal{U}(\mathcal{Q})$  for some sequence of dimensions; requiring the dimensions to be strictly increasing gives the same class of groups (Proposition 1.1).

### 3. ONE-SIDED KAZHDAN TRANSPORT

Three manifestations of one finiteness principle:

$$\begin{array}{ccc} X \subseteq Y, |X| = |Y| & V \subseteq W, \dim V = \dim W & \begin{array}{c} p \leq q, p \sim q \\ \text{in a finite } C^*\text{-algebra} \end{array} \\ \Downarrow & \Downarrow & \Downarrow \\ X = Y & V = W & p = q. \end{array}$$

In the right-hand column,  $p \sim q$  is Murray–von Neumann equivalence, and a  $C^*$ -algebra is *finite* when every isometry in it is unitary. Theorem 3.3 uses the middle implication, applied to the commutant  $C$  of the base inside a finite-dimensional endomorphism algebra: conjugation carries  $C$  into a conjugate copy of itself of the same finite dimension, so the two coincide. The proof of Theorem 3.4 uses the right-hand implication, applied to the projection  $P$  onto the Kazhdan fixed space inside a norm ultraproduct: the inclusion  $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$  implies  $P \leq VPV^*$ , the two projections are Murray–von Neumann equivalent, and the ultraproduct is finite, so  $P = VPV^*$ .

We record the two facts that the analytic proof needs, so that they do not interrupt it.

**Lemma 3.1** (comparison in a finite algebra). *Let  $A$  be a finite unital  $C^*$ -algebra and let  $p, q \in A$  be projections with  $p \leq q$  and  $p \sim q$ . Then  $p = q$ .*

*Proof.* Let  $r \in A$  satisfy  $r^*r = q$  and  $rr^* = p$ . From  $r^*r = q$  we get  $r = rq$ , and from  $\text{ran } r = \text{ran } p \subseteq \text{ran } q$  we get  $r = qr$ . Hence the cross terms in  $\sigma^*\sigma$  vanish for  $\sigma = r + (1 - q)$ , and  $\sigma^*\sigma = q + (1 - q) = 1$ , so  $\sigma$  is an isometry. Finiteness makes  $\sigma$  unitary, and  $\sigma\sigma^* = p + (1 - q) = 1$  gives  $p = q$ .  $\square$

**Lemma 3.2** (the norm ultraproduct). *Let  $(K_n)$  be finite-dimensional Hilbert spaces,  $\omega$  a free ultrafilter on  $\mathbb{N}$ ,  $K_\omega$  the Hilbert-space ultraproduct and  $B_\omega = \prod_\omega B(K_n)$  the norm ultraproduct, acting on  $K_\omega$  by  $[A_n]_\omega[\xi_n]_\omega = [A_n\xi_n]_\omega$ . Then the action is faithful,  $B_\omega$  is a finite  $C^*$ -algebra, and for projections  $P, Q \in B_\omega$  the inclusion  $\text{ran } P \subseteq \text{ran } Q$  is equivalent to  $P \leq Q$ .*

*Proof.* If  $A = [A_n]_\omega \neq 0$  then  $\lim_\omega \|A_n\| = \delta > 0$ , and unit vectors  $\xi_n$  with  $\|A_n\xi_n\| \geq \|A_n\| - 1/n$  give  $\|A[\xi_n]_\omega\| = \delta$ , so the action is faithful. Consequently  $\text{ran } P \subseteq \text{ran } Q$  makes  $QP$  and  $P$  act identically, whence  $QP = P$ , that is,  $P \leq Q$ . If  $\sigma = [\sigma_n]_\omega$  satisfies  $\sigma^*\sigma = 1$  then  $\|\sigma_n^*\sigma_n - 1\| \rightarrow 0$  along  $\omega$ ; each  $\sigma_n$  acts on a finite-dimensional space, so polar correction replaces it by a unitary  $w_n$  with  $\|\sigma_n - w_n\| \rightarrow 0$ , and  $\sigma = [w_n]_\omega$  is unitary.  $\square$

**The exact case.** The exact form of the argument needs no analysis, no property (T), and no hypothesis on the characteristic. It is the simplest case of the mechanism, and we give it first.

**Theorem 3.3** (the finite-dimensional case). *Let  $k$  be a field,  $V$  a finite-dimensional  $k$ -vector space,  $H$  a group,  $\Gamma \leq H$  a subgroup, and  $t, c \in H$ ,  $a \in \Gamma$  elements with  $t\Gamma t^{-1} \subseteq \Gamma$  and  $c\gamma = \gamma c$  for all  $\gamma \in \Gamma$ . Then every homomorphism  $\pi: H \rightarrow \text{GL}(V)$  satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

*More generally, every such  $\pi$  sends the entire subgroup  $\mathfrak{D}(H, \Gamma)$  of (9) to the identity.*

*Proof.* Let

$$C = \{x \in \text{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of  $\pi(\Gamma)$ , a linear subspace of  $\text{End}_k(V)$ , and write  $\Phi(x) = \pi(t)x\pi(t)^{-1}$  for conjugation by  $\pi(t)$ , a linear automorphism of  $\text{End}_k(V)$ .

We claim  $C \subseteq \Phi(C)$ . Let  $x \in C$  and put  $y = \pi(t)^{-1}x\pi(t)$ , so that  $x = \Phi(y)$ . For  $\gamma \in \Gamma$  we have  $t\gamma t^{-1} \in \Gamma$ , hence  $\pi(t\gamma t^{-1})$  commutes with  $x$ , and therefore

$$\pi(\gamma)y = \pi(t)^{-1}\pi(t\gamma t^{-1})x\pi(t) = \pi(t)^{-1}x\pi(t\gamma t^{-1})\pi(t) = y\pi(\gamma).$$

Thus  $y \in C$ , proving the claim. Since  $\Phi$  is a linear isomorphism,  $\dim_k \Phi(C) = \dim_k C < \infty$ , and the inclusion  $C \subseteq \Phi(C)$  forces

$$(3) \quad C = \Phi(C) = \pi(t)C\pi(t)^{-1}.$$

Now  $\pi(c) \in C$  because  $c$  centralizes  $\Gamma$ , so  $\pi(tct^{-1}) = \Phi(\pi(c)) \in C$  by (3). Since  $a \in \Gamma$ , the operator  $\pi(tct^{-1})$  commutes with  $\pi(a)$ , so  $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$  and

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1.$$

Section 12 proves the subgroup-valued assertion from the same calculation.  $\square$

Applied to  $E$ , Theorem 3.3 shows that every finite-dimensional linear representation over every field, and hence every homomorphism to a finite group, sends the marked word to the identity (Corollary 12.1).

Theorem 3.4 is the approximate analogue of this argument. An MF model is only asymptotically multiplicative, so its coordinate fixed spaces are not invariant subspaces; operator-norm control keeps the adjoint actions almost multiplicative on the Hilbert–Schmidt spaces, an almost-fixed spectral subspace of a Kazhdan average takes the place of  $C$ , and the equal-rank reversal estimate of Section B takes the place of equal dimension. Property (T) is what extends the conclusion from exact finite-dimensional representations to approximate operator-norm models. The cyclic comparison of Section 13 shows that it is needed: there is a marked group whose word is trivial in every finite-dimensional representation and nontrivial in an MF model.

**Theorem 3.4** (one-sided conjugation preserves asymptotic commutants). *Let  $\Gamma$  and  $H$  be groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, let  $s \in H$ , and let  $(d_n)$  be a sequence of positive integers. Suppose*

$$s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma).$$

*Let  $U_n: H \rightarrow \mathcal{U}(d_n)$  be an operator-norm asymptotic representation. If  $(x_n)$  is uniformly operator-norm bounded and asymptotically centralizes  $\iota(\Gamma)$  in normalized Hilbert–Schmidt norm,*

$$\|x_n U_n(\iota(\gamma)) - U_n(\iota(\gamma)) x_n\|_2 \longrightarrow 0 \quad (\gamma \in \Gamma),$$

*then the conjugated sequence does as well:*

$$\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma))]\|_2 \longrightarrow 0 \quad (\gamma \in \Gamma).$$

*Proof.* Suppose the conclusion fails: there are  $\gamma_0 \in \Gamma$ ,  $\delta > 0$ , and an infinite set  $I \subseteq \mathbb{N}$  with  $\|[U_n(s)x_n U_n(s)^*, U_n(\iota(\gamma_0))]\|_2 \geq \delta$  for  $n \in I$ . Fix a free ultrafilter  $\omega$  on  $\mathbb{N}$  with  $I \in \omega$ .

*The adjoint model.* Regard  $M_{d_n}(\mathbb{C})$  as the Hilbert space  $K_n = L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$  and let each group element act on it by conjugation,  $\text{Ad } U_n(g) \xi = U_n(g) \xi U_n(g)^*$ . Operator-norm almost multiplicativity of  $U_n$  makes  $g \mapsto \text{Ad } U_n(g)$  an operator-norm asymptotic representation on these Hilbert spaces; this is the finite-stage form of the  $\pi \otimes \bar{\pi}$  construction.

*The ultraproduct.* Let  $K_\omega$  and  $B_\omega = \prod_\omega B(K_n)$  be as in Lemma 3.2; write  $[\cdot]_\omega$  for classes in either. That lemma gives what is needed below: the action is faithful,  $B_\omega$  is finite, and an inclusion of ranges in  $K_\omega$  may be read back as an inequality of projections in  $B_\omega$ . Asymptotic multiplicativity makes

$$\pi: H \longrightarrow \mathcal{U}(B_\omega), \quad \pi(g) = [\text{Ad } U_n(g)]_\omega,$$

a homomorphism.

*The projection onto the fixed subspace.* Property (T) provides a finite symmetric generating set  $S \subseteq \Gamma$  with  $1 \in S$  and a Kazhdan constant  $\kappa \in (0, 1]$  for the pair  $(S, \kappa)$ . Let  $h$  be the Hermitian part of  $\frac{1}{|S|} \sum_{s' \in S} \pi(\iota(s'))$  and let  $\text{Fix} \subseteq K_\omega$  be the subspace of vectors fixed by every  $\pi(\iota(\gamma))$ . Both  $\text{Fix}$  and

its orthocomplement are invariant under each unitary  $\pi(\iota(s'))$ , hence under  $h$ . On  $\text{Fix}$ ,  $h = 1$ . The restriction of  $\pi \circ \iota$  to  $\text{Fix}^\perp$  has no nonzero invariant vector, so a unit vector  $\xi \in \text{Fix}^\perp$  satisfies  $\|\pi(\iota(s'))\xi - \xi\| \geq \kappa$  for some generator. For that generator  $\text{Re}\langle \pi(\iota(s'))\xi, \xi \rangle \leq 1 - \kappa^2/2$ , while every other term is at most 1. Averaging gives  $\langle h\xi, \xi \rangle \leq 1 - \kappa^2/(2|S|)$ . Consequently

$$\text{sp}(h) \subseteq \left[-1, 1 - \frac{\kappa^2}{2|S|}\right] \cup \{1\},$$

and the spectral projection  $P$  of  $h$  at the isolated point 1 belongs to  $B_\omega$  by continuous functional calculus and projects onto  $\text{Fix}$ .

*The two projections agree.* Let  $V = \pi(s)$  and  $Q = VPV^*$ . Since  $s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma)$ , every vector of  $\text{Fix}$  is fixed by each  $\pi(s\iota(\gamma)s^{-1})$ ; and  $\eta$  is fixed by all  $\pi(s\iota(\gamma)s^{-1})$  exactly when  $V^*\eta \in \text{Fix}$ . Hence  $\text{ran } P = \text{Fix} \subseteq V\text{Fix} = \text{ran } Q$ , so  $P \leq Q$  by Lemma 3.2. The element  $r = V^*Q$  satisfies  $r^*r = Q$  and  $rr^* = V^*QV = P$ , so  $P \sim Q$ . Since  $B_\omega$  is finite, Lemma 3.1 gives  $Q = P$ .

*Conclusion.* Let  $\xi = [\xi_n]_\omega \in K_\omega$ , where  $\xi_n \in K_n$  is the vector of  $x_n$ ; the uniform operator-norm bound makes  $\xi$  well defined. By unitary invariance of the normalized Hilbert–Schmidt norm, the commutator hypothesis says exactly that each  $\pi(\iota(\gamma))$  fixes  $\xi$ . Thus  $\xi \in \text{Fix}$ . Since  $Q = P$ , we obtain  $V\xi \in V\text{Fix} = \text{ran } Q = \text{ran } P = \text{Fix}$ . The vector of  $U_n(s)x_nU_n(s)^*$  is  $\text{Ad } U_n(s)\xi_n$ , so, again by unitary invariance,

$$\|[U_n(s)x_nU_n(s)^*, U_n(\iota(\gamma))]_-\|_2 = \|(\text{Ad } U_n(\iota(\gamma)) - 1) \text{Ad } U_n(s)\xi_n\| \longrightarrow 0$$

along  $\omega$ , for every  $\gamma \in \Gamma$ . Taking  $\gamma = \gamma_0$  contradicts  $I \in \omega$ . This contradicts the choice of  $\gamma_0$ ,  $\delta$  and  $I$ , so no such data exist and the conclusion holds.  $\square$

*Remark 3.5* (the finite-stage proof). The ultrafilter gives no rate. Section B proves the theorem a second time, at finite stages. There, an explicit almost-fixed spectral subspace of the averaged adjoint action replaces that projection, and the equal-rank reversal estimate replaces finiteness of  $B_\omega$ . Conditional on the robust spectral-gap input stated there, every estimate of that proof is an explicit inequality in  $\kappa$ ,  $|S|$ , the uniform bound on  $(x_n)$ , and the multiplicative defects of  $U_n$ .

The formalization follows the second route and not the first. The declarations recorded at this theorem prove it through the spectral corner and the equal-rank reversal; the apparatus of the printed proof — the Hilbert-space ultraproduct  $K_\omega$ , the norm ultraproduct  $B_\omega$ , its faithfulness and its finiteness — has no counterpart in the development, or in the library it is built on. The machine-checked statement is the one printed above; the machine-checked *proof* is the one of Section B.

#### 4. THE CENTRAL INVOLUTION OBSTRUCTION

Let  $w$  be a central involution of  $H$  and let  $\Theta$  be a homomorphism to  $\mathcal{U}(\mathcal{Q})$  with  $\Theta(w) \neq 1$ . Then  $\Theta(w)$  is a self-adjoint unitary commuting with  $\Theta(H)$ , the projection  $\frac{1}{2}(1 - \Theta(w))$  is nonzero, and compressing to it gives a matrix

model of  $H$  in which  $w$  equals  $-1$  and each block has its own normalized trace.

The corner is what turns the conclusion of Theorem 3.4 into an obstruction. That theorem bounds a commutator in normalized Hilbert–Schmidt norm, and such a bound is by itself worthless: padding an asymptotic representation with an identity block leaves every operator-norm quantity unchanged while sending all normalized Hilbert–Schmidt distances to zero (Section 16). Passing to the corner  $q_n M_{d_n}(\mathbb{C}) q_n$  removes that slack. The operator norm there is the restriction of the operator norm, so the approximation hypotheses survive; but the trace is the corner’s own normalized trace  $\text{tr}_{r_n}$ , not the restriction of  $\text{tr}_{d_n}$ . Against  $\text{tr}_{r_n}$  one has  $\|1 - (-1)\|_{2, \text{tr}_{r_n}} = 2$  however small  $r_n/d_n$  becomes.

**Theorem 4.1** (central involution obstruction). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \quad [c, \iota(\Gamma)] = 1.$$

*Put  $d = tct^{-1}$  and  $u = [d, \iota(a)]$  for some  $a \in \Gamma$ . If  $w := u^2$  is a nontrivial central involution of  $H$ , then every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  maps  $w$  to the identity, and  $H$  is not MF.*

*Proof.* Let  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  be a homomorphism and suppose  $\Theta(w) \neq 1$ .

*The negative corner.* Choose coordinate unitary lifts  $(V_{g,n})$  of  $\Theta$  (Lemma 2.2) and put  $p_n = \frac{1}{2}(1 + V_{w,n})$ , the averaging operator of the two-element subgroup  $F = \{1, w\}$ . Because the lifts are asymptotically multiplicative and  $w^2 = 1$ , the Hermitian part of  $p_n$  is asymptotically idempotent, so its spectrum clusters at  $\{0, 1\}$  and rounding at  $\frac{1}{2}$  by continuous functional calculus produces genuine projections at operator-norm distance  $o(1)$  from  $p_n$ ; centrality of  $w$  makes them asymptotically central for all of  $\Theta(H)$ . Let  $q_n$  be their complements. If  $q_n$  vanished for all large  $n$  the averages would converge to 1 and  $\Theta(w)$  would be 1; so  $q_n \neq 0$  along infinitely many coordinates, which we retain and relabel by  $\mathbb{N}$ . Compressing every lift to  $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$  and polar-correcting the compressions — each is almost unitary in the corner, by asymptotic centrality — gives an operator-norm asymptotic representation  $(W_{g,n})$  of  $H$  on nonzero blocks with

$$\|W_{w,n} + 1\| \rightarrow 0.$$

Equip each corner block with its own normalized trace  $\text{tr}_{r_n}$ ; all estimates below are relative to  $r_n$  and independent of the ratio  $r_n/d_n$ .

*Transport on the corner.* Since  $c$  centralizes  $\iota(\Gamma)$ , the sequence  $(W_{c,n})$  centralizes  $(W_{\iota(\gamma),n})$  asymptotically in operator norm, hence in normalized Hilbert–Schmidt norm. The compression relation  $t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma)$  and Theorem 3.4, applied to  $x_n = W_{c,n}$ , put the conjugated sequence  $(W_{d,n})$  in the same asymptotic commutant. Therefore  $\|W_{u,n} - 1\|_2 \rightarrow 0$ , and squaring gives  $\|W_{u,n}^2 - 1\|_2 \rightarrow 0$ .

*The contradiction.* The relation  $w = u^2$  gives  $\|W_{u,n}^2 - W_{w,n}\| \rightarrow 0$ , while  $W_{w,n} \rightarrow -1$  in operator norm and therefore also in normalized Hilbert–Schmidt norm. Hence  $\|W_{u,n}^2 + 1\|_2 \rightarrow 0$  and  $\|W_{u,n}^2 - 1\|_2 \rightarrow 0$ , so  $\|2 \cdot 1\|_2 \rightarrow 0$ , which is false. Thus  $\Theta(w) = 1$ . Since  $w \neq 1$ , no homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  is injective, and  $H$  is not MF.  $\square$

Finiteness of  $F = \{1, w\}$  entered only through the averaging operator  $p_n$ . Section 11 proves the same conclusion for an arbitrary finite normal subgroup of  $N_{\text{comp}}$  (Theorem 11.3), and then for an arbitrary normal property-(T) subgroup of it (Theorem 11.6), where a spectral projection of a Kazhdan average takes the place of the negative corner. Those forms assume neither centrality nor torsion.

## 5. THE KAZHDAN–CLIFFORD CONSTRUCTION

Let  $\Gamma$  be a finitely presented Kazhdan group, let  $\alpha$  be a proper injective endomorphism of it, and let  $a \in \Gamma \setminus \alpha(\Gamma)$ . These data give both the configuration of Theorem 4.1 and a nontrivial sign.

With  $d = tct^{-1}$  and  $u = [d, a]$ , put

$$w = u^2.$$

Since  $d$  is an involution,  $w = [d, ada^{-1}]$ , a word in the generators, so centrality of  $w$  is a relation among them; centrality then forces  $w^2 = 1$  by (4) below.

**Theorem 5.1** (Kazhdan–Clifford construction). *Let  $\Gamma$  be a finitely presented group with property (T), let  $\alpha: \Gamma \hookrightarrow \Gamma$  be an injective endomorphism, and choose  $a \in \Gamma \setminus \alpha(\Gamma)$ . Adjoin elements  $t, c$  and impose*

$$t\gamma t^{-1} = \alpha(\gamma), \quad c^2 = 1, \quad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

*Put*

$$d = tct^{-1}, \quad u = [d, a], \quad w = u^2 = [d, ada^{-1}],$$

*and impose centrality of  $w$ . More explicitly, for a finite generating set  $S_\Gamma$  it is enough to impose the conjugation and commutation relations for  $s \in S_\Gamma$ , together with*

$$[w, s] = 1 \quad (s \in S_\Gamma), \quad [w, t] = [w, c] = 1.$$

*Thus  $w$  commutes with a generating set of the entire resulting group. The group  $E(\Gamma, \alpha, a)$  is finitely presented, the canonical map  $\Gamma \rightarrow E(\Gamma, \alpha, a)$  is injective, and  $w$  is a nontrivial central involution. Every homomorphism*

$$E(\Gamma, \alpha, a) \longrightarrow \mathcal{U}(\mathcal{Q})$$

*to  $\mathcal{U}(\mathcal{Q})$  maps  $w$  to 1. Consequently  $E(\Gamma, \alpha, a)$  is not MF.*

*Proof.* Finite presentability follows from a relative finite presentation: start with the free product of  $\Gamma$  and the free group on  $t, c$ , and quotient by the finitely many displayed relations for a finite generating set of  $\Gamma$ .

The relation  $w^2 = 1$  is redundant. Put  $b = ada^{-1}$ . Both  $d$  and  $b$  are involutions,  $w = [d, b] = (db)^2$ , and direct cancellation gives

$$(4) \quad dwd^{-1} = w^{-1}.$$

Centrality also gives  $dwd^{-1} = w$ , whence  $w = w^{-1}$  and  $w^2 = 1$ .

For nontriviality, realize the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$$

concretely. Let  $\Gamma_\infty$  be the direct limit of the system  $\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \dots$ , whose  $n$ th copy of  $\Gamma$  we call its  $n$ th level; injectivity of  $\alpha$  passes to the limit, so the level-zero map embeds  $\Gamma$  in  $\Gamma_\infty$ . Applying  $\alpha$  levelwise induces an injective endomorphism  $\sigma$  of  $\Gamma_\infty$ . It is surjective, hence an automorphism, because an element at level  $k$  equals  $\sigma$  of the same element at level  $k+1$ . Put  $G = \Gamma_\infty \rtimes_\sigma \mathbb{Z}$ , with stable letter  $t$  implementing  $\sigma$ ; then  $t\gamma t^{-1} = \alpha(\gamma)$  for  $\gamma \in \Gamma$ . Write  $X = G/\Gamma$  for the coset space of the level-zero copy. Let  $G$  permute the Clifford generators  $(e_x)_{x \in X}$  and map  $c$  to  $e_\Gamma$ . Then

$$d \mapsto e_{t\Gamma}, \quad ada^{-1} \mapsto e_{at\Gamma}.$$

The two cosets are distinct: equality would give  $t^{-1}at \in \Gamma$ , equivalently  $a \in \alpha(\Gamma)$ . Hence the two Clifford generators anticommute and

$$w \mapsto (e_{t\Gamma}e_{at\Gamma})^2 = -1.$$

The Clifford sign is central, so the map respects every defining relation. It follows both that  $w \neq 1$  and that the canonical map of  $\Gamma$  is injective.

Finally,  $w = u^2$  is a nontrivial central involution. By the central involution obstruction (Theorem 4.1), every homomorphism to  $\mathcal{U}(\mathcal{Q})$  maps  $w$  to the identity. Since  $w \neq 1$ , none is faithful.  $\square$

*Remark 5.2* (a small-cancellation realization). Fournier-Facio has pointed out (personal communication, 2026) that the affine base of Sections 7–8 can be replaced by a small-cancellation construction as in [FFF]. His finitely presented torsion-free property-(T) group  $P$  contains a copy of every finitely presented torsion-free group, in particular a direct product  $P_1 \times P_2$  with  $P_i \cong P$ . Then  $\alpha: P \xrightarrow{\cong} P_1 \leq P$ , together with any  $a \in P_2 \setminus \{1\}$ , satisfies the hypotheses of Theorem 5.1. The group  $E(P, \alpha, a)$  contains involutions whatever the base. The affine base is used for the explicit presentation of Theorem A and for Theorems C and D.

## 6. A FINITELY PRESENTED NON-MF GROUP

Specialize Theorem 5.1 to

$$(5) \quad \Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \quad \alpha(v, A) = (2v, A), \quad a = (e_1, I).$$

The affine group  $\Gamma$  is finitely presented and has property (T) [BHV, Example 1.7.4(i)]; Section 7 also proves this from the presentation used below, by an exact rational sum-of-squares certificate. The endomorphism  $\alpha$  is injective because doubling is injective on  $\mathbb{Z}^3$ , it is proper because  $e_1 \notin 2\mathbb{Z}^3$ ,

and  $a \notin \alpha(\Gamma)$  for the same reason. Theorem 5.1 therefore applies, and relative to  $\Gamma$  the resulting group has the finite presentation

$$E = \left\langle \Gamma, t, c \mid \begin{array}{l} t\gamma t^{-1} = \alpha(\gamma), \quad [c, \gamma] = 1 \quad (\gamma \in S_\Gamma), \\ c^2 = 1, \quad [w, g] = 1 \end{array} \right\rangle,$$

where  $S_\Gamma$  is a finite generating set of  $\Gamma$ ,  $g$  runs over the generators, and  $w = [tct^{-1}, a(tct^{-1})a^{-1}]$ . Expanding a standard presentation of the affine group in the generators  $v_1, v_2, v_3$  for the translations and  $x, y, z$  for the linear part turns this into the eight-generator, forty-one-relator presentation below. First define the *linear presentation*

$$(6) \quad \mathcal{R} = \left\langle x, y, z \mid \begin{array}{l} x^3 = y^3 = z^3 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle,$$

and the *base*

$$(7) \quad \mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \mid \begin{array}{l} \text{the eight relations in (6),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3, \quad xv_2x^{-1} = v_1, \quad xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1, \quad yv_2y^{-1} = v_2^{-1}v_3, \quad yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle.$$

Write  $\iota: \mathcal{B} \rightarrow E$  for the homomorphism induced by the first six generators of the presentation below. This map is not assumed to be injective.

**Definition 6.1.** Use the six base generators and the relations displayed in (7). Define

$$(8) \quad E = \left\langle v_1, v_2, v_3, x, y, z, t, c \mid \begin{array}{l} \text{(B) all relations in (7),} \\ \text{(T) } tv_it^{-1} = v_i^2 \quad (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ \text{(C) } c^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ \text{(W) } [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right\rangle,$$

where  $w$  abbreviates the word  $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$ . This is an explicit finite presentation with eight generators and forty-one relators: twenty in family (B), six in family (T), seven in family (C), and eight in family (W). The four families have four distinct jobs: (B) presents the base, (T) compresses it, (C) centralizes the Clifford generator against it, and (W) makes the marked word central. Centrality together with  $c^2 = 1$  implies  $w^2 = 1$  by (4). Family (B) also gives

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \quad [c, \iota(\mathcal{B})] = 1.$$

A Tietze reduction removes two generators and nine relators (Section 7).

The marked word is a square, which is what makes the central involution of Theorem 4.1 available.

**Lemma 6.2** (the marked word is a square). *Let  $H_1$  be a group and  $d, a_1 \in H_1$  with  $d^2 = 1$ . Then*

$$[d, a_1 d a_1^{-1}] = [d, a_1]^2.$$

*Proof.* Put  $b = a_1 d a_1^{-1}$ , so  $b^2 = a_1 d^2 a_1^{-1} = 1$ ; thus  $b^{-1} = b$  and  $d^{-1} = d$ . Then  $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d (a_1 d a_1^{-1}) = db$ , and

$$[d, a_1]^2 = (db)^2 = db db = d b d^{-1} b^{-1} = [d, b]. \quad \square$$

*Proof of Theorem A.* That  $w \neq 1$  is Proposition 8.3.

Let  $(d_n)$  be given, write  $\mathcal{Q} = \mathcal{Q}((d_n))$ , and let  $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$  be a homomorphism. Apply the central involution obstruction (Theorem 4.1) with  $\Gamma = \mathcal{B}$ , the canonical homomorphism  $\iota$ , the elements  $t, c \in E$ , and  $a = v_1$ . Its hypotheses are the displayed relators: Proposition 7.1 gives property (T) of  $\mathcal{B}$ ; family (T) gives  $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ ; and family (C) gives  $[c, \iota(\mathcal{B})] = 1$ . In  $E$  we have  $d^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$ , so Lemma 6.2 with  $a_1 = \iota(v_1)$  gives

$$w = [d, \iota(v_1) d \iota(v_1)^{-1}] = [d, \iota(v_1)]^2 = u^2, \quad u = [d, \iota(v_1)].$$

Centrality of  $w$  is a defining relator; with  $c^2 = 1$  it gives  $w^2 = 1$  by (4); and  $w \neq 1$  as above. Hence  $\Theta(w) = 1$ .

An MF embedding of  $E$  would be an injective homomorphism to  $\mathcal{U}(\mathcal{Q})$ , and every homomorphism to  $\mathcal{U}(\mathcal{Q})$  maps the nontrivial element  $w$  to 1; so  $E$  is not MF. Suppose  $C_r^*(E)$  were an MF algebra, with an injective, possibly nonunital  $*$ -homomorphism  $e: C_r^*(E) \rightarrow \mathcal{Q}$ . Complement correction gives the injective unitary homomorphism  $u \mapsto e(u) + (1 - e(1))$ ; composing it with the canonical map  $E \rightarrow \mathcal{U}(C_r^*(E))$ , which is injective because  $g \mapsto \lambda_g$  is, exhibits an injective homomorphism  $E \rightarrow \mathcal{U}(\mathcal{Q})$ . The canonical map  $E \rightarrow \mathcal{U}(C_{\max}^*(E))$  is injective as well, since the left regular representation factors through  $C_{\max}^*(E)$  and sends  $u_g$  to  $\lambda_g$ , so an MF embedding of  $C_{\max}^*(E)$  produces an injective homomorphism  $E \rightarrow \mathcal{U}(\mathcal{Q})$  after the same conversion.  $\square$

The two inputs of this proof are proved from the displayed relators: property (T) of  $\mathcal{B}$  in Section 7, and the Clifford model witnessing  $w \neq 1$  in Section 8.

The obstruction already appears at a finite tolerance, on a finite set of group elements, in every dimension.

**Corollary 6.3** (uniform finite operator-norm obstruction). *There exist  $\delta > 0$  and a finite set  $F_0 \subset E$  such that, in every matrix dimension, any unitary-valued map  $\phi$  satisfying*

$$\|\phi(gh) - \phi(g)\phi(h)\| \leq \delta \quad (g, h \in F_0)$$

*also satisfies  $\|\phi(w) - 1\| < 1$ .*

*Proof.* Suppose no such pair  $(\delta, F_0)$  existed. For each  $n$ , choose a model violating the conclusion for  $\delta = 1/n$  and the  $n$ th set of an exhaustion of  $E$ . These models form an operator-norm asymptotic representation of  $E$  whose values at  $w$  stay at distance at least 1 from the identity. Its class in the quotient of sequences of unitaries is a homomorphism  $\Theta$  to  $\mathcal{U}(\mathcal{Q})$  with  $\Theta(w) \neq 1$ , contradicting Theorem A. The compactness argument is not quantitative; determining an effective modulus from the rational spectral certificate of Section 7 remains open.  $\square$

## 7. THE BASE, AND A SMALLER PRESENTATION

The base  $\mathcal{B}$  of (7) is the affine group  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$  (Remark D.2), for which property (T) is classical [BHV, Example 1.7.4(i)]. The proof given here works from the displayed relators, in three steps: an exact rational spectral certificate for an auxiliary Steinberg-type presentation, an explicit surjection onto the linear presentation, and a fixed-point bridge from the linear subgroup to the full base.

Six generators suffice. The eight-generator presentation is used because it displays the structure used in the proof: the three commuting translations, the linear generators acting on them, the stable letter  $t$ , and  $c$ . The group  $E$  is nevertheless generated by the six elements  $v_1, x, y, z, t, c$ . There is also a Tietze-equivalent presentation on these six generators with thirty-two relators. In (8), eliminate  $v_3 = xv_1x^{-1}$  and  $v_2 = x^2v_1x^{-2}$  by Tietze substitutions and delete the two defining relations; the remaining  $x$ -action relation follows from  $x^3 = 1$ , and the stable-letter,  $c$ -commutation, and centrality relations for  $v_2, v_3$  follow from those for  $v_1$  together with  $[t, x] = 1$  and  $x^3 = 1$ . Deleting these nine leaves thirty-two relators on  $v_1, x, y, z, t, c$ .

Write  $d = tct^{-1}$ . The displayed relations give  $[d, t\iota(\mathcal{B})t^{-1}] = 1$  and leave  $[d, \iota(v_1)d\iota(v_1)^{-1}]$  unconstrained; Figure 1 shows that configuration.

### 7.1. Property (T) of the base.

**Proposition 7.1.** *The twenty-relator group  $\mathcal{B}$  has property (T), in both the real-orthogonal and complex-unitary formulations.*

*Proof.* *Step 1: a certified Steinberg-type presentation.* Let  $P_{13}$  be the group on the six generators  $e_{12}, e_{13}, e_{21}, e_{23}, e_{31}, e_{32}$  with the thirteen relators

$$\begin{aligned} [e_{12}, e_{23}] &= e_{13}, & [e_{13}, e_{32}] &= e_{12}, & [e_{21}, e_{13}] &= e_{23}, \\ [e_{23}, e_{31}] &= e_{21}, & [e_{31}, e_{12}] &= e_{32}, & [e_{32}, e_{21}] &= e_{31}, \\ [e_{12}, e_{13}] &= [e_{12}, e_{32}] = [e_{13}, e_{23}] = [e_{21}, e_{23}] = [e_{21}, e_{31}] = [e_{31}, e_{32}] = 1, \\ & & (e_{12} e_{21}^{-1} e_{12})^4 &= 1, \end{aligned}$$

the elementary-matrix relations in rank three. The degree-one Hodge Laplacian  $\Delta_1$  of this presentation, a matrix over the real group ring of  $P_{13}$ , admits the exact decomposition

$$\Delta_1 = \frac{1}{250} I + Q^*Q + R,$$

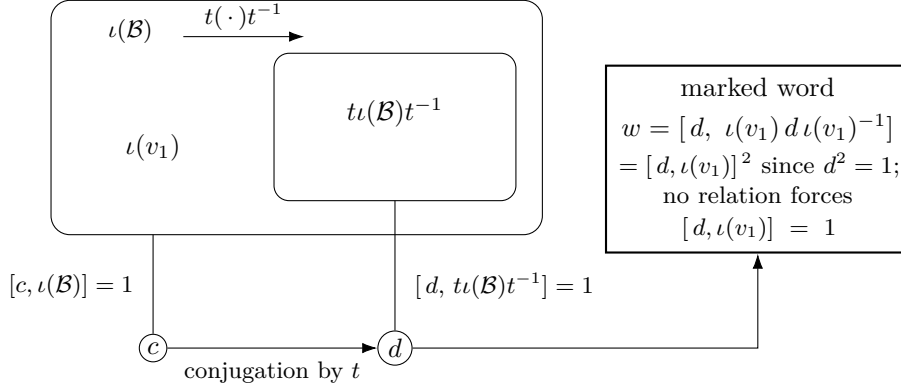


Figure 1: The one-sided conjugation configuration. The relations of (8) give  $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$  and  $[c, \iota(\mathcal{B})] = 1$ , hence  $[d, t\iota(\mathcal{B})t^{-1}] = 1$  for  $d = tct^{-1}$ ; they do not give  $[d, \iota(v_1)] = 1$ , and  $w = [d, \iota(v_1)d\iota(v_1)^{-1}] = [d, \iota(v_1)]^2$ . Section 8 proves that the inclusion is strict and that  $\iota(v_1)$  lies outside the compressed copy, as drawn; Theorem 4.1 uses only the inclusion.

in which  $Q$  and  $R$  are matrices over the group ring with rational coefficients supported on finitely many group elements, and every row and every column of  $R$  has coefficient  $\ell^1$  norm at most  $\frac{1}{500}$ . Clearing denominators makes both sides equal integer combinations of finitely many group elements, so the decomposition and the two bounds are finite identities and inequalities over  $\mathbb{Z}$ , evaluated in exact arithmetic; the certificate data are in the accompanying repository. Together they bound  $\Delta_1$  below by  $\frac{1}{500}$  in every orthogonal representation. Testing that bound on coboundaries gives the quadratic gap

$$\|\Delta\xi\|^2 \geq \frac{1}{500} \langle \Delta\xi, \xi \rangle$$

for the six-generator Laplacian  $\Delta$ . The gap places the spectrum of  $\Delta$  in  $\{0\} \cup [\frac{1}{500}, \infty)$ . The kernel of  $\Delta$  is exactly the space of invariant vectors. The localization makes 0 an isolated point of the spectrum but does not by itself exclude it. In a representation with no nonzero invariant vector the kernel is trivial, so  $\Delta$  is bounded below there. Hence a unit vector almost fixed by the six generators has small Laplacian energy and is close to an invariant vector. This yields a Kazhdan pair:  $P_{13}$  has property (T). Certificates of this kind go back to Ozawa's algebraic characterization of property (T) [Oz] and its computer-assisted implementations for  $\mathrm{SL}_3(\mathbb{Z})$  [NT].

*Step 2: the explicit surjection onto the linear presentation.* Map the six generators to the words

$$\begin{aligned} e_{12} &\mapsto y^{-1}xyz^{-1}x^{-1}, & e_{13} &\mapsto xzy^{-1}x^{-1}y, & e_{21} &\mapsto y^{-1}xyxz, \\ e_{23} &\mapsto x^{-1}y^{-1}xzy^{-1}, & e_{31} &\mapsto xy^{-1}xzy^{-1}x, & e_{32} &\mapsto x^{-1}zy^{-1}x^{-1}y^{-1}x^{-1} \end{aligned}$$

in the linear presentation  $\mathcal{R}$  of (6). Each of the thirteen relators of  $P_{13}$  reduces to the identity in  $\mathcal{R}$  using only the eight relators of (6); the thirteen reductions are finite words and are carried out in the accompanying repository. The assignment therefore defines a homomorphism  $P_{13} \rightarrow \mathcal{R}$ , and the same reductions exhibit explicit words in the six images whose values are  $z$  and  $xy$ . Those two elements generate  $\mathcal{R}$ : writing  $g = z$  and  $h = xy$ , the eight relators give the generating-pair word of [CLV]

$$y = (ghgh^{-1}g^{-1}hg^{-1}h^{-1}g^{-1}h^{-1}g^{-1}h)^2,$$

and then  $x = hy^{-1}$ . Hence the homomorphism is surjective. Property (T) passes to quotients, so  $\mathcal{R}$  has property (T).

*Step 3: the fixed-point bridge to the base.* The generators  $x, y, z$  of  $\mathcal{B}$  satisfy the relators of  $\mathcal{R}$ , so  $\mathcal{R}$  maps to the linear subgroup of  $\mathcal{B}$  they generate, and Step 2 gives a finite set  $S$  of words in  $x, y, z$  and a  $\kappa > 0$  forming a Kazhdan pair for it. Shrinking  $\kappa$  costs nothing — a Kazhdan pair stays one when its constant is decreased — so assume  $\kappa \leq 1$ ; the numerical bookkeeping below uses that normalization and nothing else about  $\kappa$ . Let  $\rho$  be an orthogonal representation of  $\mathcal{B}$  with a unit vector  $\xi$  displaced by less than  $\kappa/64$  by every element of the finite control set consisting of  $S$  together with the translations  $v_2, v_3$ . The third basis translation needs no separate control:  $v_1 = xv_2x^{-1}$  by the displayed  $x$ -action, and conjugation by the fixed unitary of  $x$  preserves displacements of vectors fixed by the linear subgroup. We construct a nonzero invariant vector near  $\xi$  as follows.

- (1) Projecting  $\xi$  to the subspace of vectors fixed by the linear subgroup gives such a fixed vector  $p$  with  $\|p - \xi\| < 1/64$ , by the moving-projection estimate for the Kazhdan pair.
- (2) Translations move  $p$  by a bound independent of the lattice vector. Three explicit words in  $x, y, z$  act on the translation lattice as elementary transvections. Consequently every lattice element is a product of two linear-subgroup conjugates of basis translations — a two-conjugate normal form proved inside the presentation. The three words, and the elimination that turns them into the normal form, are finite data of the same kind as the thirteen reductions of Step 2 and, like those, are carried out in the accompanying development rather than printed here. A conjugate  $rvr^{-1}$  with  $r$  in the linear subgroup displaces the fixed vector  $p$  exactly as much as  $v$  does. Each control translation displaces  $p$  by less than  $1/64 + \kappa/64 + 1/64 = (\kappa + 2)/64 \leq 3/64$ , its displacement bound at  $\xi$  transferring across  $\|p - \xi\|$  at both ends; the last inequality is where  $\kappa \leq 1$  is used. Hence every translation displaces  $p$  by at most  $1/8$ .
- (3) The orbit of  $p$  under the translation subgroup is bounded, so the Hilbert-space circumcenter argument gives a translation-fixed vector  $q$  with  $\|q - p\| \leq 1/8$ ; the orthogonal projection  $\eta$  of  $p$  to the translation-fixed subspace is the nearest fixed vector, so  $\|\eta - p\| \leq \|q - p\| \leq 1/8$ . The linear subgroup normalizes the translation subgroup and fixes

$p$ , so  $\eta$  is fixed by it as well. Translations and the linear generators generate  $\mathcal{B}$ , so  $\eta$  is  $\mathcal{B}$ -invariant. Finally,  $\|\eta - \xi\| \leq 1/8 + 1/64 < 1$  keeps  $\eta$  nonzero.

Thus almost-invariant vectors produce invariant vectors and  $\mathcal{B}$  has property (T) in the real-orthogonal formulation; realification and complexification identify it with the complex-unitary formulation.  $\square$

Section D identifies  $P_{13}$  with  $\mathrm{SL}_3(\mathbb{Z})$ , and the base  $\mathcal{B}$  with the affine group.

## 8. NONTRIVIALITY OF $w$

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra's hyperlinear-profile construction [Slo18, Section 2]. Below we define a homomorphism from  $E$  to a Clifford group under which  $w \mapsto -1$ .

**Construction 8.1** (Clifford groups). For a set  $X$ , use the real Clifford-algebra convention generated by  $(c_x)_{x \in X}$  with  $c_x^2 = 1$  and  $c_x c_y = -c_y c_x$  for  $x \neq y$ , and let  $\mathrm{Cl}(X)$  be the subgroup of units generated by  $-1$  and the  $c_x$ . Write  $\zeta = -1$ . In this concrete group,

$$\zeta^2 = c_x^2 = 1, \quad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign  $\zeta$  is nontrivial. Thus  $\mathrm{Cl}(X)$  is a central extension, by the sign  $\zeta$ , of the elementary abelian 2-group on the points of  $X$ : the commutator of two distinct generators is  $\zeta$ , unlike the commuting generators of a classical lamplighter. Every permutation of  $X$  acts on this group by  $c_x \mapsto c_{\sigma(x)}$  and fixes  $\zeta$ .

**Lemma 8.2** (linear model). *For  $M \in \mathrm{GL}_3(\mathbb{Q})$  and  $u \in \mathbb{Q}^3$ , write*

$$A(M, u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

*Assign  $v_i \mapsto A(I_3, e_i)$  and*

$$\begin{aligned} x &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), & y &\mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right), \\ z &\mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right). \end{aligned}$$

*Let  $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Q})$  be the subgroup generated by these six matrices, and put  $D = \mathrm{diag}(2, 2, 2, 1)$ . Direct matrix computation proves that the six matrices satisfy all twenty relators of  $\mathcal{B}$  and that*

$$\bar{\alpha}(u) = DuD^{-1}$$

*is an injective endomorphism of  $\bar{\Gamma}$  which squares the three translation generators and fixes  $x, y, z$ . Moreover the matrix assigned to  $v_1$  does not lie in  $\mathrm{range}(\bar{\alpha})$ .*

*Proof.* The twenty relator checks reduce to sixteen rational identities for each matrix equation. Conjugation by  $D$  gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in  $\mathrm{GL}_4(\mathbb{Z})$ . Hence  $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Z})$ . However  $D^{-1}\bar{v}_1 D$  has entry  $1/2$  in position  $(1, 4)$ , so it is not in  $\bar{\Gamma}$ . This proves the last assertion.  $\square$

Form the direct limit  $T = \varinjlim(\bar{\Gamma}, \bar{\alpha})$ , the direct limit of  $\bar{\alpha}$ , extend it by its shift  $\tau$ , and call the resulting group  $\Sigma = T \rtimes \mathbb{Z}$ . Let  $j: \bar{\Gamma} \hookrightarrow \Sigma$  be the level-zero map. Put  $X = \Sigma/j(\bar{\Gamma})$ , with base point  $o = j(\bar{\Gamma})$ . The final assertion of Lemma 8.2 states that  $\tau o \neq j(\bar{v}_1)\tau o$ ; the Clifford generators at these two points therefore anticommute.

**Proposition 8.3.**  $w \neq 1$  in  $E$ .

*Proof.* Form  $W := \mathrm{Cl}(X) \rtimes \Sigma$  as in Construction 8.1. Define a map on the generators of (8) by

$$\begin{aligned} g &\mapsto j(\bar{g}) \in \Sigma & (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t &\mapsto \tau, & c &\mapsto c_o. \end{aligned}$$

The exact checks in Lemma 8.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes  $o$ , so  $c_o$  commutes with every base generator, and  $c_o^2 = 1$ . Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \quad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two points are distinct by Lemma 8.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since  $\zeta$  is central in  $W$ , the remaining relations  $[w, g] = 1$  for  $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$  are also satisfied. Hence the assignment extends to a group homomorphism  $E \rightarrow W$  sending  $w$  to  $\zeta \neq 1$ , and therefore  $w \neq 1$  in  $E$ .  $\square$

When  $X$  is infinite so is  $\mathrm{Cl}(X)$ , but each of its elements involves only finitely many Clifford generators, and the remaining data of the target are rational matrices. The two points are separated by the range characterization of Lemma 8.2:  $D^{-1}\bar{v}_1 D$  has the entry  $1/2$  in position  $(1, 4)$  and therefore does not lie in  $\bar{\Gamma} \leq \mathrm{GL}_4(\mathbb{Z})$ .

## 9. CONSEQUENCES OF THE EXAMPLE

Throughout,  $d = tct^{-1}$  and  $u = [d, \iota(v_1)]$ , so that  $w = u^2$ . Write  $\mathrm{Rad}_{\mathrm{MF}}(G)$  for the *MF radical* of a countable group  $G$ : the intersection of the kernels of all its homomorphisms to  $\mathcal{U}(\mathcal{Q})$  (Definition 15.1). Thus  $G$  is MF exactly when  $\mathrm{Rad}_{\mathrm{MF}}(G) = 1$  (Proposition 15.3), and Theorem A says  $w \in \mathrm{Rad}_{\mathrm{MF}}(E)$  with  $w \neq 1$ .

**The quotient by the central involution.** Passing to the quotient by  $\langle w \rangle$  does not restore the MF property.

**Theorem B** (the quotient of  $E$  by  $\langle w \rangle$ ). *The quotient  $E/\langle w \rangle$  is not MF. The image of  $[\iota(v_1), d]$ , the inverse of the marked commutator  $u$ , is nontrivial there and lies in  $\text{Rad}_{\text{MF}}(E/\langle w \rangle)$ ; consequently  $\{1, w\}$  is a proper subgroup of  $\text{Rad}_{\text{MF}}(E)$ .*

Section 15.4 proves this from a second consequence of one-sided compression: if an involution is centralized by the compressed copy of a Kazhdan subgroup, and its conjugates under that subgroup commute with one another, then every homomorphism to  $\mathcal{U}(\mathcal{Q})$  identifies the entire conjugation orbit (Theorem 15.13). In  $E$  the pairwise commutators of those conjugates of  $d$  lie in  $\langle w \rangle$ , so in  $E/\langle w \rangle$  the element  $d$  satisfies both hypotheses, and every homomorphism  $E/\langle w \rangle \rightarrow \mathcal{U}(\mathcal{Q})$  maps  $u$  to the identity. Thus  $E/\langle w \rangle$  is a second finitely presented non-MF group, and Theorem A does not determine  $\text{Rad}_{\text{MF}}(E)$  (Section 16).

**A stably finite non-MF group algebra.** Every MF algebra is stably finite [BK]; the converse fails. Here stable finiteness comes from the faithful canonical trace (Lemma 15.20), independently of the MF question. Abstract examples follow from the failure of the Connes embedding problem [FGH]; Theorem C gives an explicit one, the reduced algebra of a finitely presented group. For  $C_r^*(\text{SL}_3(\mathbb{Z}))$  and  $C_r^*(\text{SL}_4(\mathbb{Z}))$  the MF property is undecided; only the stronger PMF property is known to fail, for  $\text{SL}_4(\mathbb{Z})$  [MdlS].

**Theorem C** (the reduced group algebra of  $E$ ).  *$C_r^*(E)$  is a unital separable  $C^*$ -algebra which has a faithful tracial state — and is therefore stably finite — but is not an MF algebra.*

*Proof.*  $C_r^*(E)$  is unital and separable because  $E$  is countable, and by Lemma 15.20 the canonical trace is faithful and the algebra is therefore stably finite. It is not an MF algebra by Theorem A.  $\square$

The algebra is not nuclear: the canonical map  $\mathcal{B} \rightarrow E$  is injective (Section 8), so  $E$  contains an infinite Kazhdan group and is nonamenable, whence  $C_r^*(E)$  is not nuclear [Lan]. The nuclear form of the Blackadar–Kirchberg problem — whether every stably finite separable nuclear  $C^*$ -algebra is quasi-diagonal — therefore remains open.

**A sofic example.** Let  $X = G/\Gamma$  be the coset space of the ascending HNN extension  $G = \Gamma_\infty \rtimes \mathbb{Z}$  used in the proof of Theorem 5.1, at the affine data (5), and let

$$W = \text{Cl}(X) \rtimes G$$

be the Clifford group over it. This is the target of the homomorphism used in that proof to show  $w \neq 1$ ; Section 8 realizes it over  $\mathbb{Q}$ , writing  $\Sigma$  for  $G$  in the explicit rational model.

**Theorem D** (a sofic non-MF group). *The Clifford quotient  $W = \text{Cl}(X) \rtimes \Sigma$  of Section 8 is a finitely generated sofic group that is not MF; since sofic groups are hyperlinear,  $W$  is likewise a hyperlinear group that is not MF. In particular, neither soficity nor hyperlinearity implies the MF property.*

Soficity comes from the finite-index image of the doubling map: every level of the direct limit then has finite orbits on  $X$ , so every finite subset of  $\text{Cl}(X) \rtimes \Gamma_\infty$  lies in a finite-by-sofic subgroup, and the remaining  $\mathbb{Z}$ -extension is sofic by Elek–Szabó [ES06]. Failure of MF follows from functoriality of the radical, since the surjection  $E \rightarrow W$  sends  $w$  to the nontrivial Clifford sign (Lemma 15.2). The same surjection shows that MF groups are not closed under quotients, and the extension  $1 \rightarrow \text{Cl}(X) \rightarrow W \rightarrow \Sigma \rightarrow 1$ , whose kernel is locally finite and whose quotient is residually finite, shows that they are not closed under extensions either (Section 15).

The kernel of  $E \rightarrow W$  is a free group of infinite rank, so soficity does not pass from  $W$  back to  $E$ . The operator-norm method of this paper decides soficity for neither group: its adjoint spectral estimates need operator-norm multiplicative control, which permutation models do not supply (Section 16). A separate argument does settle it for  $E$ , and settles it in the stronger direction.

**Theorem E** (a finitely presented sofic non-MF group). *The group  $E$  of Definition 6.1 is sofic, hence hyperlinear. Consequently  $E$  is itself a finitely presented sofic group that is not MF, and a finitely presented hyperlinear group that is not MF. It is also sofic and not LEF, and in particular not residually finite.*

The argument is independent of everything else in this paper and uses nothing about the MF property. The kernel  $E_0$  of the stable-letter exponent  $E \rightarrow \mathbb{Z}$  is locally residually finite: each of its finite subsets lies in a subgroup generated by one level of the base together with finitely many blocks of the lamp kernel, the lamp blocks are finite because the doubling map has finite index, and a finite-by-residually-finite group of that shape is residually finite. Hence  $E_0$  is LEF and therefore sofic, and  $E$  is its split extension by  $\mathbb{Z}$ , which preserves soficity exactly as in the proof of Theorem D. Soficity implies hyperlinearity; failure of MF is Theorem A; and  $E$  is not LEF because an LEF group is a marked limit of finite groups, which Theorem 15.21 excludes for a non-MF group. The full argument, including the block-by-block residual finiteness, is carried out in the accompanying development, where the two soficity routes meet the non-MF endpoint in a single premise-free statement.

Theorem E subsumes the finite-generation half of Theorem D: a finitely presented group is finitely generated, so  $E$  already witnesses that neither soficity nor hyperlinearity implies MF. Theorem D is kept because  $W$  is not an alternative example but the instrument: it is the Clifford quotient that proves  $w \neq 1$  in the first place, its soficity is proved by a short direct argument rather than by the block-tower analysis above, and it is the group

whose locally finite kernel and residually finite quotient give the permanence counterexamples of Section 15, which  $E$  does not supply.

## 10. TRANSPORT AT EVERY SCALE

The transport theorem extends to arbitrary weights  $w_n$  in place of the dimension; the normalization  $w_n = d_n$  of Theorem 3.4 is one special case.

**Theorem 10.1** (transport at every scale, and of intertwiners). *Let  $\Gamma$ ,  $H$ ,  $\iota$ , and  $s$  be as in Theorem 3.4.*

- (1) *Let  $(U_n)$  be as there, let  $(w_n)$  be nonnegative weights, and suppose there is  $C$  with  $\text{Tr}(x_n^* x_n) \leq C w_n$  for all  $n$ , and that for every  $\gamma \in \Gamma$  and every  $\varepsilon > 0$ , eventually*

$$\text{Tr} |x_n - U_n(\iota(\gamma)) x_n U_n(\iota(\gamma))^*|^2 \leq \varepsilon w_n.$$

*Then  $U_n(s) x_n U_n(s)^*$  and  $U_n(s)^* x_n U_n(s)$  satisfy both conditions as well.*

- (2) *Let  $U_n^1: H \rightarrow \mathcal{U}(d_n^1)$  and  $U_n^2: H \rightarrow \mathcal{U}(d_n^2)$  be operator-norm asymptotic representations and let  $(x_n)$  be uniformly operator-norm bounded  $d_n^1 \times d_n^2$  matrices with*

$$\frac{\text{Tr} |x_n - U_n^1(\iota(\gamma)) x_n U_n^2(\iota(\gamma))^*|^2}{d_n^1 + d_n^2} \longrightarrow 0 \quad (\gamma \in \Gamma).$$

*Then  $U_n^1(s) x_n U_n^2(s)^*$  satisfies the same convergence for every  $\gamma \in \Gamma$ .*

*Proof.* (1) Give each coordinate space with  $w_n > 0$  the inner product  $\langle x, y \rangle = \text{Tr}(y^* x)/w_n$ ; an index with  $w_n = 0$  forces  $x_n = 0$ . Conjugation by a unitary is unitary for every weight, and both hypotheses and both conclusions concern only the renormalized norms, so the ultraproduct proof of Theorem 3.4 applies with  $\text{tr}_{d_n}$  replaced by  $\text{Tr}(\cdot)/w_n$  and nothing else changed: the mass bound defines the ultraproduct vector, the adjoint actions remain operator-norm almost multiplicative because the weight does not enter the operator norm, and finiteness of the norm ultraproduct reverses the one-sided compression in both directions.

(2) The block sum  $W_n = U_n^1 \oplus U_n^2$  is an operator-norm asymptotic representation, since block-diagonal operator norms are maxima. The off-diagonal embedding of  $x_n$  has operator norm  $\|x_n\|$ , its commutator with  $W_n(\iota(\gamma))$  is the off-diagonal embedding of the intertwining defect — whose normalized Hilbert–Schmidt norm on the block space has the denominator  $d_n^1 + d_n^2$  — and conjugation by  $W_n(s)$  acts on the embedded corner by  $x_n \mapsto U_n^1(s) x_n U_n^2(s)^*$ . Theorem 3.4 applied to  $W_n$  transports the embedded sequence. Replacing the denominator  $d_n^1 + d_n^2$  by an arbitrary weight  $w_n$  and applying (1) to the block sum instead of Theorem 3.4 gives the intertwiner transport at every weight.  $\square$

The dimension weight  $w_n = d_n$  recovers Theorem 3.4 with the uniform operator-norm bound relaxed to mass boundedness; the collapse proof of

Theorem 15.13 uses a rank weight, which normalizes a defect at its own scale; and the adjoint sector of Theorem 3.4 is the diagonal case of (2).

Invariant tensors of type  $(p, q)$  may be identified with Hilbert–Schmidt intertwiners from the  $q$ th to the  $p$ th tensor power. Tensor powers of unitaries are exactly multiplicative and satisfy  $\|A^{\otimes p} - B^{\otimes p}\| \leq p \|A - B\|$ , so tensor powers of an operator-norm asymptotic representation are again operator-norm asymptotic representations, and Theorem 10.1(2) applies to every pair of them: one-sided compression of a Kazhdan image creates no asymptotic invariant tensors of any fixed type, at any weight.

## 11. OBSTRUCTIONS FROM A ONE-SIDED CONJUGATION DATUM

This section proves one theorem four times, at increasing generality, and the reader should see it as one argument. In each case a subgroup of  $H$  is shown to lie in the kernel of every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$ , by producing a corner on which the subgroup would have to be both 2-norm trivial and 2-norm large. What changes is only the hypothesis that produces the corner:

- Theorem 11.3: a *finite* normal subgroup of  $N_{\text{comp}}$ , whose averaging operator supplies the corner — the case used for the central involution of Section 4;
- Theorem 11.6: a normal property-(T) subgroup of  $N_{\text{comp}}$ , where a spectral projection of a Kazhdan average replaces the average of a finite group, and neither finiteness, centrality nor torsion is assumed;
- Theorem 11.7: the same statement with the conjugation datum discarded and its only consequence, 2-norm triviality, taken as the hypothesis;
- Corollary 11.8: that statement applied to the subgroup  $\mathfrak{D}(H, L)$  of (9), which depends on  $L$  alone and not on a choice of  $t$  and  $c$ .

The first two carry the analytic work; the last two are short deductions from the second.

**Definition 11.1** (Kazhdan conjugation datum). Let  $H$  be a countable group. A *Kazhdan conjugation datum* in  $H$  is a tuple  $(\Lambda, \iota, t, c)$ , where  $\Lambda$  is a countable group with property (T),  $\iota: \Lambda \rightarrow H$  is a group homomorphism, not assumed injective, and  $t, c \in H$ , subject to

- (M1)  $t \iota(\lambda) t^{-1} \in \iota(\Lambda)$  for every  $\lambda \in \Lambda$ ;
- (M2)  $c \iota(\lambda) = \iota(\lambda) c$  for every  $\lambda \in \Lambda$ .

Write  $d := tct^{-1}$ . The datum determines the normal closure

$$N_{\text{comp}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H \trianglelefteq H.$$

Both conditions are relations among generators, so a datum can be read off a presentation. The relations make  $c$  commute with  $\iota(\Lambda)$  and  $d$  commute with the subgroup  $t\iota(\Lambda)t^{-1}$ , but they do not constrain the commutators  $[d, \iota(\lambda)]$ ;  $N_{\text{comp}}$  is their normal closure. The theorems of this section show that every finite normal subgroup of  $H$  contained in  $N_{\text{comp}}$ , and more generally every

normal property-(T) subgroup contained in it, lies in the kernel of every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$ .

**Definition 11.2** (2-norm triviality). For an operator-norm asymptotic representation  $U = (U_n)$  of a countable group  $H$ , write

$$K_2(U) = \{g \in H : \|U_n(g) - 1\|_2 \longrightarrow 0\},$$

a normal subgroup of  $H$ ; it is the kernel of the homomorphism induced by  $U$  into the tracial ultraproduct of the matrix blocks. An element  $g$  is *2-norm trivial* if  $g \in K_2(U)$  for every such  $U$  and every ultrafilter refining the cofinite filter.

**Theorem 11.3** (finite-normal obstruction criterion). *Let  $H$  be a countable group with a Kazhdan conjugation datum  $(\Lambda, \iota, t, c)$ , and let  $F \trianglelefteq H$  be a finite normal subgroup with  $F \subseteq N_{\text{comp}}$ . Then every homomorphism  $\Theta : H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(f) = 1$  for all  $f \in F$ .*

*Proof.* *Tracial vanishing of the defect.* Let  $(U_{g,n})$  be coordinate unitary lifts of any operator-norm asymptotic representation of  $H$  (in particular, of a homomorphism to  $\mathcal{U}(\mathcal{Q})$ ), and put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \longrightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make  $K_2 \trianglelefteq H$ . Since  $c$  centralizes  $\iota(\Lambda)$ , its lifts belong to the asymptotic commutant in normalized Hilbert–Schmidt norm of  $\iota(\Lambda)$ . Applying Theorem 3.4 to  $x_n = U_{c,n}$  shows that the lifts of  $d = tct^{-1}$  lie there as well. Therefore  $[d, \iota(\lambda)] \in K_2$  for every  $\lambda \in \Lambda$ ; since  $K_2$  is normal, it contains the normal closure of these commutators:  $N_{\text{comp}} \subseteq K_2$ , and so  $F \subseteq K_2$ . Convergence of the full sequence gives triviality along every ultrafilter refining the cofinite filter, so every element of  $N_{\text{comp}}$  is 2-norm trivial (Definition 11.2).

*The coordinate averaging corner.* Suppose a homomorphism  $\Theta$  to  $\mathcal{U}(\mathcal{Q})$  is nontrivial on  $F$ ; choose coordinate unitary lifts  $(V_{g,n})$  of  $\Theta$  and form the coordinate averaging operators

$$p_n = \frac{1}{|F|} \sum_{f \in F} V_{f,n}.$$

Because the lifts are asymptotically multiplicative and  $F$  is finite,  $(p_n)$  is asymptotically self-adjoint and idempotent, and normality of  $F$  makes it asymptotically central for all of  $\Theta(H)$ . The Hermitian part of  $p_n$  is asymptotically idempotent, so its spectrum clusters at  $\{0, 1\}$  and rounding at  $1/2$  by continuous functional calculus yields genuine projections at operator-norm distance  $o(1)$  from  $p_n$ ; the rounded projections inherit the asymptotic centrality. Let  $q_n$  be their complements. Nontriviality of  $\Theta$  on  $F$  forces  $q_n \neq 0$  along infinitely many coordinates — otherwise the averages converge to 1 and  $\Theta|_F$  is trivial. Retain exactly those coordinates and relabel them by  $\mathbb{N}$ . Compress every lift to the corner  $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$  with  $r_n = \text{rank } q_n > 0$ , and polar-correct the compressed matrices: asymptotic centrality makes

each compression  $q_n V_{g,n} q_n$  almost unitary in the corner, so its polar unitary is at operator-norm distance  $o(1)$  and almost-multiplicativity is preserved by the correction. The result is an operator-norm asymptotic representation  $(W_{g,n})$  of  $H$  on the nonzero corner blocks. Since the corner is the complement of the rounded average, the finite average vanishes there:

$$(*) \quad \left\| \sum_{f \in F} W_{f,n} \right\| \longrightarrow 0 \quad \text{in operator norm.}$$

*The contradiction.* The tracial vanishing applies to the corner almost representation  $(W_{g,n})$  itself: every  $f \in F$  satisfies  $\|W_{f,n} - 1\|_2 \rightarrow 0$ . Since  $F$  is finite, the maximum of these  $|F|$  null sequences is itself null, so

$$\sum_{f \in F} W_{f,n} \longrightarrow |F|1 \quad \text{in normalized Hilbert–Schmidt norm,}$$

each block with its own normalized trace. This contradicts  $(*)$ , because operator-norm convergence dominates normalized Hilbert–Schmidt convergence. Hence no homomorphism to  $\mathcal{U}(\mathcal{Q})$  is nontrivial on  $F$ .  $\square$

The averaging-corner argument used only one property of  $N_{\text{comp}}$ : every element is 2-norm trivial in every operator-norm asymptotic representation of  $H$ . Any normal subgroup with that tracial-vanishing property therefore obstructs its finite normal subgroups in exactly the same way; Theorem 11.5 exploits this.

The central involution obstruction of the main text (Theorem 4.1) is the case  $F = \{1, w\}$  of Theorem 11.3: there the averaging operator is  $p_n = \frac{1}{2}(1 + V_{w,n})$ , its complementary corner is the negative spectral corner of the sign, and compression to that corner gives a model in which  $w = -1$  exactly.

**11.1. An intrinsic form.** The transport estimate applies to every one-sided conjugator of a subgroup, not only to the element  $t$ . For a subgroup  $L \leq H$ , let  $G_{\text{comp}}(L)$  be generated by the elements  $s$  satisfying  $sLs^{-1} \subseteq L$  — the *one-sided conjugators* of  $L$  — and put

$$(9) \quad \mathfrak{D}(H, L) = \left\langle \left\langle [gzg^{-1}, \ell] : g \in G_{\text{comp}}(L), z \in C_H(L), \ell \in L \right\rangle \right\rangle_H,$$

the subgroup  $\mathfrak{D}(H, L)$ . It depends only on the pair  $(H, L)$ , not on a choice of datum.

**Corollary 11.4** (transport along all one-sided conjugators). *Let  $(r_n)$  and  $V_{g,n} \in \mathcal{U}(r_n)$  be an operator-norm asymptotic representation of  $H$ , let  $\Lambda$  have property (T), let  $\iota : \Lambda \rightarrow H$  be a homomorphism, and put  $L = \iota(\Lambda)$ . Suppose that  $(x_n \in M_{r_n}(\mathbb{C}))$  is uniformly bounded in operator norm and belongs to the asymptotic commutant in normalized Hilbert–Schmidt norm of  $L$ :*

$$\left\| x_n - V_{\iota(\lambda),n} x_n V_{\iota(\lambda),n}^* \right\|_2 \longrightarrow 0 \quad (\lambda \in \Lambda).$$

Then every  $g \in G_{\text{comp}}(L)$  acts in both directions on that asymptotic commutant. Explicitly, both

$$V_{g,n}x_nV_{g,n}^* \quad \text{and} \quad V_{g,n}^*x_nV_{g,n}$$

satisfy the preceding convergence for every  $\lambda \in \Lambda$ .

*Proof.* For a one-sided conjugator  $s$ , Theorem 3.4 gives forward transport. Reverse transport follows from the same proof: the equality  $P = V P V^*$  obtained there gives  $V^* P V = P$  as well, so the fixed subspace is preserved by conjugation in both directions. In the finite-stage proof of Section B, the equal-rank reversal estimate reverses the one-sided containment of the spectral subspaces; the same argument then runs with the adjoint unitary, with the displacement and concentration steps unchanged. The elements acting in both directions form a subgroup of  $H$ , and hence contain the subgroup generated by all one-sided conjugators, namely  $G_{\text{comp}}(L)$ .  $\square$

**Theorem 11.5** (the subgroup  $\mathfrak{D}$  and the MF radical). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and put  $L = \iota(\Gamma)$ . If  $F \trianglelefteq H$  is finite and  $F \leq \mathfrak{D}(H, L)$ , then every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  maps every element of  $F$  to the identity.*

*Proof.* Let  $(U_{h,n})$  be coordinate lifts of an operator-norm asymptotic representation of  $H$  and put  $K_2 = \{h : \|U_{h,n} - 1\|_2 \rightarrow 0\}$ , a normal subgroup of  $H$ . If  $z \in C_H(L)$ , then  $(U_{z,n})$  lies in the asymptotic commutant of  $L$ , and Corollary 11.4 places  $(U_{gzg^{-1},n})$  there for every  $g \in G_{\text{comp}}(L)$ . Hence each generator  $[gzg^{-1}, \ell]$ ,  $\ell \in L$ , displayed in (9) belongs to  $K_2$ ; since  $K_2 \trianglelefteq H$ , it contains their normal closure:  $\mathfrak{D}(H, L) \leq K_2$ . Every element of  $\mathfrak{D}(H, L)$  is therefore 2-norm trivial in every operator-norm asymptotic representation of  $H$ . The coordinate averaging corner of Theorem 11.3 uses only that triviality, so it applies to any finite normal  $F \leq \mathfrak{D}(H, L)$ .  $\square$

**11.2. The normal-Kazhdan form of the obstruction.** Finiteness of  $F$  enters Theorem 11.3 only through the averaging corner. A finite group has property (T), and its averaging operator is its fixed-space average; for an arbitrary normal property-(T) subgroup, the spectral projection of a Kazhdan average takes its place. The resulting form of the obstruction has no finiteness, centrality, or torsion hypothesis; it is used in Section 16.

**Theorem 11.6** (normal-Kazhdan obstruction). *Let  $H$  be a countable group with a Kazhdan conjugation datum  $(\Lambda, \iota, t, c)$ , and let  $K \trianglelefteq H$  be a normal subgroup with property (T) such that  $K \subseteq N_{\text{comp}}$ . Then every homomorphism  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(k) = 1$  for all  $k \in K$ .*

Finite groups have property (T), so  $K = F$  recovers Theorem 11.3. In Theorem 11.6,  $K$  may be infinite, noncentral, and torsion-free.

*Proof.* The containment  $N_{\text{comp}} \subseteq K_2$  from the proof of Theorem 11.3 uses no finiteness, so every element of  $K$  is 2-norm trivial in every operator-norm asymptotic representation of  $H$ . It remains to replace the averaging corner

by a corner adapted to  $K$ ; the spectral projection of a Kazhdan average provides it.

Suppose a homomorphism  $\Theta$  to  $\mathcal{U}(\mathcal{Q})$  is nontrivial on  $K$ . Its image  $\bar{H} = \Theta(H)$  is countable, hence MF, and Proposition 1.1, applied along an exhaustion of  $\bar{H}$ , gives an operator-norm asymptotic representation  $(V_{g,n})$  in which every fixed nontrivial element eventually stays at operator-norm distance at least 1 from the identity. These models bound only operator-norm distances; at growing dimension they give no lower bound on normalized Hilbert–Schmidt distances, and the corner constructed below provides that lower bound. The datum and the subgroup pass to their  $\Theta$ -images with normality and property (T) preserved, and  $\bar{K} = \Theta(K)$  is nontrivial and lies in the subgroup  $N_{\text{comp}}$  determined by the image datum; it suffices to derive a contradiction for  $\bar{H}$ .

*The spectral projection  $q_n$ .* Fix a finite symmetric generating Kazhdan set  $S \subseteq \bar{K}$  with constant  $\varepsilon_0$ , at each coordinate let  $h_n$  be the Hermitian part of the average of the models of  $S$ , and cut the spectrum of  $h_n$  at  $\theta = 1 - \varepsilon_0^2/(4|S|)$ . Let  $q_n$  be the spectral projection of  $h_n$  for  $[-1, \theta]$ . On the complementary range, where the spectrum of  $h_n$  lies above  $\theta$ , every model of every element of  $\bar{K}$  is close to the identity in operator norm, by the Kazhdan-pair displacement estimate. If  $q_n = 0$  for all large  $n$ , every element of  $\bar{K}$  would satisfy  $\|V_{k,n} - 1\| \rightarrow 0$ , contradicting the operator-norm separation of any fixed nontrivial element of  $\bar{K}$ . So  $q_n \neq 0$  along an infinite set of coordinates, which we retain.

*The tracial deficit on the corner.* Equip each retained corner block  $q_n M_{d_n}(\mathbb{C}) q_n$  with its renormalized trace. By the definition of the cut,  $q_n h_n q_n \leq \theta q_n$ , so the average over  $S$  of the real parts of the renormalized traces of the compressed models is at most  $\theta$ . Hence at every retained coordinate some generator in  $S$  has compressed model with real renormalized trace at most  $\theta$ , and therefore normalized Hilbert–Schmidt distance at least  $\sqrt{2(1-\theta)} - o(1)$  from the corner identity after polar correction; passing to a subsequence fixes a single such generator  $s_0$ .

*Compression.* Normality makes the cut asymptotically central. Conjugating by an ambient model  $V_{g,n}$  carries  $h_n$  to the Hermitian average over the conjugate generating set  $gSg^{-1}$  of  $\bar{K}$ , which has the same Kazhdan constant. The two estimates of Section B now apply to either average symmetrically: a vector on which one average has spectrum above  $\theta$  is almost fixed by each generator of the other set (displacement, transported across the finitely many conjugations by almost-multiplicativity), hence almost lies in the corresponding range for the other average (the concentration inequality). Two-sided almost-containment of the two subspaces implies  $\|q_n - V_{g,n} q_n V_{g,n}^*\| \rightarrow 0$ , that is,  $\|[q_n, V_{g,n}]\| \rightarrow 0$  for every  $g \in \bar{H}$ . Compressing every model to the corner  $q_n M_{d_n}(\mathbb{C}) q_n$  and polar-correcting yields an operator-norm asymptotic representation of all of  $\bar{H}$  on nonzero blocks, in which  $s_0$  keeps a definite normalized Hilbert–Schmidt distance from the identity.

*The contradiction.* The corner models form an operator-norm almost representation of  $\bar{H}$ , and  $\bar{K}$  lies in the subgroup  $N_{\text{comp}}$  determined by the image datum, so the tracial vanishing applies to them: every element of  $\bar{K}$  — in particular  $s_0$  — is sent to the identity by the corner representation, contradicting the tracial deficit of  $s_0$ .  $\square$

The proofs above use the conjugation datum through one property only: every element of  $N_{\text{comp}}$  is 2-norm trivial in every operator-norm asymptotic representation. That property alone suffices.

**Theorem 11.7** (obstruction from 2-norm triviality). *Let  $H$  be a countable group and let  $D \leq H$  be a subgroup every element of which is 2-norm trivial (Definition 11.2). Let  $K \trianglelefteq H$  be a normal subgroup with property (T) such that  $K \leq D$ . Then every homomorphism  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  satisfies  $\Theta(k) = 1$  for all  $k \in K$ .*

*Proof.* The proof of Theorem 11.6 used its datum only to make every element of  $N_{\text{comp}}$  2-norm trivial in every operator-norm asymptotic representation; here that triviality is the hypothesis. Suppose a homomorphism  $\Theta$  to  $\mathcal{U}(\mathcal{Q})$  is nontrivial on  $K$ . As there, the image  $\bar{H} = \Theta(H)$  is countable and MF, and Proposition 1.1 provides an operator-norm asymptotic representation of  $\bar{H}$  that separates every fixed nontrivial element. Every operator-norm asymptotic representation of  $\bar{H}$  composes with  $\Theta$  to one of  $H$ , so every element of  $\Theta(D)$  inherits the hypothesis, while  $\bar{K} = \Theta(K) \leq \Theta(D)$  is normal, Kazhdan, and nontrivial. Three steps of the proof of Theorem 11.6 — the spectral projection  $q_n$ , the tracial deficit, and the compression — use only that  $\bar{K}$  is normal, Kazhdan and nontrivial, so they apply to  $\bar{K}$  here; the contradiction step then uses the inherited triviality in place of the defect containment.  $\square$

Applied to the intrinsic defect, Theorem 11.7 gives the normal-Kazhdan companion of Theorem 11.5:

**Corollary 11.8** (normal-Kazhdan form of the intrinsic radical). *Let  $H$  and  $\Gamma$  be countable groups, let  $\Gamma$  have property (T), let  $\iota: \Gamma \rightarrow H$  be a homomorphism, and put  $L = \iota(\Gamma)$ . If  $K \trianglelefteq H$  is a normal subgroup with property (T) and  $K \leq \mathfrak{D}(H, L)$ , then every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  maps every element of  $K$  to the identity.*

*Proof.* The proof of Theorem 11.5 shows that every element of  $\mathfrak{D}(H, L)$  is 2-norm trivial in every operator-norm asymptotic representation of  $H$ . Theorem 11.7 with  $D = \mathfrak{D}(H, L)$  concludes.  $\square$

The subgroup generated by *all* normal property-(T) subgroups of  $\mathfrak{D}(H, L)$  therefore also lies in the kernel of every homomorphism to  $\mathcal{U}(\mathcal{Q})$ .

## 12. THE FINITE-DIMENSIONAL OBSTRUCTION

Theorem 3.3, stated and proved in Section 3, needs finite dimensionality and the one-sided conjugation and centralizer relations alone. Its conclusion

covers the entire subgroup (9), which is what this section proves, and it has two consequences for  $E$  and one abstract reading.

Applied to  $H = E$ , Theorem 3.3 sends  $w$  to the identity in every finite-dimensional representation of  $E$  over every field. So finite-dimensional linear representations do not separate the points of  $E$ , no finite quotient witnesses  $w \neq 1$ , and the Clifford quotient of Section 8 has to be infinite. In the terminology of [BDL, Definition 1.2], the central involution  $w$ , invisible in every finite quotient, makes  $E$  a group of *Deligne type*. Such groups were proposed there as the natural candidate counterexamples for all three remaining approximation problems, and  $E$  realizes that proposal for the operator norm.

The same calculation proves the final, subgroup-valued assertion of Theorem 3.3. The elements of  $H$  whose conjugation action preserves  $C$  in both directions form a subgroup. The argument above places every one-sided conjugator in that subgroup, and therefore all of  $G_{\text{comp}}(\Gamma)$  in it. Thus, for  $g \in G_{\text{comp}}(\Gamma)$  and  $z \in C_H(\Gamma)$ , the operator  $\pi(gzg^{-1})$  still belongs to  $C$  and commutes with every  $\pi(\ell)$ ,  $\ell \in \Gamma$ . Hence every displayed generator in (9) lies in  $\ker \pi$ ; since  $\ker \pi \trianglelefteq H$ , the whole normal closure  $\mathfrak{D}(H, \Gamma)$  does as well.

**Corollary 12.1.** *The group  $E$  of Definition 6.1 admits no injective homomorphism into  $\text{GL}(V)$  for any finite-dimensional vector space  $V$  over any field. More strongly, every finite-dimensional linear representation maps  $w$  to the identity, and every homomorphism to a finite group maps  $w$  to the identity.*

*Proof.* Apply Theorem 3.3 with  $H = E$ , the image of  $\mathcal{B}$ , and the elements  $t, c, v_1$ ; the hypotheses hold by the defining relations (8). Every finite-dimensional representation maps  $w$  to the identity, and  $w \neq 1$  by Proposition 8.3. For a finite target, compose with its faithful left regular representation on the group algebra over  $\mathbb{Q}$ . The composite is a finite-dimensional linear representation, so it maps  $w$  to the identity. Evaluating the resulting identity matrix at the basis vector of the identity element returns the image of  $w$  in the finite group.  $\square$

*Remark 12.2* (the invariant-size principle). The proof of Theorem 3.3 used one feature of finite dimensionality: if  $V \subseteq W$  and  $\dim V = \dim W < \infty$ , then  $V = W$ . Abstractly, let a conjugation-invariant size  $\sigma$  with values in  $\mathbb{N}$  be defined on the subgroups of a group  $Q$ , and suppose  $K \leq L$  with  $\sigma(L) \leq \sigma(K)$  implies  $K = L$ . If  $gKg^{-1} \subseteq K$  then the two have equal size, so the inclusion is an equality: such a target admits no strict one-sided compression. Cardinality on the subgroups of a finite group qualifies, which is the finite-quotient argument of Section 1.

It is worth being clear that the finite-dimensional case is *not* that principle applied to subgroups. The natural candidate size, the dimension  $\dim_k \text{End}_H(V)$  of the commutant, is order-reversing rather than order-preserving, and it does not separate nested subgroups:  $A_4 < S_4$  in the standard three-dimensional representation have the same commutant  $k \cdot 1$ .

Theorem 3.3 applies the principle to the commutant instead, where the inclusion  $C \subseteq uCu^{-1}$  between subspaces of equal finite dimension is an equality (3); it is that equality, and not an equality of subgroup images, that sends the defect to the identity. Theorem A is the analytic counterpart at infinite dimension.

*Remark 12.3* (chain conditions suffice). Finiteness of the invariant can be weakened to a chain condition. A conjugation-equivariant monotone invariant whose values satisfy the ascending or descending chain condition takes the same value on  $\Gamma$  as on  $t\Gamma t^{-1}$ , because an order isomorphism of such a poset fixes every point it moves weakly upward. Three instances: the fixed subspaces of  $\Gamma$  and  $t\Gamma t^{-1}$  agree in every finite-dimensional representation over every field; the same holds for the invariants of the induced representations on  $V^{\otimes p} \otimes (V^*)^{\otimes q}$ , of which the commutant identity (3) is the case  $\text{End}(V) \cong V \otimes V^*$ ; and the Zariski closures of  $\sigma(\Gamma)$  and  $\sigma(t\Gamma t^{-1})$  coincide, the descending chain condition for Zariski-closed subgroups of  $\text{GL}(V)$  coming from the Hilbert basis theorem. Theorem 10.1(2) is the asymptotic counterpart at fixed tensor type.

### 13. THE CYCLIC COMPARISON

The finite-dimensional calculation of Section 12 does not by itself determine whether its marked word is trivial:

**Theorem 13.1** (the cyclic comparison). *Define the marked presentation*

$$E_{\text{BS}} = \left\langle \gamma_0, t, c \left| \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\text{BS}}^2 = 1, \quad [w_{\text{BS}}, \gamma_0] = [w_{\text{BS}}, t] = [w_{\text{BS}}, c] = 1 \end{array} \right. \right\rangle,$$

where  $w_{\text{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$ . Then  $E_{\text{BS}}$  is finitely presented and  $w_{\text{BS}} \neq 1$ . There is an explicit surjective homomorphism from  $E_{\text{BS}}$  onto its realized Clifford quotient that maps  $w_{\text{BS}}$  to the nontrivial central sign. Nevertheless every finite-dimensional representation of  $E_{\text{BS}}$  over every field maps  $w_{\text{BS}}$  to the identity.

Thus being sent to the identity by every exact finite-dimensional representation does not by itself force triviality of the marked word. The MF conclusion in Theorem A is the separate uniform assertion proved by the fixed-space argument.

*Proof of Theorem 13.1.* As with  $w$ , the relator  $w_{\text{BS}}^2 = 1$  is redundant:  $tct^{-1}$  and  $\gamma_0(tct^{-1})\gamma_0^{-1}$  are involutions, so centrality forces it by (4). Deleting it gives a six-relator presentation of the same group. The stable relation gives  $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$ , and  $c$  centralizes this cyclic base. The same commutant argument as in Theorem 3.3 therefore maps  $w_{\text{BS}}$  to the identity in every finite-dimensional representation over every field.

For nontriviality, use the affine realization of  $\text{BS}(1, 2)$  and its coset action. The base point is fixed by the cyclic base, while the two points represented

by  $t$  and  $\gamma_0 t$  are distinct. Attach Clifford generators to these points and map  $c$  to the generator at the base point. All seven relators hold, and

$$w_{\text{BS}} \mapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus  $w_{\text{BS}} \neq 1$ . Restricting the target to the subgroup generated by the three displayed images gives the asserted surjective realized quotient.  $\square$

Sharpness of the Kazhdan hypothesis. The realized Clifford quotient in Theorem 13.1 is amenable, hence MF, and its quotient map sends  $w_{\text{BS}}$  to a nontrivial element. Composing with a faithful homomorphism from that quotient to  $\mathcal{U}(\mathcal{Q})$  gives a homomorphism  $\Theta: E_{\text{BS}} \rightarrow \mathcal{U}(\mathcal{Q})$  with  $\Theta(w_{\text{BS}}) \neq 1$ , so property (T) cannot be omitted from the central involution obstruction. In the radical language of Section 15,  $w_{\text{BS}} \notin \text{Rad}_{\text{MF}}(E_{\text{BS}})$ . The affine group  $\text{BS}(1, 2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$  is solvable. The Clifford group on its coset space is locally finite, so their semidirect product is amenable. The realized quotient is a subgroup of this amenable group and is therefore amenable, hence MF by [TWW]. Its marked word is the nontrivial Clifford sign by the calculation above.

The homomorphism to  $\mathcal{U}(\mathcal{Q})$  just constructed is necessarily approximate: neither the group  $E_{\text{BS}}$  nor its realized Clifford quotient is LEF, and the latter is not residually finite. In every finite group the doubling relation makes the image of  $\langle\gamma_0^2\rangle$  equal to the image of  $\langle\gamma_0\rangle$ ; the generator therefore commutes with the required “halved” element and the marked word maps to the identity, so neither group is residually finite. The same equations transport through any exact local embedding of a finite test set containing the defining relators and the marked word, so an exact finite-dimensional representation would already send the marked word to the identity: neither group is LEF.

## 14. THE SCALING FAMILY

The same construction works for every exponent. For  $m \geq 2$ , let  $E_m$  be the same eight-generator presentation, replacing the three relations  $tv_it^{-1} = v_i^2$  by  $tv_it^{-1} = v_i^m$  and leaving every other relator unchanged; thus  $E_2 = E$ . Write  $w_m$  for the image of the same marked word  $w$ .

**Corollary 14.1** (the scaling family). *For every natural number  $m \geq 2$ , the group  $E_m$  is finitely presented and  $w_m$  is a nontrivial central involution mapped to the identity by every homomorphism  $E_m \rightarrow \mathcal{U}(\mathcal{Q})$ . Consequently  $E_m$  is not MF, and neither  $C_{\max}^*(E_m)$  nor  $C_r^*(E_m)$  is an MF  $C^*$ -algebra.*

*Proof.* Replace the affine dilation  $\text{diag}(2, 2, 2, 1)$  by  $\text{diag}(m, m, m, 1)$  (both acting as in Lemma 8.2). It conjugates each integral translation  $v_i$  to  $v_i^m$  and fixes the three linear generators. Its inverse sends  $v_1$  to translation by  $e_1/m$ , which is not integral; hence the two Clifford generators remain distinct and  $w_m$  maps to  $-1$ . The Kazhdan base, the one-sided conjugation argument, and the central involution obstruction are independent of  $m$ .  $\square$

## 15. CONSEQUENCES

## 15.1. The MF radical.

**Definition 15.1.** For a countable group  $G_1$ , the *MF radical* is

$$\text{Rad}_{\text{MF}}(G_1) = \bigcap \{ \ker \Theta : \Theta \text{ a homomorphism } G_1 \rightarrow \mathcal{U}(\mathcal{Q}) \},$$

a normal subgroup of  $G_1$ . The intersection does not change if the dimension sequences are allowed to be arbitrary positive rather than strictly increasing (Proposition 1.1). This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST].

**Lemma 15.2** (functoriality). *For every homomorphism  $\varphi: G_1 \rightarrow G_2$  of countable groups,*

$$\varphi(\text{Rad}_{\text{MF}}(G_1)) \subseteq \text{Rad}_{\text{MF}}(G_2) :$$

*the MF radical is functorial. In particular, if  $\varphi: E \rightarrow G_2$  satisfies  $\varphi(w) \neq 1$ , then  $G_2$  is not MF.*

*Proof.* If  $x \in \text{Rad}_{\text{MF}}(G_1)$  and  $\Theta$  is a homomorphism  $G_2 \rightarrow \mathcal{U}(\mathcal{Q})$ , then  $\Theta \circ \varphi$  is a homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ , so  $\Theta(\varphi(x)) = 1$ . For the second statement,  $w \in \text{Rad}_{\text{MF}}(E)$  by Theorem A, so  $1 \neq \varphi(w) \in \text{Rad}_{\text{MF}}(G_2)$ , and an injective homomorphism  $G_2 \rightarrow \mathcal{U}(\mathcal{Q})$  would map  $\varphi(w)$  to 1, a contradiction.  $\square$

**Proposition 15.3** (the universal MF quotient). *Let  $G_1$  be a countable group and  $R = \text{Rad}_{\text{MF}}(G_1)$ .*

- (1)  *$R$  is the kernel of a single homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ .*
- (2)  *$G_1/R$  is MF.*
- (3) *Every homomorphism from  $G_1$  to an MF group factors through  $G_1/R$ .*

In particular  $G_1$  is MF if and only if  $R = \{1\}$ .

*Proof.* (3) If  $\varphi: G_1 \rightarrow M$  with  $M$  MF and  $j: M \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$  is an injective homomorphism to  $\mathcal{U}(\mathcal{Q})$ , then  $j \circ \varphi$  is a homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ , so  $j(\varphi(R)) = 1$  and, by injectivity of  $j$ ,  $\varphi(R) = 1$ .

(1) and (2). Work on  $Q = G_1/R$ . Every homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$  maps  $R$  to 1, and for each  $g \notin R$  there is one with  $\Theta(g) \neq 1$ , which descends; so for every  $\bar{g} \neq 1$  in  $Q$  there is a homomorphism  $\Theta_{\bar{g}}: Q \rightarrow \mathcal{U}(\mathcal{Q})$  with  $\Theta_{\bar{g}}(\bar{g}) \neq 1$ . Enumerate the nonidentity elements and these representations, pass to coordinate unitary lifts, and diagonalize: at stage  $k$ , select in each of the first  $k$  of them a coordinate at which multiplicativity holds within  $1/k$  on the first  $k$  group elements and each of the first  $k$  marked elements stays separated by its fixed positive constant — possible because each  $\Theta_{\bar{g}}$  separates its own element along infinitely many coordinates — and take the block sum. Block operator norms are maxima, so the resulting sequence is an almost representation separating every nonidentity element from the identity; its class, composed with the polar-correction isomorphism of Lemma 2.2, is a faithful homomorphism  $Q \rightarrow \mathcal{U}(\mathcal{Q})$ . This proves (2), and composing with  $G_1 \twoheadrightarrow Q$  proves (1). Finally, if  $G_1$  is MF then an injective homomorphism to  $\mathcal{U}(\mathcal{Q})$  gives  $R = \{1\}$ ; if  $R = \{1\}$  then (2) says  $G_1$  is MF.  $\square$

**Corollary 15.4** (exact radical from a candidate quotient). *Let  $N \trianglelefteq G_1$  satisfy  $N \leq \text{Rad}_{\text{MF}}(G_1)$ . Every homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$  factors uniquely through  $G_1/N$ . If in addition  $G_1/N$  is MF, then*

$$\text{Rad}_{\text{MF}}(G_1) = N.$$

*Proof.* The containment  $N \leq \ker \Theta$  gives the unique quotient factorization for every such  $\Theta$ . For the reverse containment in the displayed equality, compose the quotient map  $G_1 \rightarrow G_1/N$  with an injective homomorphism  $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$ : its kernel is exactly  $N$ , while the MF radical lies in the kernel of every homomorphism to  $\mathcal{U}(\mathcal{Q})$ .  $\square$

The MF hypothesis on the quotient can be dropped entirely; what remains is an unconditional reduction.

**Corollary 15.5** (radical reduction to the quotient). *Let  $N \trianglelefteq G_1$  satisfy  $N \leq \text{Rad}_{\text{MF}}(G_1)$  and let  $q: G_1 \rightarrow G_1/N$  be the quotient map. Then*

$$\text{Rad}_{\text{MF}}(G_1) = q^{-1}(\text{Rad}_{\text{MF}}(G_1/N)).$$

*Proof.* For the forward containment, Lemma 15.2 gives  $q(\text{Rad}_{\text{MF}}(G_1)) \subseteq \text{Rad}_{\text{MF}}(G_1/N)$ . Conversely, suppose  $q(x) \in \text{Rad}_{\text{MF}}(G_1/N)$  and let  $\Theta$  be a homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$ . Every homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$  maps  $N$  to 1 by hypothesis, so  $\Theta$  factors as  $\bar{\Theta} \circ q$  with  $\bar{\Theta}$  a homomorphism  $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$ ; hence  $\Theta(x) = \bar{\Theta}(q(x)) = 1$ .  $\square$

Corollary 15.4 is the case of trivial quotient radical. Corollary 15.5 needs no hypothesis on the quotient: every homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$  factors uniquely through  $G_1/N$ , whether or not  $G_1/N$  is MF.

By Corollary 15.5, the passage from  $N$  to the full preimage of the quotient radical is a closure operation on the normal subgroups of a fixed group.

**Proposition 15.6** (the MF closure of relations). *For a countable group  $G_1$  and  $N \trianglelefteq G_1$  with quotient map  $q: G_1 \rightarrow G_1/N$ , put*

$$\text{cl}_{\text{MF}}(N) = q^{-1}(\text{Rad}_{\text{MF}}(G_1/N)).$$

*An element belongs to  $\text{cl}_{\text{MF}}(N)$  exactly when every homomorphism  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$  that maps  $N$  to the identity maps it to the identity. The operation  $N \mapsto \text{cl}_{\text{MF}}(N)$  is extensive, monotone, and idempotent on normal subgroups;  $G_1/\text{cl}_{\text{MF}}(N)$  is MF; and  $\text{cl}_{\text{MF}}(N) = N$  exactly when  $G_1/N$  is MF.*

*Proof.* Homomorphisms  $G_1 \rightarrow \mathcal{U}(\mathcal{Q})$  that map  $N$  to 1 correspond to homomorphisms  $G_1/N \rightarrow \mathcal{U}(\mathcal{Q})$ , which gives the membership characterization; extensivity, monotonicity, and idempotence are read off from it. The quotient clause is Proposition 15.3(2) transported along the third isomorphism theorem, and the fixed-point clause is the final equivalence of Proposition 15.3.  $\square$

For a presentation  $G = \langle X \mid R \rangle = F/N$ , the closure of  $N$  consists of the relations satisfied by every solution of  $R$  in every MF group, and the group is MF exactly when  $N$  is closed. Proposition 15.22 displays one finite fragment of this closure, and the marked element of Theorem A is an element of  $\text{cl}_{\text{MF}}(N) \setminus N$  for the forty-one relators.

**Proposition 15.7** (iterated sound rules). *Consider the subgroups of  $G_1$  reachable from the trivial subgroup by two operations: from a reached normal subgroup  $N$ , pass to any subgroup of  $\text{cl}_{\text{MF}}(N)$ ; and from any family of reached subgroups, pass to their join, the subgroup generated by their union. Every subgroup so reached lies in  $\text{Rad}_{\text{MF}}(G_1)$ ; a reached normal subgroup with MF quotient equals  $\text{Rad}_{\text{MF}}(G_1)$ ; and if the whole group is reached, every homomorphism from  $G_1$  to a countable MF group is trivial.*

*Proof.* Induct over the two operations. If a reached normal subgroup  $N$  satisfies  $N \leq \text{Rad}_{\text{MF}}(G_1)$ , then Corollary 15.5 gives  $\text{cl}_{\text{MF}}(N) = \text{Rad}_{\text{MF}}(G_1)$ , so every subgroup chosen in the first operation is again inside the radical — normal or not. For the second operation, the radical is itself a subgroup, hence an upper bound in the subgroup lattice for any family contained in it; so the join  $\bigvee_i H_i = \langle \bigcup_i H_i \rangle$  lies in the radical too. The final two assertions are Corollary 15.4 and Proposition 15.3; the second of them needs  $N$  normal, which is where the quotient  $G_1/N$  is formed.  $\square$

Theorems 11.3, 11.6, and 15.13 are sound rules in this sense: each places a subgroup inside the radical of every countable group with its data, hence inside the radical of every reached quotient. Theorem 15.19 is the case

$D_{\text{coll}}(L, s) = H$ . A rule instance can hold at a later stage without holding at an earlier one: modulo the earlier stage an element may acquire finite order, an orbit may begin to commute, or a subgroup may become normal.

**Corollary 15.8.** *Let  $A$  be a unital  $C^*$ -algebra admitting an injective, possibly nonunital,  $*$ -homomorphism into a matrix quotient. Then the group  $E$  admits no injective homomorphism into  $\mathcal{U}(A)$ .*

*Proof.* Let  $j: A \rightarrow \mathcal{Q}$  be the possibly nonunital inclusion into the ambient matrix quotient and let  $\phi: E \rightarrow \mathcal{U}(A)$ . Complement correction gives

$$\hat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(\mathcal{Q}).$$

This is an injective homomorphism on  $\mathcal{U}(A)$ , so  $\hat{j} \circ \phi$  is a homomorphism  $E \rightarrow \mathcal{U}(\mathcal{Q})$ . By Theorem A,  $(\hat{j} \circ \phi)(w) = 1$ ; injectivity of  $\hat{j}$  gives  $\phi(w) = 1$ . Proposition 8.3 has  $w \neq 1$ , so  $\phi$  cannot be injective.  $\square$

## 15.2. Permanence and its failure.

**Lemma 15.9** (permanence). *Let all groups be countable.*

- (1) *Every subgroup of an MF group is MF.*
- (2) *Every residually finite group is MF.*
- (3) *Every locally finite group is MF.*

*Proof.* (1) Restrict an injective homomorphism to  $\mathcal{U}(\mathcal{Q})$ . (2) For a finite test set  $F$ , a finite quotient injective on  $F$ , composed with its left regular permutation action, is an exact finite-dimensional representation; distinct permutations differ in some column of a permutation matrix, giving operator-norm separation at least 1, and the local-to-global direction of Proposition 1.1 concludes. (3) A finite test set lies in a finite subgroup  $K$ ; a map that is the identity on all of  $K$ , composed with the left regular permutation action of  $K$ , is an exact finite-dimensional representation on the test set — products of test elements stay in  $K$  — separating as in (2).  $\square$

**Corollary 15.10** (quotients). *MF groups are not closed under quotients: the free group  $F_8$  is MF, and the explicit group  $E$  is its quotient but is not MF.*

*Proof.* Map the eight free generators onto the eight displayed generators of  $E$ . The resulting homomorphism is surjective because those generators define the displayed presentation. The group  $F_8$  is residually finite, hence MF by Lemma 15.9(2). Theorem A shows that  $E$  is not MF.  $\square$

**15.3. Matricial stability and the radical.** Following Dadarlat [Dad], a countable group  $G$  is *matricially stable* if for every operator-norm asymptotic representation  $(V_n)$  of  $G$  there are genuine finite-dimensional unitary representations  $\pi_n$ , in the same matrix dimensions, with  $\|V_n(g) - \pi_n(g)\| \rightarrow 0$  for every  $g \in G$ . Write  $\text{Rad}_{\text{fd}}(G)$  for the intersection of the kernels of all finite-dimensional unitary representations of  $G$ .

**Proposition 15.11** (stability identifies the radicals). *For every countable group,  $\text{Rad}_{\text{MF}}(G) \leq \text{Rad}_{\text{fd}}(G)$ . If  $G$  is matricially stable, then  $\text{Rad}_{\text{MF}}(G) = \text{Rad}_{\text{fd}}(G)$ ; in particular a stable group with  $\text{Rad}_{\text{fd}}(G) \neq 1$  is not MF.*

*Proof.* The first inclusion needs no stability. A finite-dimensional unitary representation  $\pi$  on  $\mathbb{C}^d$  amplifies to  $\pi^{\oplus n}$  on  $\mathbb{C}^{nd}$ ; the dimensions  $nd$  are strictly increasing, so  $(\pi^{\oplus n})_n$  is a homomorphism to  $\mathcal{U}(\mathcal{Q})$  in the sense of (2). Since  $\|\pi^{\oplus n}(x) - 1\| = \|\pi(x) - 1\|$  for every  $n$ , that sequence is null exactly when  $\pi(x) = 1$ . Hence an element of  $\text{Rad}_{\text{MF}}(G)$  is sent to the identity by every finite-dimensional representation.

For the converse inclusion, assume  $G$  matricially stable, let  $x \in \text{Rad}_{\text{fd}}(G)$ , and suppose toward a contradiction that a homomorphism  $\Theta$  to  $\mathcal{U}(\mathcal{Q})$  has  $\Theta(x) \neq 1$ . Choose unitary coordinate lifts  $U_n(g)$  of  $\Theta(g)$ . Since  $\Theta(x) - 1 \neq 0$  in the norm quotient, after passing to a subsequence there is  $\delta > 0$  with  $\|U_n(x) - 1\| \geq \delta$  for every  $n$ . The lifts form an operator-norm asymptotic representation, so stability gives genuine representations  $\pi_n$  on the same blocks with  $\|U_n(g) - \pi_n(g)\| \rightarrow 0$  for each fixed  $g$ . But  $x \in \text{Rad}_{\text{fd}}(G)$  gives  $\pi_n(x) = 1$  for every  $n$ , so  $\|U_n(x) - 1\| = \|U_n(x) - \pi_n(x)\| \rightarrow 0$ , against the uniform lower bound.  $\square$

For a stable group the two radicals therefore agree, so the homomorphisms to  $\mathcal{U}(\mathcal{Q})$  see exactly what the finite-dimensional representations see. With Theorem 3.3 this gives a second route to non-MF groups: every matricially stable group with  $\text{Rad}_{\text{fd}}(G) \neq 1$  fails to be MF.

**15.4. Orbit collapse.** The criteria of Section 11 bound the radical from below by torsion or rigidity already present in  $N_{\text{comp}}$ . This subsection gives a second application of one-sided compression, under a commuting-orbit hypothesis in place of centrality: every homomorphism to  $\mathcal{U}(\mathcal{Q})$  identifies an entire conjugation orbit of an involution, so a whole family of commutators enters the radical, and by Corollary 15.5 the radical computation reduces to the quotient by that family.

**Definition 15.12** (involutive compression witness). Let  $H$  be a group,  $L \leq H$  a subgroup, and  $s \in H$ . An element  $k \in H$  is an *involutive compression witness* for  $(L, s)$  if

- (W1)  $k^2 = 1$ ;
- (W2)  $s\gamma s^{-1}$  commutes with  $k$  for every  $\gamma \in L$ ;
- (W3) the conjugates  $\gamma_1 k \gamma_1^{-1}$  and  $\gamma_2 k \gamma_2^{-1}$  commute for all  $\gamma_1, \gamma_2 \in L$ .

The pair  $(L, s)$  determines the normal subgroup

$$D_{\text{coll}}(L, s) = \langle\langle [\gamma, k] : \gamma \in L, k \text{ a witness for } (L, s) \rangle\rangle_H.$$

In a permutational wreath product  $K^{(X)} \rtimes G$  with  $L \leq G$  compressed by  $s$  and  $x_0 \in X$  a point fixed by  $sLs^{-1}$ , every involution of the copy of  $K$  at  $x_0$  is a witness: copies at distinct points commute, and the compressed copy

fixes the point. The theorem below then identifies those copies along the whole  $L$ -orbit of  $x_0$  in every homomorphism to  $\mathcal{U}(\mathcal{Q})$ .

**Theorem 15.13** (involutive collapse). *Let  $H$  be countable, let  $L \leq H$  be a subgroup with property (T), and let  $s \in H$  satisfy  $s\gamma s^{-1} \in L$  for every  $\gamma \in L$ . Then*

$$D_{\text{coll}}(L, s) \leq \text{Rad}_{\text{MF}}(H).$$

*Proof.* Fix a witness  $k$  and suppose, toward a contradiction, that some homomorphism  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  has  $\Theta([\gamma_0, k]) \neq 1$  for some  $\gamma_0 \in L$ . Coordinate unitary lifts (Lemma 2.2) give an operator-norm asymptotic representation  $(U_{g,n})$  on blocks  $M_{d_n}(\mathbb{C})$ , and after passing to a subsequence there is  $\eta > 0$  with  $\|U_{[\gamma_0, k], n} - 1\| \geq \eta$  for every  $n$ .

*Step 1: approximation by commuting self-adjoint involutions.* Write  $v_\gamma = \gamma k \gamma^{-1}$  and let  $O = \{v_\gamma : \gamma \in L\}$  be the conjugation orbit of  $k$ ; by (W1), (W3), and asymptotic multiplicativity the models of the elements of  $O$  asymptotically commute and square to 1. The spectral sign of the Hermitian part of a unitary  $u$  is an exact self-adjoint involution at operator distance at most  $\|u^2 - 1\|$  from  $u$ . Enumerate  $O$  as  $v^{(1)}, v^{(2)}, \dots$  and correct in that order, carrying the induction hypothesis that after stage  $m$  the operators  $V_n(v^{(1)}), \dots, V_n(v^{(m)})$  are exact self-adjoint involutions commuting pairwise, and that no later stage alters them. That second clause is what makes the induction close, and it is what the construction supplies: at stage  $m$  the model  $U_{v^{(m)}, n}$  fails to commute with  $V_n(v^{(j)})$ ,  $j < m$ , by at most  $\| [U_{v^{(m)}, n}, U_{v^{(j)}, n}] - \| + 2 \| V_n(v^{(j)}) - U_{v^{(j)}, n} \|$ , the first term an asymptotic-commutation error of (W3) and the second the error already accumulated, so the total is a sum of  $m$  quantities each tending to 0. The joint commutant of  $V_n(v^{(1)}), \dots, V_n(v^{(m-1)})$  is a unital  $*$ -subalgebra — a finite intersection of joint spectral corners of exact commuting involutions — so compressing  $U_{v^{(m)}, n}$  into it moves it by at most that amount, and taking the spectral sign of the Hermitian part *inside that subalgebra* yields an exact self-adjoint involution commuting exactly with all earlier ones while leaving them untouched. The cost is the accumulated sum, linear in  $m$ ; each  $m$  is fixed before  $n$  grows, so the limits displayed next are taken at fixed  $m$ . The induction yields exact pairwise-commuting self-adjoint involutions  $V_n(v)$ ,  $v \in O$ , with

$$\|V_n(v) - U_{v,n}\| \longrightarrow 0, \quad \left\| U_{g,n} V_n(v) U_{g,n}^* - V_n(gv g^{-1}) \right\| \longrightarrow 0$$

for each fixed  $v \in O$  and  $g \in L$ . Write  $V_n(\gamma)$  for  $V_n(v_\gamma)$ ; it depends on  $\gamma$  only through  $v_\gamma$ .

*Step 2: a rank pseudometric on the orbit.* For commuting exact involutions  $e, f$  the self-adjoint displacement  $T = e - f$  satisfies  $T^3 = 4T$ , so its eigenvalues lie in  $\{0, \pm 2\}$ , the operator  $T^2/4$  is the orthogonal projection onto  $\text{ran } T$ , and

$$(10) \quad \text{Tr } |e - f|^2 = 4 \text{rank}(e - f) :$$

rank is an exact support count for the Hilbert–Schmidt displacement here. Hence  $d_n(\gamma, \mu) = \text{rank}(V_n(\gamma) - V_n(\mu))$  is an integer-valued pseudometric. It is eventually translation invariant,  $d_n(g\gamma, g\mu) = d_n(\gamma, \mu)$  for all large  $n$  and each fixed triple: the comparison is made between the projections  $T^2/4$ , whose ranks agree once they lie at operator distance below 1, and the displacements have operator norm at most 2, so a covariance error below  $1/8$  moves the associated projections by at most  $1/4$ .

*Step 3: normalization at the generators.* Property (T) makes  $L$  finitely generated; fix a Kazhdan pair  $(S, \kappa)$  with  $S$  finite symmetric and put  $k_n = \sum_{a \in S} d_n(1, a)$ . For each fixed  $\gamma$ , translation invariance and the triangle inequality along one fixed word for  $\gamma$  give  $d_n(1, \gamma) \leq |\gamma|_S k_n$  for all large  $n$ , where  $|\cdot|_S$  is the word length. If  $k_n = 0$  along a subsequence, the whole orbit collapses there to a single exact involution and  $\|U_{[\gamma_0, k], n} - 1\| \rightarrow 0$  along it, against the separation; so  $k_n \geq 1$  for all large  $n$ .

*Step 4: an approximate 1-cocycle.* This step and the next work with the *unnormalized* Hilbert–Schmidt, or Frobenius, norm  $\|x\|_F = \text{Tr}(x^*x)^{1/2}$ , so that  $\|x\|_F^2 = d_n \|x\|_2^2$  on  $M_{d_n}(\mathbb{C})$ ; the normalization is by the rank  $k_n$  below rather than by  $d_n$ . Set  $b_n(\gamma) = k_n^{-1/2}(V_n(\gamma) - V_n(1))$ . By (10) and Steps 2 and 3,  $\|b_n(\gamma)\|_F^2 = 4d_n(1, \gamma)/k_n \leq 4|\gamma|_S$  at every large stage, with total generator mass

$$(11) \quad \sum_{a \in S} \|b_n(a)\|_F^2 = 4$$

whenever  $k_n \geq 1$ . Conjugation by  $U_{a,n}$  is a Frobenius isometry. Fix  $a, \gamma \in L$ , put  $\mu = a^{-1}\gamma$ , and let  $k_n^{-1/2}D_n$  be the defect in the cocycle identity, so that

$$D_n = (V_n(\gamma) - V_n(1)) - U_{a,n}(V_n(\mu) - V_n(1))U_{a,n}^* - (V_n(a) - V_n(1)).$$

As written,  $D_n$  is a sum of three displacements of the commuting family of Step 1, one of them conjugated; rank is subadditive and invariant under unitary conjugation, so Step 2 with the bound of Step 3 gives

$$\text{rank } D_n \leq d_n(\gamma, 1) + d_n(\mu, 1) + d_n(a, 1) \leq (|\gamma|_S + |\mu|_S + |a|_S)k_n$$

eventually. Regrouped as

$$D_n = (V_n(\gamma) - U_{a,n}V_n(\mu)U_{a,n}^*) + (U_{a,n}V_n(1)U_{a,n}^* - V_n(a)),$$

the same  $D_n$  is a sum of two covariance errors of Step 1, so  $\|D_n\| \rightarrow 0$ . Both bounds are needed, because operator-norm control alone does not bound rank: two exact involutions at small operator distance can differ in full rank. Combining them in  $\text{Tr } |D|^2 \leq \text{rank}(D) \cdot \|D\|^2$  bounds the normalized defect mass by  $(|\gamma|_S + |\mu|_S + |a|_S) \|D_n\|^2 \rightarrow 0$ : the defect mass vanishes at the scale  $k_n$ . Only differences of the involutions enter what follows.

*Step 5: the limiting cocycle is a coboundary.* Fix a free ultrafilter  $\omega$  on  $\mathbb{N}$ , give  $M_{d_n}(\mathbb{C})$  the inner product  $\langle x, y \rangle = \text{Tr}(y^*x)$ , and let  $K_\omega$  be the Hilbert-space ultraproduct of these spaces along  $\omega$ . Conjugation is unitary for that inner product, and the dimension-free estimate  $\|\text{Ad } u_1 - \text{Ad } u_2\| \leq$

$2\|u_1 - u_2\|$  makes  $g \mapsto \text{Ad } U_{g,n}$  asymptotically multiplicative in the operator norm of  $B(M_{d_n}(\mathbb{C}))$ , so

$$\pi(g) = [\text{Ad } U_{g,n}]_\omega$$

is a unitary representation of  $H$  on  $K_\omega$ , exactly as in the proof of Theorem 3.4. The bound  $\|b_n(\gamma)\|_F^2 \leq 4|\gamma|_S$  of Step 4 makes  $b(\gamma) = [b_n(\gamma)]_\omega$  a well-defined vector of  $K_\omega$  for every  $\gamma \in L$ , and the vanishing of the normalized defect mass makes

$$b(a\mu) = b(a) + \pi(a)b(\mu) \quad (a, \mu \in L),$$

so  $b: L \rightarrow K_\omega$  is a 1-cocycle for  $\pi|_L$ . Since  $\lim_\omega$  is linear, (11) passes to the limit as the identity

$$(12) \quad \sum_{a \in S} \|b(a)\|^2 = 4.$$

By the Delorme–Guichardet theorem [BHV, Theorem 2.12.4], property (T) of  $L$  makes every 1-cocycle of every unitary representation a coboundary, so there is  $y \in K_\omega$  with

$$b(\gamma) = y - \pi(\gamma)y \quad (\gamma \in L).$$

*Step 6: transport by the conjugator  $s$ .* By (W2) the element  $sas^{-1}$  commutes with  $k$  for every  $a \in L$ , so  $v_{sas^{-1}} = k = v_1$ ; the assignment  $\gamma \mapsto v_\gamma$  of Step 1 then gives  $V_n(sas^{-1}) = V_n(1)$  at every stage, and hence  $b(sas^{-1}) = 0$ . Therefore  $y = \pi(sas^{-1})y$  for every  $a \in L$ : the vector  $y$  is fixed by  $\pi(sLs^{-1})$ .

What is needed is the reverse of the inclusion that  $sLs^{-1} \leq L$  supplies, namely

$$\text{Fix } \pi(sLs^{-1}) \subseteq \text{Fix } \pi(L),$$

and it is what the proof of Theorem 3.4 establishes. That proof builds, from the Kazhdan gap alone, the projection  $P$  of  $K_\omega$  onto  $\text{Fix } \pi(L)$ ; identifies the range of  $Q = \pi(s)P\pi(s)^*$  with  $\text{Fix } \pi(sLs^{-1})$ ; observes  $P \leq Q$ ; and concludes  $Q = P$  from Lemmas 3.1 and 3.2, the norm ultraproduct being finite. No sequence being transported enters that chain, and Lemma 3.2 is stated for arbitrary finite-dimensional Hilbert spaces, so rescaling the inner product on each coordinate — here from  $\text{tr}_{d_n}$  to  $\text{Tr}$  — changes nothing in it.

Hence  $y \in \text{Fix } \pi(L)$ , so  $b(a) = y - \pi(a)y = 0$  for every  $a \in L$ , contradicting (12). Therefore  $\Theta([\gamma, k]) = 1$  for every  $\gamma \in L$  and every witness  $k$ , and normality of these kernels extends the conclusion to all of  $D_{\text{coll}}(L, s)$ .  $\square$

*Remark 15.14* (the quantitative form). Steps 5 and 6 take an ultraproduct, and the transport and the contradiction do not need it: both can be assembled at one coordinate, with every tolerance fixed in advance. The one input that is still read off the limit is the constant  $C$  below. Property (T) then enters only through a bound uniform in  $\gamma$ ,

$$(13) \quad \|b_n(\gamma)\|_F \leq C \quad (\gamma \in L),$$

past some stage, with  $C$  depending on the Kazhdan pair alone; Step 4 supplies only the  $\gamma$ -dependent bound  $2|\gamma|_S^{1/2}$ , and uniformity is the whole content of (13). Uniformity is what (T) gives, in the quantitative form

of the Delorme–Guichardet argument [BHV, Section 2.12]: for a Kazhdan pair  $(S, \kappa)$  there is a constant  $C_0 = C_0(S, \kappa)$  such that every 1-cocycle  $b$  of every unitary representation of  $L$  satisfies  $\|b(\gamma)\| \leq C_0 \max_{a \in S} \|b(a)\|$  for all  $\gamma \in L$  — the fixed vector produced from the Kazhdan projection lies within  $C_0 \max_{a \in S} \|b(a)\|$  of the origin, and  $b(\gamma) = y - \pi(\gamma)y$  is at most twice that. Read on the limiting cocycle of Step 5 with (11), this gives  $C = 4C_0$ . That reading is the one place where the present assembly still consults the limit rather than a single stage: what is ultraproduct-free below is the transport and the contradiction, not the provenance of  $C$ . Put

$$\rho = \frac{1}{\sqrt{|S|} + 1}, \quad q = \frac{\rho}{24(C + 1)}, \quad \tau \leq \min\left\{\frac{\rho}{2}, \frac{(1 - \theta)\rho^2}{256(C + 1)}\right\},$$

with  $\theta < 1$  the spectral threshold below, and fix a single stage  $n$ , past the finitely many thresholds of Steps 1–4, at which the covariance errors of Step 1 are below  $\tau$ . Let  $P_n$  be the spectral projection of  $|S|^{-1} \sum_{a \in S} \text{Ad } U_{a,n}$  above  $\theta$  — the *Kazhdan corner* — and let  $W$  be the compression of  $b_n$  into it. Two estimates hold:

$$(14) \quad \|(\text{Ad } U_{a,n} - 1)P_n\| \leq q \quad (a \in S),$$

$$(15) \quad \|(1 - P_n)Q_n\| \leq q, \quad Q_n = \text{Ad } U_{s,n} P_n \text{Ad } U_{s,n}^*.$$

The first says the corner is almost fixed, by the Kazhdan gap. The second is the finite-stage form of  $P = \pi(s)P\pi(s)^*$ , and it comes in the two moves of Section B. First the easy containment  $\|(1 - Q_n)P_n\| \leq \varepsilon$ : for  $a \in S$  the one-sided relation gives  $sas^{-1} = \delta$  for a single  $\delta \in L$ , and

$$(1 - \text{Ad } U_{a,n}) \text{Ad } U_{s,n}^* P_n = \text{Ad } U_{s,n}^* (1 - \text{Ad } U_{\delta,n}) P_n,$$

whose norm the Kazhdan gap makes small. Each  $\delta$  is a word in  $S$ , and the gap controls the displacement it induces one letter at a time, so this is one more threshold per generator —  $|S|$  of them, all fixed before the stage is chosen. So the corner lies almost inside its image under  $s$ . Then the equal-rank reversal estimate of Section B turns that containment around:  $Q_n$  is a unitary conjugate of  $P_n$  and so has the same rank, and for two projections of equal finite rank a containment within  $\varepsilon < 1$  in one direction gives one within  $\varepsilon/\sqrt{1 - \varepsilon^2}$  in the other. That estimate is finiteness again and not a norm trick: the leakage bound makes the compression  $\text{ran } P_n \rightarrow \text{ran } Q_n$  injective, and equal rank promotes an injective endomorphism of a finite-dimensional space to a surjective one. Equal rank is what replaces the absence of a proper isometry, and the one-sided inclusion  $sLs^{-1} \leq L$  is the only group-theoretic input.

Compressing into the corner costs a capture term  $c = (C + 1)2\tau/(1 - \theta)$ , and one stage of transport bounds the displacement of  $W$  by each generator:

$$\text{Tr}|W - U_{a,n}WU_{a,n}^*|^2 \leq 18q^2C^2 + 16c \leq \left(\frac{\rho}{2}\right)^2,$$

the second inequality being the arithmetic of the choice of  $q$  and  $\tau$ . Since  $\|b_n(a) - W\|_F \leq \tau \leq \rho/2$ , this gives  $\|b_n(a)\|_F \leq \rho$  for every  $a \in S$  at once,

while (11) reads  $\sum_{a \in S} \|b_n(a)\|_F^2 = 4$  and  $\sum_{a \in S} \rho^2 = |S|/(\sqrt{|S|} + 1)^2 < 1$ : the single stage  $n$  already fails.

The commutation hypothesis (W3) is essential: it gives the joint spectral calculus behind the rank identity of Step 2. Anticommuting Clifford coordinates have no joint spectral labels; their collapse mechanism is the central sign of Theorem A. In  $\mathcal{Q}$ , Theorem 15.15 below removes the restriction to involutions, and Theorem 15.16 extends the collapse to witnesses of every finite order through their discrete Fourier idempotents.

The witness of Theorem 15.13 is a group element. The same collapse holds for arbitrary projections of  $\mathcal{Q}$ .

**Theorem 15.15** (projection collapse). *Let  $H$  be countable, let  $L \leq H$  have property (T), let  $s \in H$  satisfy  $s\gamma s^{-1} \in L$  for every  $\gamma \in L$ , and let  $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$  be a homomorphism. Let  $p \in \mathcal{Q}$  be a projection such that  $p$  commutes with  $\Theta(s\gamma s^{-1})$  for every  $\gamma \in L$  and the projections  $\Theta(\gamma)p\Theta(\gamma)^*$ ,  $\gamma \in L$ , commute pairwise. Then  $p$  commutes with  $\Theta(L)$ .*

*Proof.* The symmetry  $\sigma = 1 - 2p$  is a self-adjoint unitary of  $\mathcal{Q}$ ; it is written  $\sigma$  because  $w$  denotes the marked word of  $E$  throughout. The subgroup  $H' \leq \mathcal{U}(\mathcal{Q})$  generated by  $\Theta(H)$  and  $\sigma$  is countable. Inside  $H'$ , the image  $\Theta(L)$  has property (T),  $\Theta(s)$  compresses it,  $\sigma$  is an involution centralizing the compressed copy, and the  $\Theta(L)$ -conjugates of  $\sigma$  commute because the conjugated projections do. So  $\sigma$  is an involutive compression witness in  $H'$ , and Theorem 15.13 places every commutator  $[\Theta(\gamma), \sigma]$ ,  $\gamma \in L$ , in  $\text{Rad}_{\text{MF}}(H')$ . The inclusion  $H' \rightarrow \mathcal{U}(\mathcal{Q})$  is itself a homomorphism  $H' \rightarrow \mathcal{U}(\mathcal{Q})$ , so it maps each of these commutators to the identity:  $[\Theta(\gamma), \sigma] = 1$  in  $\mathcal{Q}$ , and  $\sigma = 1 - 2p$  gives  $[\Theta(\gamma), p]_- = 0$ .  $\square$

The witness here need not come from a group element: the projection is an arbitrary element of  $\mathcal{Q}$ . The relative commutant  $\Theta(sLs^{-1})' \cap \mathcal{Q}$  therefore cannot exceed  $\Theta(L)' \cap \mathcal{Q}$  by a projection with commuting  $L$ -orbit.

With those projections available, functional calculus removes the involutivity of the witness altogether.

**Theorem 15.16** (torsion collapse). *Let  $H$  be countable, let  $L \leq H$  have property (T), let  $s \in H$  satisfy  $s\gamma s^{-1} \in L$  for every  $\gamma \in L$ , and let  $k \in H$  have finite order. If the compressed copy  $sLs^{-1}$  centralizes  $k$  and the conjugates  $\gamma k \gamma^{-1}$ ,  $\gamma \in L$ , commute pairwise, then every commutator  $[\gamma, k]$ ,  $\gamma \in L$ , is 2-norm trivial in every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$ ; the normal closure of all such commutators, over all finite-order witnesses for  $(L, s)$ , lies in  $\text{Rad}_{\text{MF}}(H)$ , with equality whenever the corresponding quotient of  $H$  is MF.*

*Proof.* Let  $\Theta$  be a homomorphism to  $\mathcal{U}(\mathcal{Q})$ ,  $m$  the order of  $k$ , and  $u = \Theta(k)$ , a unitary with  $u^m = 1$ . Its discrete Fourier idempotents  $p_j = m^{-1} \sum_{t \in \mathbf{Z}/m} e^{-2\pi i j t / m} u^t$ , for  $j \in \mathbf{Z}/m$ , are self-adjoint idempotents of  $\mathcal{Q}$ . Each  $p_j$  is a polynomial in  $u$ , so whatever commutes with  $u$  commutes with every  $p_j$ : the represented compressed copy centralizes each  $p_j$ , and

the  $\Theta(L)$ -conjugates of each  $p_j$  commute pairwise because the conjugated witnesses do. Theorem 15.15 therefore makes every  $p_j$  commute with  $\Theta(L)$ , and Fourier reconstruction expresses  $u = \sum_j e^{2\pi i j/m} p_j$ , so  $u$  commutes with  $\Theta(L)$  and  $\Theta([\gamma, k]) = 1$  for every  $\gamma \in L$ .  $\square$

In the quotient of the Kazhdan–Clifford group of Theorem A by its central involution all lamps commute, and the moved base-point generator is itself an involutive witness.

**Theorem 15.17** (the commuting-generator quotient). *The quotient  $W/\langle\zeta\rangle$  of the Kazhdan–Clifford group by its central involution is sofic and is not MF. More precisely, the image of the commutator  $[v_1, tct^{-1}]$ , the inverse of the marked commutator  $u$ , in  $W/\langle\zeta\rangle$  is nontrivial and lies in  $\text{Rad}_{\text{MF}}(W/\langle\zeta\rangle)$ ; that last conclusion holds with  $W$  replaced by the ambient affine–Clifford group of any injective endomorphism  $\alpha$  of a countable property-(T) group and any base element outside the image of  $\alpha$ .*

*Proof.* Write  $q: W \rightarrow W/\langle\zeta\rangle$  and  $k = q(tct^{-1})$ , the moved base-point generator. Since  $q(\zeta) = 1$ , any two lamps commute in the quotient, so the  $q(L)$ -conjugation orbit of  $k$  commutes pairwise, where  $L = q(\bar{\Gamma})$  is the image of the base copy;  $k$  remains an involution, and the compressed copy  $q(t)Lq(t)^{-1}$  centralizes  $k$  because the compressed base fixes the moved point. Property (T) passes to the image  $L$ , so Theorem 15.13 places every commutator  $[q(\bar{v}_1), k]$  in the MF radical of the quotient. The commutator remains nontrivial in the sign quotient: in the signed finite-support model of the lamp kernel, the commutator is the product of the two distinct lamp basis elements at  $\tau o$  and  $j(\bar{v}_1)\tau o$ , whose configuration coordinate is nonzero, while every element of  $\langle\zeta\rangle$  has configuration coordinate zero.

The lamp kernel becomes abelian in the quotient:  $\text{Cl}(X)/\langle\zeta\rangle$  is the elementary abelian group  $\bigoplus_X \mathbb{Z}/2$  under the same  $\Sigma$ -action, so  $W/\langle\zeta\rangle \cong (\bigoplus_X \mathbb{Z}/2) \rtimes \Sigma$ . That kernel is again locally finite with finite level orbits, so the argument of Theorem D gives soficity unchanged.  $\square$

Thus  $W/\langle\zeta\rangle$  is a second finitely generated sofic group that is not MF. Theorem 15.13 applies to the finitely presented group  $E/\langle w \rangle$  in the same way.

*Proof of Theorem B.* Write  $q: E \rightarrow E/\langle w \rangle$ ,  $\bar{d} = q(d)$ , and  $\bar{L} = q(\iota(\mathcal{B}))$ . The element  $\bar{d}$  is an involution because  $c^2 = 1$ , and  $q(t)\bar{L}q(t)^{-1}$  centralizes  $\bar{d}$  because  $t\iota(\mathcal{B})t^{-1}$  commutes with  $c$  in  $E$ . Property (T) passes to  $\bar{L}$ , and  $q(t)$  compresses it. Conditions (W1) and (W2) of Definition 15.12 therefore hold. It remains to verify (W3).

The rotation letters commute with  $d$ , and  $t$  carries each  $v_i$  to  $v_i^2$ , so  $\iota(v_i^2)$  commutes with  $d$  as well. Hence  $\iota(g)d\iota(g)^{-1}$  depends only on the exponent vector of the translation part of  $g$  modulo 2, which leaves eight conjugates, indexed by  $(\mathbb{Z}/2)^3$ . By Theorem 5.1 the marked word is  $w = [d, \iota(v_1)d\iota(v_1)^{-1}]$ , the commutator attached to the class of  $v_1$ . The six

rotation words  $x, x^2, z, yz, yx, y^2x$  carry that class onto the six remaining nonzero classes, and each commutes with  $d$ ; conjugating the displayed identity by the corresponding  $\iota(r)$  exhibits every nonzero parity commutator as a conjugate of  $w$ , while the zero class contributes  $[d, d] = 1$ . Centrality of  $w$  places all eight in  $\langle w \rangle$ , and one further conjugation reduces an arbitrary pair of conjugates to this case. Thus  $\bar{d}$  is an involutive compression witness for  $(\bar{L}, q(t))$ , and Theorem 15.13 places  $[q(\iota(v_1)), \bar{d}]$  in  $\text{Rad}_{\text{MF}}(E/\langle w \rangle)$ .

Nontriviality passes through the homomorphism  $E \rightarrow W$  of Proposition 8.3: it carries  $[\iota(v_1), d]$  to the product of the Clifford generators at the two distinct points  $\tau o$  and  $j(\bar{v}_1)\tau o$ , and carries  $\langle w \rangle$  into  $\langle \zeta \rangle$ , whose configuration coordinate is zero. Hence  $[\iota(v_1), d] \notin \langle w \rangle$ , its image in  $E/\langle w \rangle$  is nontrivial, and Proposition 15.3 makes  $E/\langle w \rangle$  non-MF. Corollary 15.5 identifies  $\text{Rad}_{\text{MF}}(E)$  with the  $q$ -preimage of  $\text{Rad}_{\text{MF}}(E/\langle w \rangle)$ , so  $u$  belongs to it.  $\square$

Thus  $E/\langle w \rangle$  is a second finitely presented non-MF group.

**Corollary 15.18** (collapse computes the radical). *In the setting of Theorem 15.13, write  $q: H \rightarrow H/D_{\text{coll}}(L, s)$  for the quotient map. Then, unconditionally,*

$$\text{Rad}_{\text{MF}}(H) = q^{-1}(\text{Rad}_{\text{MF}}(H/D_{\text{coll}}(L, s))),$$

and if  $H/D_{\text{coll}}(L, s)$  is MF, then  $\text{Rad}_{\text{MF}}(H) = D_{\text{coll}}(L, s)$ .

*Proof.* Combine Theorem 15.13 with Corollaries 15.5 and 15.4.  $\square$

In the second case  $H/D_{\text{coll}}(L, s)$  is the universal MF quotient of  $H$  (Proposition 15.3): every homomorphism  $H \rightarrow \mathcal{U}(\mathcal{Q})$  factors through it, and it is MF.

**Theorem 15.19** (when the collapse subgroup is everything). *In the setting of Theorem 15.13:*

- (1) *Every homomorphism  $f$  from  $H$  to a countable MF group maps every element of  $D_{\text{coll}}(L, s)$  to the identity.*
- (2) *If  $D_{\text{coll}}(L, s) = H$ , then  $\text{Rad}_{\text{MF}}(H) = H$ ; every homomorphism from  $H$  to a countable MF group is trivial; and if  $H$  is nontrivial, then  $H$  is not MF.*

*Proof.* (1) By Theorem 15.13 and Lemma 15.2,

$$f(D_{\text{coll}}(L, s)) \leq f(\text{Rad}_{\text{MF}}(H)) \leq \text{Rad}_{\text{MF}}(f(H)).$$

The subgroup  $f(H)$  of the MF target is MF (Lemma 15.9(1)), so its radical is trivial by Proposition 15.3. (2) The radical statement is Theorem 15.13 with  $D_{\text{coll}}(L, s) = H$ ; triviality of every homomorphism follows from (1); and a nontrivial group with full radical is not MF by Proposition 15.3.  $\square$

A group as in (2) admits no nontrivial finite-dimensional unitary representation, no nontrivial MF quotient, and no nontrivial homomorphism to any countable residually finite group. Whether a small-cancellation construction produces a finitely presented group with  $D_{\text{coll}}(L, s) = H$ , or with a normal-Kazhdan defect equal to the whole group, is open.

**15.5. The reduced group algebra of  $E$ .** The stable finiteness in Theorem C comes from the following two standard facts.

**Lemma 15.20.** *Let  $G_1$  be a countable discrete group.*

- (1) *The canonical tracial state  $\tau$  on  $C_r^*(G_1)$  is faithful.*
- (2) *Every unital  $C^*$ -algebra  $A$  with a faithful tracial state is stably finite: for every  $k \geq 1$ , every isometry in  $M_k(A)$  is a unitary.*

*Proof.* (1) Realize  $C_r^*(G_1) \subseteq B(\ell^2 G_1)$  as the norm closure of the span of the left translation unitaries  $\lambda_g$ . Let  $\rho_g$  be the right translation unitaries,  $(\rho_g \xi)(h) = \xi(hg)$ . Each  $\rho_g$  commutes with each  $\lambda_{g'}$ , hence with every element of  $C_r^*(G_1)$ . The canonical trace is  $\tau(x) = \langle x\delta_e, \delta_e \rangle$ . If  $\tau(x^*x) = 0$  then  $\|x\delta_e\| = 0$ , so  $x\delta_e = 0$ , and then  $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$  for every  $g \in G_1$ . The vectors  $\delta_g$  are total in  $\ell^2 G_1$ , so  $x = 0$ .

(2) On  $M_k(A)$  the functional  $\tau_k([x_{ij}]) = \frac{1}{k} \sum_i \tau(x_{ii})$  is a state with the trace property  $\tau_k(xy) = \frac{1}{k} \sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$ , and it is faithful: the  $i$ -th diagonal entry of  $x^*x$  is  $\sum_j x_{ji}^*x_{ji}$ , so  $\tau_k(x^*x) = 0$  implies  $\tau(x_{ji}^*x_{ji}) = 0$ , hence  $x_{ji} = 0$ , for all  $i, j$ . If  $v \in M_k(A)$  satisfies  $v^*v = 1$ , then  $y := 1 - vv^*$  is a projection with  $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$ ; since  $y = y^*y$ , faithfulness gives  $y = 0$ , i.e.  $vv^* = 1$ .  $\square$

**15.6. A sofic non-MF group.** The same quotient that proves  $w \neq 1$  also provides the sofic non-MF group of Theorem D.

*Proof of Theorem D.* Write  $T = \varinjlim (\bar{\Gamma}, \bar{\alpha})$ , so  $\Sigma = T \rtimes \mathbb{Z}$ , and let  $B \leq \Sigma$  be the level-zero copy of  $\bar{\Gamma}$ . The image of  $\bar{\alpha}$  has finite index (at most eight, by the parity classes of the translation lattice). Thus every level  $\bar{\Gamma}_n \leq T$  is commensurable with  $B$ . The stable letter conjugates  $B$  onto  $\bar{\alpha}(\bar{\Gamma})$ , so all of  $\Sigma$  commensurates  $B$ . Orbit-stabilizer now shows that every  $\bar{\Gamma}_n$ -orbit in  $X = \Sigma/B$  is finite.

Consider a finite subset of  $\text{Cl}(X) \rtimes T$ . Its  $T$ -coordinates lie in one level  $\bar{\Gamma}_n$ . Each of its Clifford coordinates has finite  $\bar{\Gamma}_n$ -orbit: this holds for the central sign and for each generator  $c_x$ , hence for every Clifford word. The finitely many resulting orbits generate a finite  $\bar{\Gamma}_n$ -invariant subgroup  $L \leq \text{Cl}(X)$ , because  $\text{Cl}(X)$  is locally finite. Our original finite set therefore lies in  $L \rtimes \bar{\Gamma}_n$ , which is sofic (finite-by-sofic). A group is sofic as soon as each of its finitely generated subgroups is, so  $\text{Cl}(X) \rtimes T$  is sofic.

Reassociating the semidirect products gives

$$W = \text{Cl}(X) \rtimes (T \rtimes \mathbb{Z}) \cong (\text{Cl}(X) \rtimes T) \rtimes \mathbb{Z}.$$

The final group is sofic because a split extension of a sofic group by  $\mathbb{Z}$  is sofic. Truncate the integer levels to  $\mathbb{Z}/N$  and let the shift act cyclically. Induce the kernel's sofic approximations over this finite level set. The cocycle identity that makes the finite-quotient case exact now fails only on the wrap-around boundary. For a window shifted by  $k$ , that boundary occupies at most  $2|k|/N$  of the levels, which vanishes as  $N$  grows. This is the  $\mathbb{Z}$ -extension case of

the Elek–Szabó amenable-extension theorem [ES06]. Here  $\bar{\Gamma}$  is sofic: its six generators lie in  $\mathrm{GL}_4(\mathbb{Z})$ , the resulting integral subgroup is residually finite through its congruence quotients, and it is isomorphic to  $\bar{\Gamma}$ .

The map  $E \rightarrow W$  of Proposition 8.3 is surjective: its image contains  $\Sigma$  and the generator  $c_o$  at the base point, hence every Clifford generator by transitivity of the  $\Sigma$ -action on  $X$ . Since  $E$  is generated by  $v_1, x, y, z, t, c$  (by the Tietze calculation above), their six images generate  $W$ . The map sends  $w \in \mathrm{Rad}_{\mathrm{MF}}(E)$  to the nontrivial sign  $\zeta$ ; functoriality (Lemma 15.2) makes  $\zeta \in \mathrm{Rad}_{\mathrm{MF}}(W)$ . Hence  $W$  is not MF.  $\square$

Failure of extension permanence. MF groups are not closed under extensions. More precisely,  $W$  fits into an exact sequence

$$1 \longrightarrow \mathrm{Cl}(X) \longrightarrow W \longrightarrow \Sigma \longrightarrow 1$$

whose kernel and quotient are MF, while  $W$  is not. The concrete lamp kernel is locally finite, LEF, sofic, and MF, while the total witness is finitely generated and non-MF.

*Proof.* By Lemma 15.9(3), the Clifford group is locally finite, hence MF. The group  $\Sigma$  identifies with the subgroup of  $\mathrm{GL}_4(\mathbb{Q})$  generated by  $\bar{\Gamma}$  and  $D$ : the levels are  $D^{-n}\bar{\Gamma}D^n$ , and the determinant gives the  $\mathbb{Z}$ -coordinate. It is residually finite. At every odd modulus the determinant 8 of the dilation is invertible, so the congruence reduction of the integral affine model descends to conjugation by the dilation and extends over the direct limit and its shift. These reductions separate any nontrivial element; compare Mal'cev's theorem [Mal] for finitely generated linear groups. Thus  $\Sigma$  is MF by Lemma 15.9(2), whereas  $W$  is not MF by the preceding construction.  $\square$

Extensions by  $\mathbb{Z}$ . Composing the two canonical projections gives a surjection  $W \rightarrow \mathbb{Z}$  whose kernel is  $\mathrm{Cl}(X) \rtimes T$ , and this kernel is LEF, hence MF: every finite subset lies in  $L \rtimes \bar{\Gamma}_n$  for a finite invariant lamp subgroup  $L$  and one level, and that group is residually finite because the level acts on the finite group  $L$  through a finite quotient. Thus  $W$  is an extension of an LEF group by  $\mathbb{Z}$  and is not MF, while soficity is preserved by every extension with amenable quotient, as in Elek–Szabó [ES06]. The  $\mathbb{Z}$ -quotient case used above is the one proved in this paper, in the proof of Theorem D; the general permanence theorem is proved — from the invariant-mean definition of amenability, rather than quoted — in the accompanying development, and no argument in this paper depends on the general form.

An exact stably finite non-MF algebra. The same witness gives a concrete separable exact example:  $C_r^*(W)$  is exact, has a faithful tracial state, is stably finite, and is not MF. Indeed, the locally finite lamp kernel is amenable and hence exact, while the linear group  $\Sigma \leq \mathrm{GL}_4(\mathbb{Q})$  is exact [GHW]. Exact groups are closed under extensions [KW2], so  $W$  is exact; for discrete groups this is equivalent to exactness of the reduced group algebra [KW1]. The canonical trace and stable finiteness follow as in Lemma 15.20. Finally, an

MF embedding of  $C_r^*(W)$  would restrict to a faithful homomorphism to  $\mathcal{U}(\mathcal{Q})$  on the canonical group unitaries, contradicting that  $W$  is not MF.

A simple sofic envelope. There is a countable simple sofic group  $S_{\text{simp}}$  with

$$\text{Rad}_{\text{MF}}(S_{\text{simp}}) = S_{\text{simp}}.$$

Equivalently, every homomorphism from  $S_{\text{simp}}$  to an MF group is trivial.

*Proof.* By Elek–Szabó, every countable sofic group embeds in a countable simple sofic group [ES05, Theorem 1]. Embed  $W$  in such a group. Functoriality of the MF radical sends  $\zeta$  to a nontrivial element of the MF radical of the simple group. That radical is normal, so it is the whole group. Every homomorphism to an MF group maps the MF radical to 1.  $\square$

**15.7. Marked-limit closure.** The explicit clopen cylinder constructed in the next subsection is an instance of a general phenomenon: at every fixed number of marked generators, the MF property is closed under limits of marked groups, and its failure always has a finite certificate.

**Theorem 15.21** (the MF locus is closed). *Fix  $k \geq 1$  and consider the space of  $k$ -marked groups — quotients of the free group  $F_k$ , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators.*

- (1) *The marked groups with MF quotient form a closed set.*
- (2) *Equivalently, the non-MF marked groups form an open set.*
- (3) *Every non-MF  $k$ -marked group admits a finite-radius certificate: there is a radius  $R$  such that every  $k$ -marked group whose relations agree with it on the reduced words of length at most  $R$  is non-MF.*

*Proof.* A failure of MF for the quotient is a failure of normalized local approximation, and such a failure is witnessed on one finite subset of the quotient. Expressing the finitely many multiplications and inequalities of that subset by words in the marked generators produces finitely many relation constraints; every marked group agreeing on those constraints inherits the same failed local-model problem, so an entire basic clopen neighborhood fails MF. Enclosing the constraint words in a reduced-word ball gives the radius form.  $\square$

Thus failure of MF is always certified by one finite ball of the marked Cayley diagram.

**15.8. A finite universal Horn obstruction.** Theorem A can be read as one first-order fact about all MF groups at once: a finite quasi-identity, that is, a single universal implication between finitely many equations (a universal Horn sentence). Write  $R$  for the forty-one relators of (8), regarded as words in eight variables, and  $w$  for the marked word in those variables.

**Proposition 15.22** (finite quasi-identity). *Every MF group  $G$  satisfies the finite universal Horn sentence*

$$\forall x_1, \dots, x_8: \bigwedge_{r \in R} r(x_1, \dots, x_8) = 1 \implies w(x_1, \dots, x_8) = 1,$$

while the canonical generating tuple of  $E$  satisfies every relator in  $R$  and has  $w \neq 1$ .

*Proof.* A tuple satisfying the relators induces a homomorphism  $E \rightarrow G$  sending the marked word of  $E$  to  $w(x_1, \dots, x_8)$ . If  $G$  is MF, that image is trivial:  $w \in \text{Rad}_{\text{MF}}(E)$  by Theorem A, and the radical is functorial (Lemma 15.2). The canonical tuple of  $E$  satisfies  $R$  by definition, and its marked word is nontrivial by Theorem A.  $\square$

**Corollary 15.23** (a basic clopen set of non-MF marked groups). *In the space of eight-generator marked groups — quotients of the free group  $F_8$ , topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators — the marked groups satisfying all relators of  $R$  and  $w \neq 1$  form a nonempty clopen set consisting entirely of non-MF groups.*

*Proof.* Both conditions constrain finitely many values of the marked multiplication, so the set is clopen; it contains the marked group  $E$  by Theorem A, and Proposition 15.22 excludes every MF point.  $\square$

Thus the obstruction of Theorem A is finite, local, and locally constant on a basic clopen neighborhood in the marked-group topology.

**15.9. Undecidability of MF recognition.** Lemma 15.9(1) and Theorem A make MF a Markov property of finitely presented groups: the trivial group is MF, and by subgroup heredity the finitely presented group  $E$  embeds in no MF group. The Adian–Rabin theorem [Ra] therefore applies.

**Corollary 15.24** (undecidability). *In the recursive coding of finite presentations fixed above, suppose given a computable map sending each instance of the halting problem to a presentation code whose group is MF exactly when that instance halts. Then no algorithm decides, from a finite presentation, whether the presented group is MF, and the set of presentation codes of non-MF groups is not recursively enumerable.*

The coding is the fixed recursive one, not an arbitrary semantics, and that matters: a semantics is only required to attach a finitely presented group to each code, not to do so effectively, so an ineffective choice would make the statement true for reasons having nothing to do with groups. Pinning the coding removes that reading — the group is read syntactically off the code, and the only way to supply the hypothesis is to build presentations from machines.

The corollary reduces from the halting problem, whose undecidability and whose non-enumerable complement are proved in the accompanying

formalization, so the classical fact that some finitely presented group has unsolvable word problem is not needed: the reduction interface asks for a computable map and a correctness equivalence, and does not require its source to be a word problem. One object remains hypothetical, the map itself, and the corollary quantifies it rather than citing it. Constructing it is the Adian–Rabin argument [Ra] run over a group built from a machine; classically one takes the word problem of a finitely presented group with unsolvable word problem and applies Adian–Rabin to the Markov property “is MF”. That construction fixes one finite presentation  $P_0$  whose group has undecidable word problem, and gives a computable map taking a word  $w$  in the generators of  $P_0$  — the instance varies, the presentation does not — to a finite presentation whose group is trivial — hence MF — when  $w = 1$  in  $P_0$ , and contains  $E$  — hence non-MF, by subgroup heredity — when  $w \neq 1$ . The words representing 1 form a recursively enumerable set which, the word problem of  $P_0$  being undecidable, is not recursive; hence the words with  $w \neq 1$  form a set that is *not* recursively enumerable. An enumeration of the non-MF presentations would enumerate that set, making the word problem decidable. The argument uses undecidability of the word problem and takes the preimage of a set that is not recursively enumerable under the computable Adian–Rabin reduction. Theorem 15.21 certifies single non-MF groups by finite balls; the corollary says no algorithm classifies the positive and negative cases from presentations.

The phenomenon has infinite multiplicity in two senses. The groups  $E \times \mathbb{Z}^k$  for  $k \geq 0$  are finitely presented, pairwise nonisomorphic, and non-MF, each containing  $E$  (Lemma 15.9(1)). They are pairwise nonisomorphic because their abelianizations have distinct torsion-free ranks; the number of homomorphisms to  $\mathbb{Z}/2$  separates them as well. Beyond finite presentation, pairing  $E$  with a continuum of pairwise nonisomorphic finitely generated groups yields  $2^{\aleph_0}$  pairwise nonisomorphic finitely generated non-MF groups: each product  $E \times N$  is non-MF, and a fixed countable group  $X$  satisfies  $X \cong E \times N$  for at most countably many of the continuum many isomorphism types  $N$  — each such  $N$  embeds in  $X$  as a finitely generated subgroup, a countable group has only countably many finitely generated subgroups, and the types  $N$  are pairwise nonisomorphic. A map with countable fibres from a continuum-sized family has continuum-sized image. Products of pairwise nonisomorphic groups with a fixed factor can still be isomorphic, which is why the isomorphism types of the products are counted rather than read off the factors. The deduction uses only that the  $N$  are finitely generated, and such a continuum is constructed in the accompanying development, as a lamplighter over coset spaces of the free group rather than as a subdirect product of alternating groups. Neumann’s sharper two-generator form [Ne] is not needed here.

## 16. LIMITATIONS OF THE OPERATOR-NORM METHOD

The proof is specific to the operator norm. If multiplication is controlled only in normalized Hilbert–Schmidt norm, the induced adjoint operators on the Hilbert–Schmidt spaces need not be close in operator norm. Both transport proofs require exactly that operator-norm control: the ultraproduct proof needs it to obtain an honest representation on the ultraproduct, and the finite-stage proof of Section B needs it for the spectral estimates. Normalized Hilbert–Schmidt control therefore does not give the transport used here. The failure is exact and dimension-free: against the operator-norm estimate  $\|\text{Ad}(U) - \text{Ad}(V)\| \leq 2\|U - V\|$ , there are unitaries of arbitrarily small normalized Hilbert–Schmidt distance whose adjoints stay 2 apart in operator norm — the same constant, in the opposite direction.

Conversely, amplification by an identity block separates the two normalizations exactly. Padding a model with an identity block preserves every operator-norm quantity — the multiplicative defects and the pairwise separations are unchanged — while the normalized Hilbert–Schmidt distances decay with the padding ratio: below any prescribed bound, every approximate unitary representation can be replaced by one all of whose pairwise Hilbert–Schmidt distances are smaller than the bound. 2-norm triviality of an element alone therefore can never prove a group non-MF: an obstruction must produce an invariant substructure with its own scale, renormalized on its own range so that the normalization is unchanged by amplification, rather than reading a normalized trace off the ambient blocks. What such a substructure needs is its own normalization, not a large relative rank. Padding leaves the averaging corner  $q_n M_{d_n}(\mathbb{C}) q_n$  untouched while  $d_n$  grows, so its relative rank  $r_n/d_n$  may tend to zero while the argument uses the corner’s own normalized trace; the spectral projection  $q_n$  of Theorem 11.6 is normalized on its own range in the same way, and the collapse argument of Theorem 15.13 normalizes by the rank  $k_n$ .

This is a limitation of the method, not a tracial nonapproximation theorem for  $E$ . Alekseev–Thom ask whether commutants arising in sofic approximations of Kazhdan groups can be recovered from finite-dimensional coordinate subalgebras [AT, Open Problem 6.2]. Such a coordinate description is separate from the operator-norm theorem proved here.

Finite dimensionality of the coordinates is essential in a second, sharper sense: it cannot be replaced by stable finiteness of the coordinate algebras together with faithful traces. In a general faithfully traced unital  $C^*$ -algebra the transport argument would act on the trace Hilbert space by the left and right regular actions, and the von Neumann algebra these two actions generate can fail every finiteness property — for an infinite-dimensional  $\text{II}_1$  factor in standard form it is all bounded operators on the trace space. Constant coordinates  $A_n = C_r^*(E)$  give a counterexample to any such generalization: the left regular representation embeds  $E$  exactly and injectively into the corresponding matrix quotient, the element  $c$  centralizes the Kazhdan base

there, and conjugation by  $t$  produces  $tct^{-1}$ , whose commutator with the base is the element  $u$  with  $u^2 = w \neq 1$  of Theorem A. A faithfully-traced coordinate transport theorem would make that commutator trivial in the matrix quotient; no such theorem can hold. Precisely: Theorem 3.4 becomes false when its matrix coordinates are replaced by finite faithfully traced unital  $C^*$ -coordinates and its normalized Hilbert–Schmidt norm by the trace 2-norm, with all its remaining hypotheses unchanged.

**Questions.** The finite-normal criterion has a precise torsion limitation: every finite subgroup of a torsion-free group is trivial. Theorem 11.6 removes it: a normal property-(T) subgroup of  $N_{\text{comp}}$  obstructs homomorphisms to  $\mathcal{U}(\mathcal{Q})$  with no torsion anywhere. What remains for a torsion-free example is group-theoretic — a torsion-free finitely presented group whose subgroup  $N_{\text{comp}}$  contains a normal subgroup with property (T) — and Remark 5.2 records a small-cancellation route to such a group, over the torsion-free finitely presented Kazhdan group of [FFF], which removed the analogous torsion obstacle for nonsocicity.

- (1) What is  $\text{Rad}_{\text{MF}}(E)$ ? It properly contains  $\{1, w\}$  by Theorem B, and two reductions apply: Corollary 15.18, applied to  $E/\langle w \rangle$  with the witness of that proof, reduces the computation to the quotient by the resulting subgroup  $D_{\text{coll}}$ , and Corollary 15.4 then identifies the radical with the preimage of  $D_{\text{coll}}$  as soon as that quotient is MF. Whether it is MF is open: a faithful homomorphism from that quotient to  $\mathcal{U}(\mathcal{Q})$  could exist only if both the sign collapse of Theorem A and the commuting-orbit collapse failed to apply to it. By Theorem 3.3, whatever part of  $\mathfrak{D}(E, \iota(\mathcal{B}))$  survives in that quotient lies in the kernel of every finite-dimensional linear representation, and hence in the kernel of every homomorphism to a finite group; such a homomorphism would have to be injective on a subgroup that all of them send to the identity.
- (2) Is there a torsion-free finitely presented non-MF group? The analytic obstruction no longer requires torsion (Theorem 11.6); the remaining input is group-theoretic, and it suffices to construct a quotient  $q: E' \twoheadrightarrow Q$  of the ascending HNN extension  $E'$  formed as in Theorem 5.1 over Fournier-Facio’s torsion-free finitely presented property-(T) group [FFF], subject to five conditions:  $Q$  is finitely presented;  $Q$  has property (T);  $Q$  is torsion-free;  $Q$  is generated by two elements; and  $q(N_{\text{comp}}) = Q$ . Given such a  $q$ , the group  $Q$  is two-generated, finitely presented, torsion-free, Kazhdan, and equal to its own MF radical, so every homomorphism from  $Q$  to the unitary group of a matrix quotient is trivial, and every nontrivial quotient of  $Q$  likewise fails to be MF. By Theorem 15.19, a group that equals its own collapse defect admits no nontrivial homomorphism to any countable MF group at all. Meeting the five conditions at once is the open part; no unconditional torsion-free finitely presented non-MF group is

proved here. Hull’s small-cancellation theorem for acylindrically hyperbolic groups [Hu, Theorem 1.5] would deliver four of the five from a two-generated *suitable* subgroup  $S$  of  $N_{\text{comp}}$  — one whose induced action is non-elementary and which normalizes no finite subgroup — since the quotient it returns is obtained by adjoining finitely many relators, preserves element orders, and restricts to a surjection from  $S$ , while property (T) descends from a torsion-free Kazhdan lattice through a common quotient. What is not available is a suitable such  $S$ : one infinite-order element of the defect generating a hyperbolically embedded cyclic subgroup would supply it [Hu, Lemma 5.8], but the Bass–Serre action of an ascending HNN extension fixes an end of its Bass–Serre tree, and an acylindrical action is never quasi-parabolic [Os, Theorem 1.1], so that action does not certify it.

- (3) Soficity and hyperlinearity of  $E$  are not decided by the operator-norm method: the adjoint spectral estimates require operator-norm multiplicative control, which permutation and Hilbert–Schmidt models do not supply. That is a limitation of the method, not an open question: Theorem E settles both by an independent argument. What the operator-norm method does leave open for  $E$  is the size of its MF radical, item (1).
- (4) Does MF imply hyperlinearity? A faithful embedding into a matrix quotient need not make any coordinate trace faithful, so the defining MF representation does not immediately give a hyperlinear model. This question concerns the CDE convention fixed above; for stronger trace-controlled MF conventions the implication is built into the definition.
- (5) By Proposition 15.7, every family of theorems placing subgroups in the MF radical generates, by iteration through quotients, a closure contained in  $\text{Rad}_{\text{MF}}$ . For which groups do the rules of Theorems 11.3, 11.6, 15.13, and 15.15 compute the radical exactly?

#### APPENDIX A. THE MF EQUIVALENCES

An *approximate unitary representation* for a countable group  $G$ , a finite set  $F \subseteq G$ , a separation constant  $\delta > 0$ , and a tolerance  $\varepsilon > 0$  is a map  $V: G \rightarrow \mathcal{U}(r)$ , for some positive dimension  $r$ , such that

$$\begin{aligned} \|V(g)V(h) - V(gh)\| &\leq \varepsilon \quad (g, h \in F), \\ \|V(g) - V(h)\| &\geq \delta \quad (g \neq h \in F). \end{aligned}$$

The local formulation in Proposition 1.1 asks for such a model with  $\delta = 1$  for every finite  $F$  and every  $\varepsilon > 0$ .

*Proof of Proposition 1.1.* Lemma 2.2 identifies the unitary-sequence quotient with the unitary group of the  $C^*$ -quotient. Replacing an arbitrary finite coordinate set by a standard set of the same cardinality merely conjugates

that coordinate by a permutation unitary. This proves the first equivalence once the dimension convention is normalized.

*Dimension normalization.* Given positive dimensions  $d_n$ , replace the  $n$ th sequence of unitary lifts by

$$\tilde{V}_n(g) = I_{X_0} \oplus \cdots \oplus I_{X_{n-1}} \oplus V_n(g)$$

on  $X_0 \oplus \cdots \oplus X_n$ , of strictly increasing dimension  $D_n = d_0 + \cdots + d_n$ . Products, multiplicative defects, and pairwise operator-norm distances are unchanged, because block-diagonal operator norms are maxima and every earlier block is the same identity for all group elements. The construction therefore preserves and reflects the quotient relation and injectivity, and forgetting strict increase proves the converse.

Suppose first that  $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}((d_n)))$  is injective, and choose unitary lifts  $u_n(g)$  of  $\rho(g)$ . Fix a finite set  $F \subseteq G$  and  $\varepsilon > 0$ . For each ordered pair  $g \neq h$  in  $F$ , injectivity says that  $(u_n(g)^{-1}u_n(h))_n$  does not converge to the identity in operator norm. Hence there are  $\delta_{g,h} > 0$  and arbitrarily large coordinates at which

$$\|u_n(g) - u_n(h)\| \geq \delta_{g,h}.$$

Choose an integer  $N_{g,h}$  with  $8 < N_{g,h} \delta_{g,h}^2$ . An explicit tensor-power estimate then applies: whenever two unitaries are  $\delta_{g,h}$ -separated, some tensor power of exponent  $p \leq N_{g,h}$  separates them by more than 1 — amplifying a spectral gap of the relative unitary until it crosses the unit circle takes at most  $N_{g,h}$  steps once  $8 < N_{g,h} \delta_{g,h}^2$ . The quotient homomorphism law makes all multiplicative defects for pairs in  $F$  arbitrarily small at every sufficiently large coordinate. Choose, for each  $g \neq h$ , a coordinate with the displayed separation and with every required defect at most  $\varepsilon/N_{g,h}$ . Then the  $p$ th tensor power separates the designated pair by more than 1, while the tensor-product telescoping identity bounds each new defect by  $p\varepsilon/N_{g,h} \leq \varepsilon$ . Finally take the block sum of these finitely many pair-specific models. Operator norms of block diagonal matrices are maxima, so every pair in  $F$  is separated by at least 1 and every required product has defect at most  $\varepsilon$ . This gives the normalized approximate representation.

Conversely, approximate unitary representations with separation constant 1 for all finite sets and tolerances make the group *weakly* MF. Choose models along an exhaustion with errors tending to zero. Then each fixed pair  $g, h$  is eventually represented multiplicatively, and each fixed pair  $g \neq h$  is eventually separated by 1. Hence the classes define an injective homomorphism into the unitary-sequence quotient, with no condition on the dimension sequence. Weak MF in turn makes the MF radical of Definition 15.1 trivial, and the block-sum diagonalization in the proof of Proposition 15.3(2) then assembles the separating sequences into a single faithful homomorphism to  $\mathcal{U}(\mathcal{Q})$ . That diagonalization uses the dimension normalization above and nothing else from the present proposition, so the two arguments are not circular.  $\square$

## APPENDIX B. THE FINITE-STAGE TRANSPORT PROOF

This appendix proves Theorem 3.4 a second time, at finite stages. Conditional on the robust spectral-gap input of (i), every estimate below is an explicit inequality in  $\kappa$ ,  $|S|$ , the uniform bound on  $(x_n)$ , and the multiplicative defects of  $U_n$ . That input is the one place where a limit still enters: property (T) constrains exact unitary representations, while at a finite stage the matrix models only generate a representation of the free group on  $S$ , so no threshold in  $\kappa$ ,  $|S|$  and the defects on a fixed finite set can supply it. Making that step effective as well would need a sum-of-squares certificate for property (T) in the sense of [Oz].

*Finite-stage proof of Theorem 3.4.* Property (T) provides a finite symmetric generating set  $S \subseteq \Gamma$  with  $1 \in S$  and a Kazhdan constant  $\kappa \in (0, 1]$  for the pair  $(S, \kappa)$ . As in the proof of Section 3, each group element acts by conjugation,  $\text{Ad } U_n(g) \xi = U_n(g) \xi U_n(g)^*$ , on  $L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$ , and operator-norm almost multiplicativity of  $U_n$  makes  $g \mapsto \text{Ad } U_n(g)$  an operator-norm asymptotic representation on these Hilbert spaces.

*The almost-fixed spectral subspace.* Let  $H_n$  be the Hermitian part of the average  $\frac{1}{|S|} \sum_{s' \in S} \text{Ad } U_n(\iota(s'))$ , fix the threshold

$$\theta = \frac{1}{2} \left( \left( 1 - \frac{\kappa^2}{4|S|} \right) + 1 \right) < 1,$$

and let  $P_n$  be the spectral projection of  $H_n$  for the part of the spectrum above  $\theta$ . Two spectral estimates tie  $P_n$  to the group.

- (i) *Displacement.* For every  $\gamma \in \Gamma$ ,  $\|(\text{Ad } U_n(\iota(\gamma)) - 1) P_n\| \rightarrow 0$ . The mechanism is a robust spectral gap: for every pair  $a, b$  with  $1 - \kappa^2/(4|S|) < a$  and  $b < 1$ , once the multiplicative defects of  $U_n$  on the finitely many group elements involved are small enough, no eigenvalue of  $H_n$  lies in  $[a, b]$  — an eigenvector with intermediate eigenvalue would be a unit vector that the Kazhdan pair forbids: too fixed to be moved by  $\kappa$ , too moved to be fixed. Hence the spectrum of  $H_n$  above  $\theta$  is eventually within any prescribed  $\delta$  of 1, so a unit vector  $\eta \in \text{ran } P_n$  has  $\langle (1 - H_n)\eta, \eta \rangle \leq \delta$ ; since  $1 - H_n$  is the average of the positive operators  $1 - \text{Re } \text{Ad } U_n(\iota(s'))$ , each generator satisfies  $\frac{1}{2} \|(\text{Ad } U_n(\iota(s')) - 1)\eta\|_2^2 \leq |S| \delta$ , and a word in  $S$  of length  $\ell$  moves  $\eta$  by at most  $\ell\sqrt{2|S|\delta}$  plus the accumulated multiplicative defects. The projection  $P_n$  is the finite-stage substitute for the fixed-subspace projection.
- (ii) *Spectral concentration.* Write  $\xi_n \in L^2(M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$  for the vector  $x_n$ . The commutator hypothesis says each  $\text{Ad } U_n(\iota(s'))$  almost fixes  $\xi_n$ , so the residual  $(1 - H_n)\xi_n$  is Hilbert–Schmidt null. On  $\text{ran}(1 - P_n)$  the operator  $1 - H_n$  is bounded below by  $1 - \theta$ , and it commutes with  $P_n$ , so every vector  $\eta$  satisfies

$$\|(1 - P_n)\eta\|_2 \leq (1 - \theta)^{-1} \|(1 - H_n)\eta\|_2.$$

Applied to  $\eta = \xi_n$ , this shows  $\xi_n$  is asymptotically concentrated in  $\text{ran } P_n$ .

*Reversing the one-sided projection estimate.* Put  $T_n = \text{Ad } U_n(s)$  and  $Q_n = T_n P_n T_n^*$ , a projection of the same rank as  $P_n$ . The conjugation relation enters exactly once: for  $\gamma \in \Gamma$  the element  $s^{-1}$  pulls the almost-fixedness of a vector of  $\text{ran } P_n$  under  $\iota(\Gamma) \supseteq s\iota(\Gamma)s^{-1}$  back to almost-fixedness under each  $\iota(s')$ ,  $s' \in S$ . By the concentration estimate, the pulled-back vector is again almost in the subspace. Thus  $\text{ran } P_n$  is almost contained in its conjugate,  $\|(1 - Q_n)P_n\| \rightarrow 0$  — the finite-stage form of the one-sided inclusion  $P \leq VPV^*$ . Since  $\text{rank } P_n = \text{rank } Q_n$ , the equal-rank reversal estimate — for projections  $p, q$  of equal finite rank,  $\|(1 - q)p\| \leq \varepsilon < 1$  implies  $\|(1 - p)q\| \leq \varepsilon/\sqrt{1 - \varepsilon^2}$  — reverses the containment:

$$\|(1 - P_n)Q_n\| \rightarrow 0.$$

This is the middle implication of the display at the head of Section 3, and it is the finite-dimensional substitute for finiteness of a norm ultraproduct: a one-sided compression cannot enlarge the subspace.

*Combining the estimates.* Fix  $\gamma$  and  $\varepsilon > 0$ , let  $M \geq 1$  bound  $\|x_n\|$ , and put  $q = \min(1, \varepsilon/(40(M^2 + 1)))$ . For all large  $n$  the three estimates hold at tolerance  $q$ :

$$\|(1 - P_n)\xi_n\|_2^2 \leq q, \quad \|(1 - P_n)Q_n\| \leq q, \quad \|(A_n - 1)P_n\| \leq q,$$

where  $A_n = \text{Ad } U_n(\iota(\gamma))$ . Decompose the transported vector along  $P_n$  and  $Q_n$ :

$$T_n \xi_n = P_n Q_n T_n \xi_n + (1 - P_n)Q_n T_n \xi_n + (1 - Q_n)T_n \xi_n.$$

Since  $Q_n = T_n P_n T_n^*$  and  $T_n$  is conjugation by a unitary,

$$\|(1 - Q_n)T_n \xi_n\|_2 = \|T_n(1 - P_n)\xi_n\|_2 = \|(1 - P_n)\xi_n\|_2 \leq q^{1/2};$$

the middle term obeys  $\|(1 - P_n)Q_n T_n \xi_n\|_2 \leq \|(1 - P_n)Q_n\| \|\xi_n\|_2 \leq qM$ , and the displacement estimate moves the first term by at most  $q \|\xi_n\|_2 \leq qM$ . Applying  $A_n - 1$ , whose operator norm is at most 2, to the last two terms and using the displacement bound on the first gives

$$\|(A_n - 1)T_n \xi_n\|_2^2 \leq (qM + 2qM + 2q^{1/2})^2 \leq 18q^2 M^2 + 16q \leq \varepsilon.$$

Since  $T_n \xi_n$  is the vector of  $U_n(s)x_n U_n(s)^*$ , this proves the asserted convergence. Once the threshold of (i) is fixed, every bound used above is explicit in  $\kappa$ ,  $|S|$ ,  $M$ , and the multiplicative defects of  $U_n$  on the finitely many group elements involved.  $\square$

APPENDIX C. THE MAXIMAL GROUP  $C^*$ -ALGEBRA

For a countable group  $H$ , the maximal group  $C^*$ -algebra  $C_{\max}^*(H)$  is the completion of the group ring  $\mathbb{C}[H]$  in the norm

$$\|x\|_{\max} = \sup_{\pi} \|\pi(x)\|,$$

the supremum taken over unitary representations of  $H$ ; the supremum is finite because  $\|\pi(\sum_h \lambda_h h)\| \leq \sum_h |\lambda_h|$ . Write  $u_h$  for the canonical unitary of  $h \in H$ . We use only the universal property.

**Proposition C.1** (the universal property). *The canonical map  $h \mapsto u_h$  is injective, and for every unital  $C^*$ -algebra  $B$  and every homomorphism  $\rho: H \rightarrow \mathcal{U}(B)$  there is a unique unital  $*$ -homomorphism  $C_{\max}^*(H) \rightarrow B$  with  $u_h \mapsto \rho(h)$  for every  $h \in H$ .*

Injectivity comes from the left regular representation, uniqueness from density of  $\mathbb{C}[H]$ . (The formalization builds the same algebra as a subalgebra of a product indexed by a set of representations, since “all representations” is a proper class; that device is invisible here.)

The maximal and reduced algebras separate already at the level of finiteness, by an algebraic mechanism:

**Proposition C.2** (proper compression forbids finiteness). *Let  $A$  be a unital  $C^*$ -algebra containing a projection  $p$  and a unitary  $u$  such that the conjugate  $upu^*$  satisfies  $p(upu^*) = (upu^*)p = p$ , equivalently  $p \leq upu^*$ , and  $upu^* \neq p$ . Then  $A$  contains an isometry that is not a unitary; consequently  $A$  is not stably finite and has no faithful tracial state.*

*Remark C.3* (the maximal algebra under strict Kazhdan compression). For a group with a strict Kazhdan compression  $t\Gamma t^{-1} \subsetneq \Gamma$  — the base of  $E$  is the running example — the spectral projection of the averaged generator of the Kazhdan pair, built as in Section 3, exists in the maximal algebra, conjugation by  $u_t$  dominates it, and the domination is strict: at the quasi-regular coordinate  $\ell^2(G/t\Gamma t^{-1})$  the point mass at the base coset is fixed by the compressed subgroup and moved by every element outside it, so the two projections differ there. By Proposition C.2 the maximal algebra of such a group is then not stably finite, has no faithful trace, and in particular is neither residually finite-dimensional nor MF. The assembly runs through the spectral projection, in the maximal algebra, of the averaged generator built from the Kazhdan pair: the spectral gap places the projection, the quasi-regular coordinate separates it from its conjugate, and Proposition C.2 applies. For the reduced algebra of  $E$ , by contrast, the canonical trace is faithful and the algebra is stably finite (Theorem C), so its failure of MF is not a failure of finiteness; for the maximal algebra, infiniteness is forced by the strict compression before any MF considerations arise.

APPENDIX D. THE THIRTEEN-RELATOR PRESENTATION OF  $\mathrm{SL}_3(\mathbb{Z})$ 

A reader willing to take property (T) of  $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$  from the literature [BHV, Example 1.7.4(i)] may skip this appendix: nothing in the main line depends on it beyond that property, and the presentations themselves go back to [CRW, Theorem 2]. It is included because Theorem A is a statement about one explicitly displayed finite presentation, and we want every hypothesis of the obstruction verified from those relators rather than from an identification of the group they present. Proposition 7.1 derives property (T) for the twenty relators of (7) through the thirteen-relator presentation  $P_{13}$  below, so the completeness of  $P_{13}$  is what closes that route. The argument is elementary and self-contained, and the formalization follows it step for step.

**Theorem D.1** (completeness of the thirteen-relator presentation). *The canonical homomorphism  $P_{13} \rightarrow \mathrm{SL}_3(\mathbb{Z})$ , sending each generator  $e_{ij}$  to the corresponding elementary matrix, is an isomorphism.*

*Proof.* Write  $\pi: P_{13} \rightarrow \mathrm{SL}_3(\mathbb{Z})$  for the map in question, and

$$\omega_{12} = e_{12}e_{21}^{-1}e_{12}, \quad \omega_{13} = e_{13}e_{31}^{-1}e_{13}, \quad \omega_{23} = e_{23}e_{32}^{-1}e_{23}$$

for the three signed swaps;  $\omega_{12}$  is the Weyl element of the upper-left  $2 \times 2$  block, and the thirteenth relator is  $\omega_{12}^4 = 1$ . Let  $e_3$  be the third basis column of  $\mathbb{Z}^3$ , let  $\|\cdot\|_1$  be the  $\ell^1$  norm on  $\mathbb{Z}^3$ , and put

$$H_{\mathrm{par}} = \langle e_{12}, e_{21}, e_{31}, e_{32} \rangle \leq P_{13},$$

the subgroup on the four root letters whose matrices fix  $e_3$ .

*Surjectivity.* This is Euclidean elimination. Division with remainder in the first column of  $A \in \mathrm{SL}_3(\mathbb{Z})$  is a product of integer powers of elementary matrices, and iterating it on columns and then on rows reduces  $A$  to a diagonal sign matrix. Each such sign matrix is the square of a signed swap, and each signed swap is the displayed three-transvection word.

*Injectivity: a normal form.* Write an element of  $P_{13}$  as a word  $W = \ell_k \cdots \ell_1$  in the letters  $e_i^{\pm 1}$ . Given a column  $b \in \mathbb{Z}^3$ , the *trajectory* of  $W$  over  $b$  is the sequence  $b, \pi(\ell_1)b, \pi(\ell_2\ell_1)b, \dots, \pi(W)b$ , and  $W$  is *norm-monotone* over  $b$  when the  $\ell^1$  norms along that sequence do not decrease. Using only the thirteen relators, every  $W$  can be rewritten as

$$W = Gm, \quad m \in \langle \omega_{13}, \omega_{23} \rangle, \quad G \text{ norm-monotone over } b := \pi(m)e_3.$$

The rewriting is a descent, in the lexicographic order on  $\mathbb{N} \times \mathbb{N}$ , of the pair consisting of the largest height at which monotonicity fails and the position of the topmost such failure. One step splits  $W$  at that position and rewrites the offending pair of adjacent letters; the thirty-six ordered pairs of root families fall into four classes — cancelling, commuting, Steinberg, and braid — each closed by one of the displayed relators, and each strictly decreasing the pair. The thirty-six cases are enumerated in the accompanying development.

*Injectivity: a kernel word has constant trajectory.* Suppose  $\pi(W) = 1$ . The subgroup  $\langle \omega_{13}, \omega_{23} \rangle$  maps into the signed permutation matrices, so

$b = \pi(m)e_3$  is a signed basis column and  $\|b\|_1 = 1$ . The trajectory of  $G$  over  $b$  starts at  $b$  and ends at  $\pi(G)b = \pi(m)^{-1}b = e_3$ , again of norm one, and it is nondecreasing; hence it is constant. A root letter that preserves the  $\ell^1$  norm of a signed basis column fixes that column, so every letter of  $G$  fixes  $b$  — and, taking the whole word,  $\pi(m)e_3 = \pi(G)b = e_3$ , so  $\pi(m)$  fixes  $e_3$ .

*Injectivity: into the parabolic.* Conjugation by  $\omega_{13}$  and by  $\omega_{23}$  permutes the twelve letters  $e_i^{\pm 1}$ , by two explicit tables derived from the relators. Applying them letter by letter turns  $m^{-1}Gm$  into a word in root letters, each of which fixes  $\pi(m)^{-1}b = e_3$ . The root letters whose matrices fix  $e_3$  are exactly  $e_{12}^{\pm 1}, e_{21}^{\pm 1}, e_{31}^{\pm 1}, e_{32}^{\pm 1}$ , so  $m^{-1}Gm \in H_{\text{par}}$ .

*Injectivity: the subgroup  $\langle \omega_{13}, \omega_{23} \rangle$ .* The group  $\langle \omega_{13}, \omega_{23} \rangle$  acts on the six signed basis columns, and the orbit of  $e_3$  is all six of them, with coset representatives

$$1, \quad \omega_{13}, \quad \omega_{23}, \quad \omega_{13}^2, \quad \omega_{13}^3, \quad \omega_{23}\omega_{13}^2$$

carrying  $e_3$  to  $e_3, e_1, e_2, -e_3, -e_1, -e_2$  respectively. Twelve Schreier relations, one for each generator and coset, have the form  $\omega_{\bullet} r_j = r_j \omega_{12}^k$ : every Schreier element is a power of the block Weyl element  $\omega_{12}$ ; the twelve are listed in the accompanying development. The remaining Schreier relation is  $\omega_{23}^2 = \omega_{12}^2 \omega_{13}^2$ , the residue of the classical computation  $\{-1, -1\} = \omega_{12}^4$  in  $K_2(3, \mathbb{Z})$ , and it is available exactly because the thirteenth relator gives  $\omega_{12}^4 = 1$ . Hence every element of  $\langle \omega_{13}, \omega_{23} \rangle$  is  $r_j \omega_{12}^k$ ; if in addition its matrix fixes  $e_3$ , then  $r_j e_3 = e_3$  forces  $j = 0$ , leaving  $m = \omega_{12}^k \in H_{\text{par}}$ . With the previous step,

$$W = m \cdot (m^{-1}Gm) \in H_{\text{par}}.$$

*Injectivity: separation inside the parabolic.* Let  $N = \langle e_{31}, e_{32} \rangle$  and  $\Lambda = \langle e_{12}, e_{21} \rangle$ . The Steinberg relations make  $N$  abelian and normalized by  $\Lambda$ , and every element of  $H_{\text{par}}$  is  $e_{31}^a e_{32}^b \gamma$  with  $\gamma \in \Lambda$ . On  $\Lambda$  the displayed relators reduce to the braid presentation  $\langle p, q \mid pqp = qpq, (pqp)^4 = 1 \rangle$ , which maps isomorphically onto  $\text{SL}_2(\mathbb{Z})$ : it surjects onto  $\mathbb{Z}/2 * \mathbb{Z}/3$ , that quotient embeds by the Möbius ping-pong argument, and the residual central element is trivial by  $(pqp)^4 = 1$ . So  $\pi$  is injective on  $\Lambda$ . Now let  $W = e_{31}^a e_{32}^b \gamma \in H_{\text{par}}$  have  $\pi(W) = 1$ . The upper-left  $2 \times 2$  block of  $\pi(W)$  is the matrix of  $\gamma$ , so  $\gamma = 1$ ; the bottom row of  $\pi(W)$  is then  $(a, b, 1)$ , so  $a = b = 0$ . Hence  $W = 1$ , and  $\ker \pi$  is trivial.  $\square$

The separation step is applied only after the factor  $m$  has itself been placed in  $H_{\text{par}}$  and the two factors recombined: applied directly to  $m^{-1}Gm$  it would meet  $\pi(m^{-1}Gm) = \pi(m)^{-1}$  rather than 1.

*Remark D.2* (the classical identification). The eight relations (6) present  $\text{SL}_3(\mathbb{Z})$ : Theorem D.1 composed with the surjection of Step 2 identifies  $\mathcal{R}$  with  $\text{SL}_3(\mathbb{Z})$ , and the base  $\mathcal{B}$  is then the affine group  $\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$  — the surjection  $\mathcal{B} \rightarrow \bar{\Gamma}$  of Lemma 8.2 onto the matrix realization is an isomorphism. The presentations go back to [CRW, Theorem 2], with the matrix-labelled

generators also recorded in [CLV], and property (T) of the affine group is classical [BHV, Example 1.7.4(i)].

#### ACKNOWLEDGEMENTS

We thank Caleb Eckhardt for discussions clarifying the finite-quotient analogy that organizes Section 1. We thank Francesco Fournier-Facio for formulating the group-theoretic input as a Kazhdan group that is not co-Hopfian, together with its ascending HNN extension. We also thank him for the small-cancellation construction recorded in Remark 5.2. We further thank him for helpful discussions on the distinct roles of the operator and Hilbert–Schmidt norms. We thank Alon Lubotzky for pointing out that the interaction between those two norms also occurs in the earlier approach of Bachner–Dogon–Lubotzky [BDL].

#### REFERENCES

- [AT] V. Alekseev and A. Thom, *Centralizers of sofic approximations of Kazhdan groups*, preprint, 2026, [arXiv:2608.05362](#).
- [BDL] B. Bachner, A. Dogon, and A. Lubotzky, *On  $L^1$ -approximation of groups*, J. Algebra **702** (2026), 235–243, [doi:10.1016/j.jalgebra.2026.04.041](#).
- [Be] M. E. B. Bekka, *Amenable unitary representations of locally compact groups*, Invent. Math. **100** (1990), no. 2, 383–401, [doi:10.1007/BF01231192](#).
- [BHV] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan’s property (T)*, New Mathematical Monographs, vol. 11, Cambridge University Press, Cambridge, 2008.
- [BV] M. E. B. Bekka and A. Valette, *Kazhdan’s property (T) and amenable representations*, Math. Z. **212** (1993), no. 2, 293–299, [doi:10.1007/BF02571659](#).
- [BK] B. Blackadar and E. Kirchberg, *Generalized inductive limits of finite-dimensional  $C^*$ -algebras*, Math. Ann. **307** (1997), no. 3, 343–380, [doi:10.1007/s002080050039](#).
- [CDE] J. R. Carrión, M. Dadarlat, and C. Eckhardt, *On groups with quasidiagonal  $C^*$ -algebras*, J. Funct. Anal. **265** (2013), no. 1, 135–152, [doi:10.1016/j.jfa.2013.04.004](#), [arXiv:1210.4050](#).
- [CLV] M. D. E. Conder, G. Liversidge, and M. R. Vsemirnov, *Generating pairs for  $SL(n, \mathbb{Z})$* , J. Algebra **662** (2025), 123–137, [doi:10.1016/j.jalgebra.2024.08.008](#).
- [CRW] M. D. E. Conder, E. F. Robertson, and P. Williams, *Presentations for 3-dimensional special linear groups over integer rings*, Proc. Amer. Math. Soc. **115** (1992), no. 1, 19–26, [doi:10.2307/2159559](#).
- [Dad] M. Dadarlat, *Obstructions to matricial stability of discrete groups and almost flat  $K$ -theory*, Adv. Math. **384** (2021), Paper No. 107722, [doi:10.1016/j.aim.2021.107722](#).
- [DGLT] M. De Chiffre, L. Glebsky, A. Lubotzky, and A. Thom, *Stability, cohomology vanishing, and nonapproximable groups*, Forum Math. Sigma **8** (2020), Paper No. e18, [doi:10.1017/fms.2020.5](#).
- [ES05] G. Elek and E. Szabó, *Hyperlinearity, essentially free actions and  $L^2$ -invariants: the sofic property*, Math. Ann. **332** (2005), no. 2, 421–441, [doi:10.1007/s00208-005-0640-8](#), [arXiv:math/0408400](#).
- [ES06] G. Elek and E. Szabó, *On sofic groups*, J. Group Theory **9** (2006), no. 2, 161–171, [doi:10.1515/JGT.2006.011](#), [arXiv:math/0305352](#).
- [FFF] F. Fournier-Facio, *A torsion-free non-sofic group*, preprint, 2026, [arXiv:2608.02025](#).

- [FGH] A. Fox, I. Goldbring, and B. Hart, *Locally universal  $C^*$ -algebras with computable presentations*, J. Funct. Anal. **287** (2024), no. 12, Paper No. 110652, [doi:10.1016/j.jfa.2024.110652](#), [arXiv:2303.02301](#).
- [Fri] T. Fritz, *On infinite-dimensional state spaces*, J. Math. Phys. **54** (2013), no. 5, 052107, [doi:10.1063/1.4807079](#).
- [GKEMP] D. Gao, S. Kunnawalkam Elayavalli, A. Manzoor, and G. Patchell, *A new source of purely finite matricial fields*, preprint, 2026, [arXiv:2603.24502](#).
- [Gl] L. Glebsky, *Extension of a residually finite group by a residually finite group is weakly sofic*, preprint, 2019, [arXiv:1910.08631](#).
- [GR] L. Glebsky and L. M. Rivera, *Sofic groups and profinite topology on free groups*, J. Algebra **320** (2008), no. 9, 3512–3518, [doi:10.1016/j.jalgebra.2008.08.008](#).
- [GHW] E. Guentner, N. Higson, and S. Weinberger, *The Novikov conjecture for linear groups*, Publ. Math. Inst. Hautes Études Sci. **101** (2005), 243–268, [doi:10.1007/s10240-005-0030-5](#).
- [HT] U. Haagerup and S. Thorbjørnsen, *A new application of random matrices:  $\text{Ext}(C_{\text{red}}^*(F_2))$  is not a group*, Ann. of Math. (2) **162** (2005), no. 2, 711–775, [doi:10.4007/annals.2005.162.711](#), [arXiv:math/0212265](#).
- [Hu] M. Hull, *Small cancellation in acylindrically hyperbolic groups*, Groups Geom. Dyn. **10** (2016), 1077–1119, [doi:10.4171/GGD/377](#), [arXiv:1308.4345](#).
- [JNVWY] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen,  $\text{MIP}^* = \text{RE}$ , Commun. ACM **64** (2021), no. 11, 131–138, [doi:10.1145/3485628](#), [arXiv:2001.04383](#).
- [KW1] E. Kirchberg and S. Wassermann, *Exact groups and continuous bundles of  $C^*$ -algebras*, Math. Ann. **315** (1999), no. 2, 169–203, [doi:10.1007/s002080050364](#).
- [KW2] E. Kirchberg and S. Wassermann, *Permanence properties of  $C^*$ -exact groups*, Doc. Math. **4** (1999), 513–558, [doi:10.4171/DM/67](#).
- [Ku16] G. Kun, *On sofic approximations of property (T) groups*, preprint, 2016, [arXiv:1606.04471](#).
- [KT19] G. Kun and A. Thom, *Inapproximability of actions and Kazhdan’s property (T)*, preprint, 2019, [arXiv:1901.03963](#).
- [KT26] G. Kun and A. Thom, *Nonsofic wreath products of residually finite groups*, preprint, 2026, [arXiv:2608.06222](#).
- [Lan] E. C. Lance, *On nuclear  $C^*$ -algebras*, J. Funct. Anal. **12** (1973), 157–176.
- [Lo] T. A. Loring, *Lifting solutions to perturbing problems in  $C^*$ -algebras*, Fields Institute Monographs, vol. 8, American Mathematical Society, Providence, RI, 1997.
- [LuO] A. Lubotzky and I. Oppenheim, *Non  $p$ -norm approximated groups*, preprint, 2018, [arXiv:1807.06790](#).
- [Mal] A. I. Mal’cev, *On isomorphic matrix representations of infinite groups*, Mat. Sb. **8(50)** (1940), 405–422; English translation, Amer. Math. Soc. Transl. (2) **45** (1965), 1–18.
- [MdIS] M. Magee and M. de la Salle,  *$\text{SL}_4(\mathbb{Z})$  is not purely matricial field*, C. R. Math. Acad. Sci. Paris **362** (2024), 903–910, [doi:10.5802/crmath.617](#), [arXiv:2312.03220](#).
- [NST] N. Nikolov, J. Schneider, and A. Thom, *Some remarks on finitarily approximable groups*, J. Éc. polytech. Math. **5** (2018), 239–258, [doi:10.5802/jep.69](#).
- [Ne] B. H. Neumann, *Some remarks on infinite groups*, J. London Math. Soc. **12** (1937), 120–127, [doi:10.1112/jlms/s1-12.46.120](#).
- [NT] T. Netzer and A. Thom, *Kazhdan’s property (T) via semidefinite optimization*, Exp. Math. **24** (2015), no. 3, 371–374, [doi:10.1080/10586458.2014.999149](#), [arXiv:1411.2488](#).
- [OAI] OpenAI, *Nonsofic groups exist*, in *Ten advances in mathematics and theoretical computer science*, Chapter 3, 78–95, 2026, <https://cdn.openai.com/pdf/ten-proofs-oai.pdf>.

- [Os] D. Osin, *Acylindrically hyperbolic groups*, Trans. Amer. Math. Soc. **368** (2016), no. 2, 851–888, [arXiv:1304.1246](#).
- [Oz] N. Ozawa, *Noncommutative real algebraic geometry of Kazhdan’s property (T)*, J. Inst. Math. Jussieu **15** (2016), no. 1, 85–90, [doi:10.1017/S1474748014000309](#), [arXiv:1312.5431](#).
- [Ra] M. O. Rabin, *Recursive unsolvability of group theoretic problems*, Ann. of Math. (2) **67** (1958), 172–194.
- [Sc] C. Schafhauser, *Finite dimensional approximations of certain amalgamated free products of groups*, preprint, 2023, [arXiv:2306.02498](#).
- [Sh] T. Shulman, *The MF property for amalgamated free products*, preprint, 2026, [arXiv:2603.13564](#).
- [Slo17] W. Slofstra, *The set of quantum correlations is not closed*, Forum Math. Pi **7** (2019), Paper No. e1, [doi:10.1017/fmp.2018.3](#).
- [Slo18] W. Slofstra, *A group with at least subexponential hyperlinear profile*, preprint, 2018, [arXiv:1806.05267](#).
- [SV] W. Slofstra and T. Vidick, *Entanglement in non-local games and the hyperlinear profile of groups*, Ann. Henri Poincaré **19** (2018), no. 10, 2979–3005, [doi:10.1007/s00023-018-0718-y](#).
- [Th] A. Thom, *Finitary approximations of groups and their applications*, Proceedings of the International Congress of Mathematicians — Rio de Janeiro 2018, [arXiv:1712.01052](#).
- [TWW] A. Tikuisis, S. White, and W. Winter, *Quasidiagonality of nuclear  $C^*$ -algebras*, Ann. of Math. (2) **185** (2017), no. 1, 229–284, [doi:10.4007/annals.2017.185.1.4](#), [arXiv:1509.08318](#).